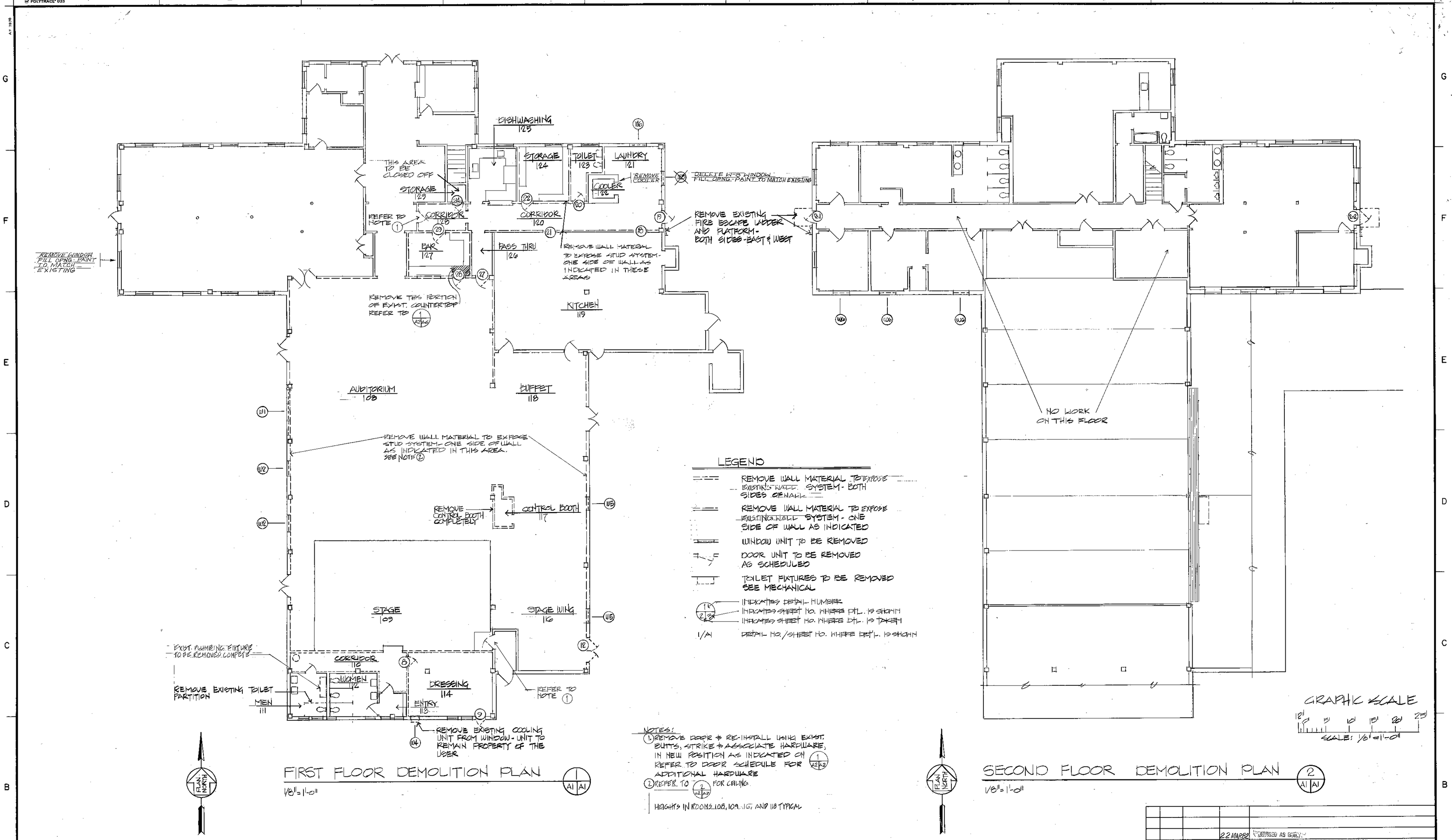


SYM	NO.	ACTION	DATE	DESCRIPTION OF REVISION
			22 MAR 82	REVISED AS BUILT

KILLEBREW/RUCKER/ASSOC., INC.
600 PETROLEUM BLDG.
WICHITA FALLS, TEXAS

U.S. ARMY ENGINEER DISTRICT, FORT WORTH
CORPS OF ENGINEERS
FORT WORTH, TEXAS

DESIGNED BY: TEY
DRAWN BY: TEY, KW
CHECKED BY: SHG
SUBMITTED BY:
ENGINEER:

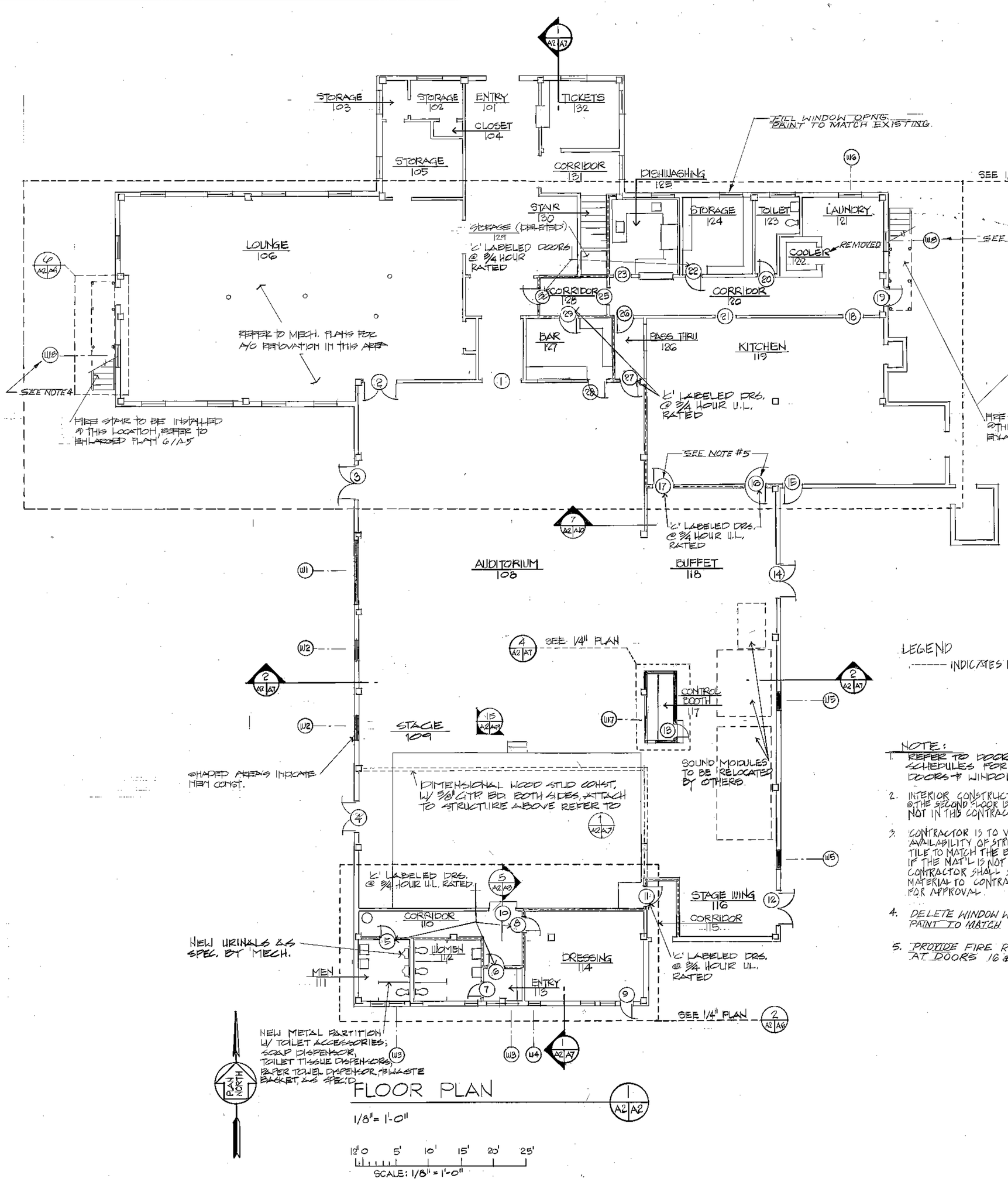
FORT SAM HOUSTON
SAN ANTONIO, TEXAS

UPGRADE BUILDING 2652
REPAIR BUILDING 2652

DEMOLITION PLANS

INV. NO. DACA63-81-B-0071 DATED: 2/13/81
CONTR. NO. DACA63-81-C-0080
DRAWING NUMBER
SHEET NO. A1 OF 10
SEQUENCE NO. 2

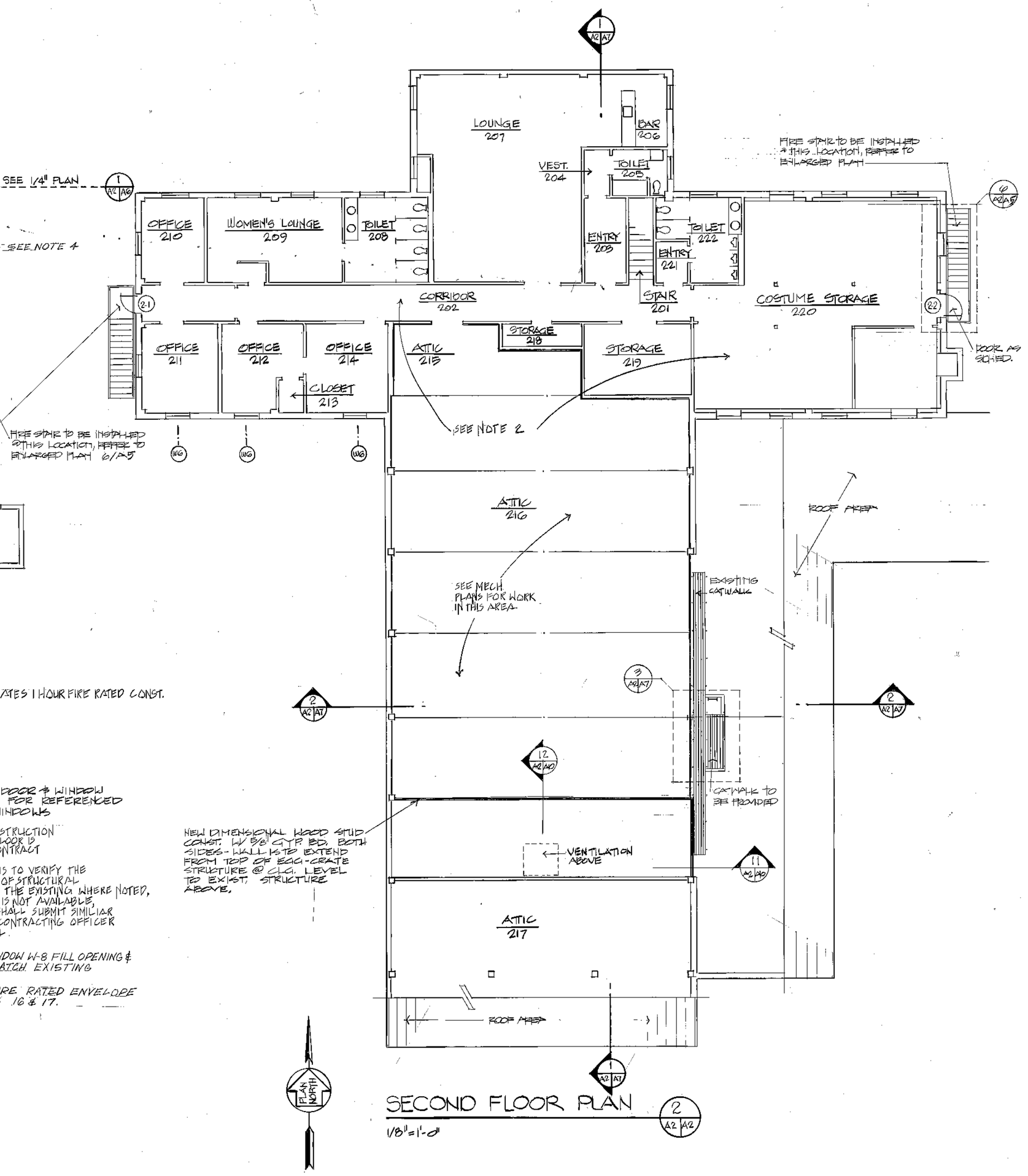
CONTR. NO. DACA63-81-C-0080



FLOOR PLAN

1/8" = 1'-0"

SCALE: 1/8" = 1'-0"



SECOND FLOOR PLAN

1/8" = 1'-0"

LEGEND

INDICATES 1 HOUR FIRE RATED CONST.

NOTE:

1. REFER TO DOOR & WINDOW SCHEDULES FOR REFERENCED DOORS & WINDOWS
2. INTERIOR CONSTRUCTION @ THE SECOND FLOOR IS NOT IN THIS CONTRACT
3. CONTRACTOR IS TO VERIFY THE AVAILABILITY OF STRUCTURAL TILE TO MATCH THE EXISTING. WHERE NOTED, IF THE MAT'L IS NOT AVAILABLE, CONTRACTOR SHALL SUBMIT SIMILAR MATERIAL TO CONTRACTING OFFICER FOR APPROVAL.
4. DELETE WINDOW W-8 FILL OPENING & PAINT TO MATCH EXISTING
5. PROVIDE FIRE RATED ENVELOPE AT DOORS 16 & 17.

SYM.	DG. NO.	ACTION	DATE	DESCRIPTION OF REVISION
			22 MAR 82	ISSUED AS BIDS

KILLEBREW/RUCKER/ASSOC., INC.
600 PETROLEUM BLDG.
WICHITA FALLS, TEXAS

U.S. ARMY ENGINEER DISTRICT, FORT WORTH
CORPS OF ENGINEERS
FORT WORTH, TEXAS

DESIGNED BY: TEY
DRAWN BY: TEY
CHECKED BY: SHG

FORT SAM HOUSTON
SAN ANTONIO, TEXAS

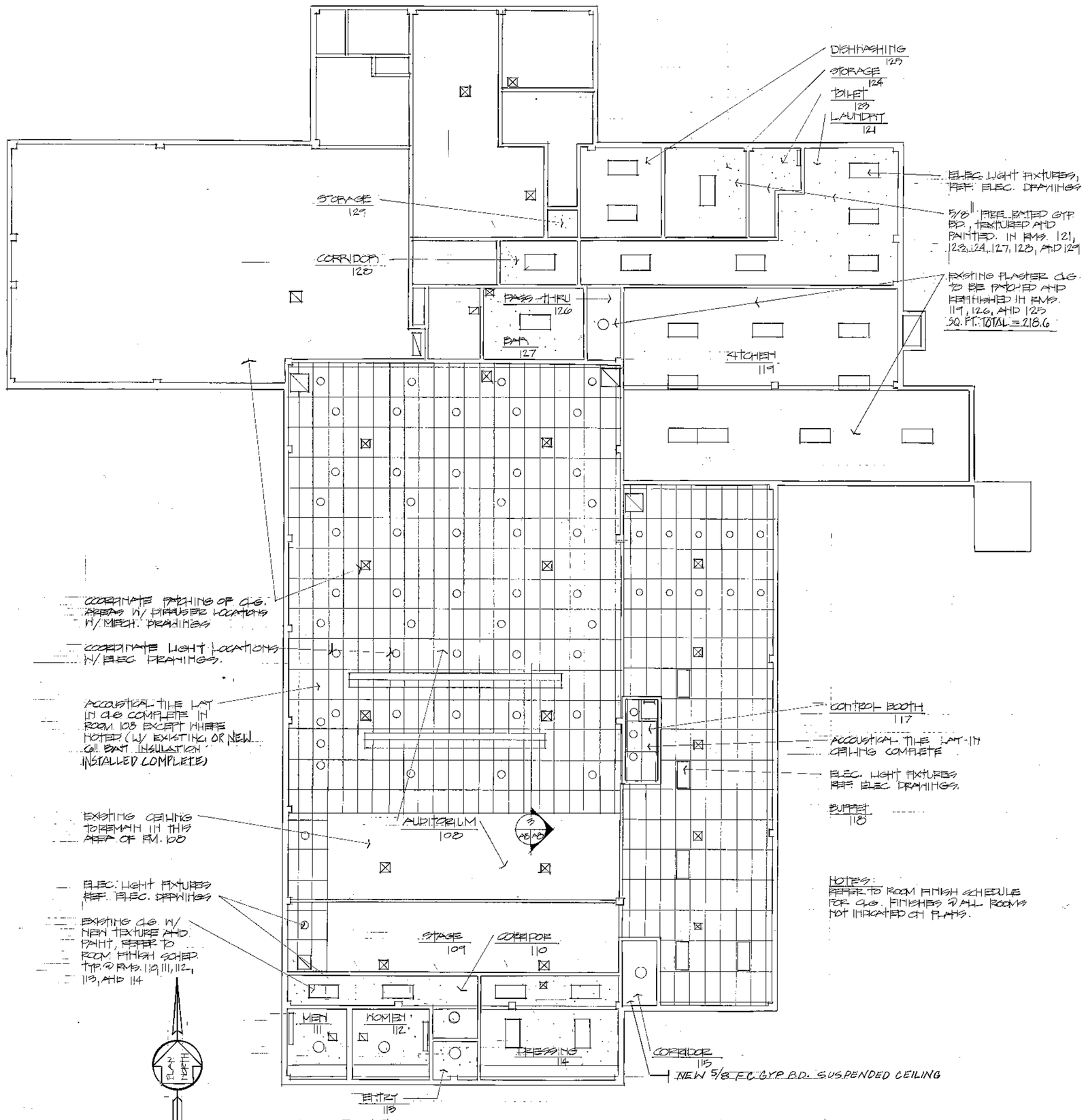
UPGRADE BUILDING 2652
REPAIR BUILDING 2652

RENOVATION PLANS

SUBMITTED BY:
ENGINEER:

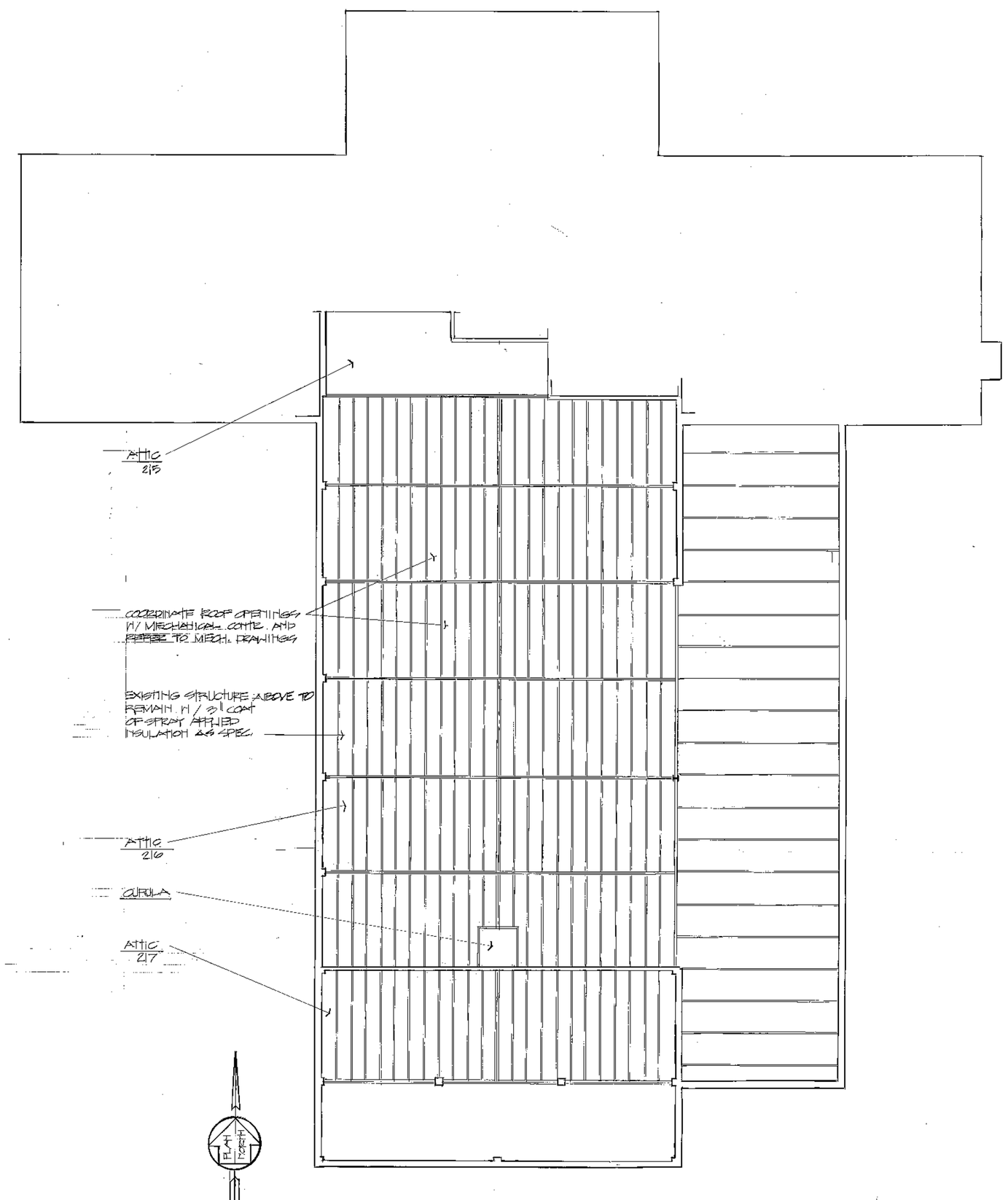
INV. NO. DACA 63-81-B-0071 DATED: 2/13/81
CONTR. NO. DACA 63-81-C-0080
DRAWING NUMBER
SHEET NO. A2 OF 10
SEQUENCE NO. 7

CONTR. NO. DACA63-81-C-0080



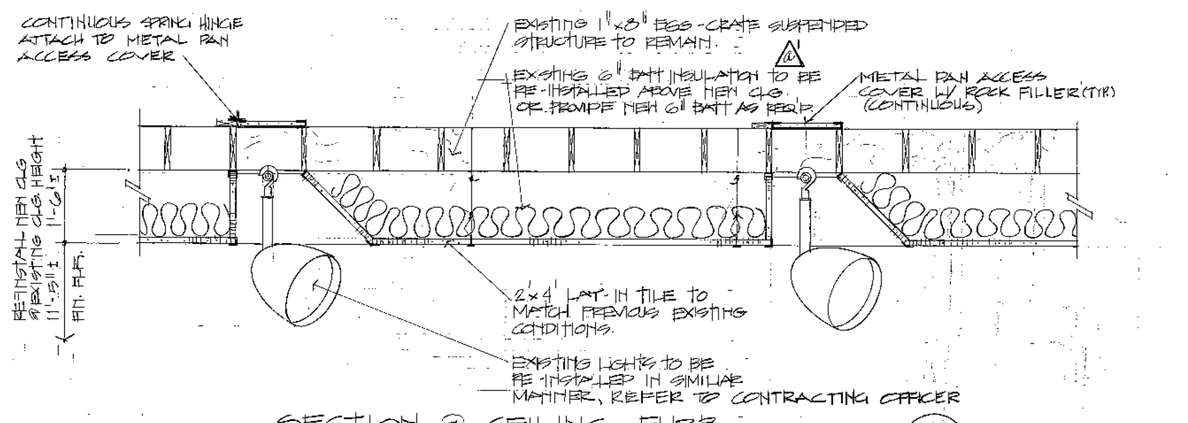
FIRST FLOOR REFLECTED CEILING PLAN (1 / A8/A8)

1/8"=1'-0"



SECOND FLOOR - REFLECTED CEILING PLAN (2 / A8/A8)

1/8"=1'-0"



SECTION @ CEILING FURR (3 / A8/A8)

3/4"=1'-0"

GRAPHIC SCALES

SCALE: 1/8"=1'-0"

SCALE: 3/4"=1'-0"

SYM.	SQ. NO.	ACTION	DATE	DESCRIPTION OF REVISION
			22 MAR 82	"DRAWING AS BUILT"
	AM #0001		13 MAR. 81	CORRECTED NOTE TO INCLUDE "NEW 6" BATT INSUL."

KILLEBREW/RUCKER/ASSOC., INC.
600 PETROLEUM BLDG.
WICHITA FALLS, TEXAS

U.S. ARMY ENGINEER DISTRICT, FORT WORTH
CORPS OF ENGINEERS
FORT WORTH, TEXAS

DESIGNED BY: TBT
DRAWN BY: TBT, KEH
CHECKED BY: SHG
SUBMITTED BY:
ENGINEER:

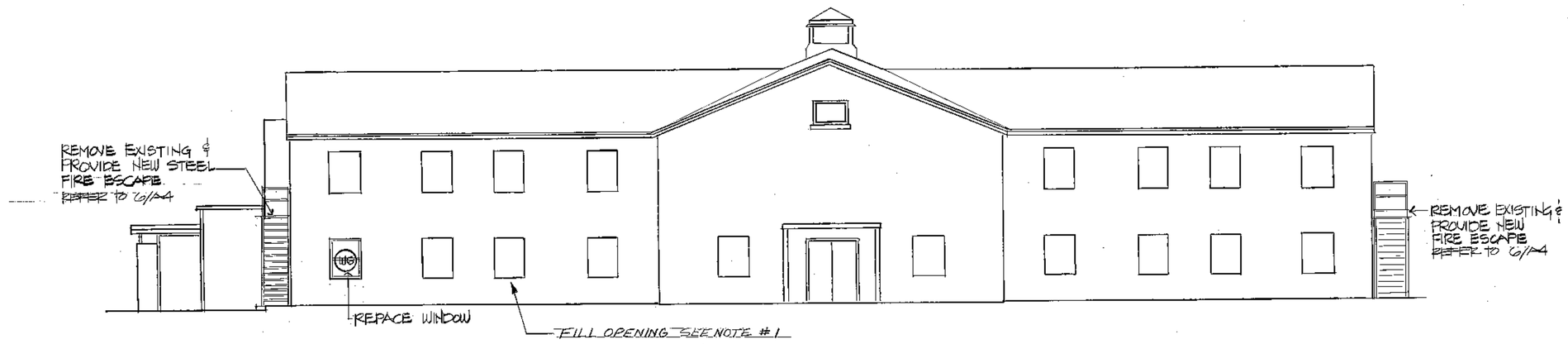
FORT SAM HOUSTON
SAN ANTONIO, TEXAS

UPGRADE BUILDING 2652
REPAIR BUILDING 2652

REFLECTED CEILING PLAN

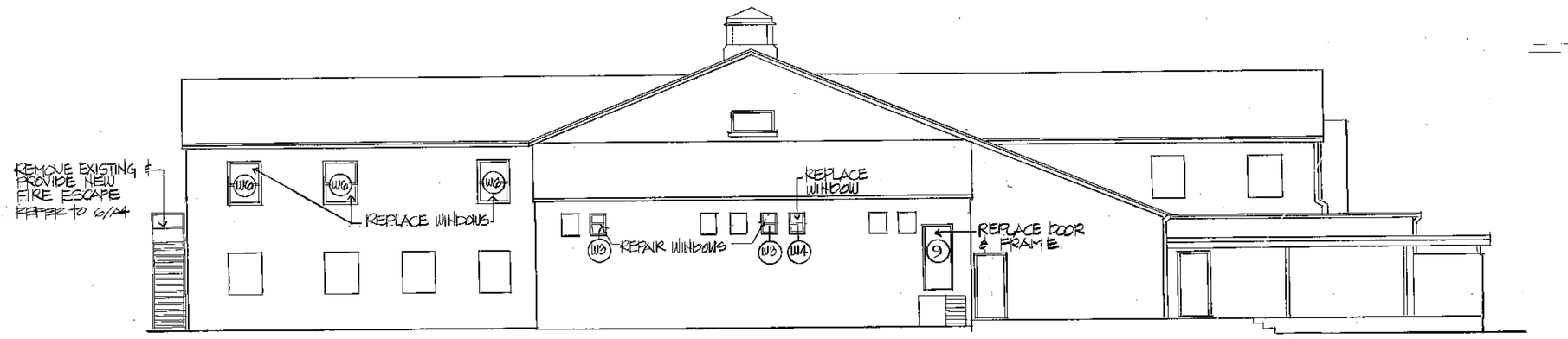
INV. NO. DACA63-81-B-0071 DATED: 2/13/81
CONTR. NO. DACA63-81-C-0080
DRAWING NUMBER
SHEET NO. A8 OF 10
SEQUENCE NO. 9

CONTR. NO. DACA63-81-C-0080



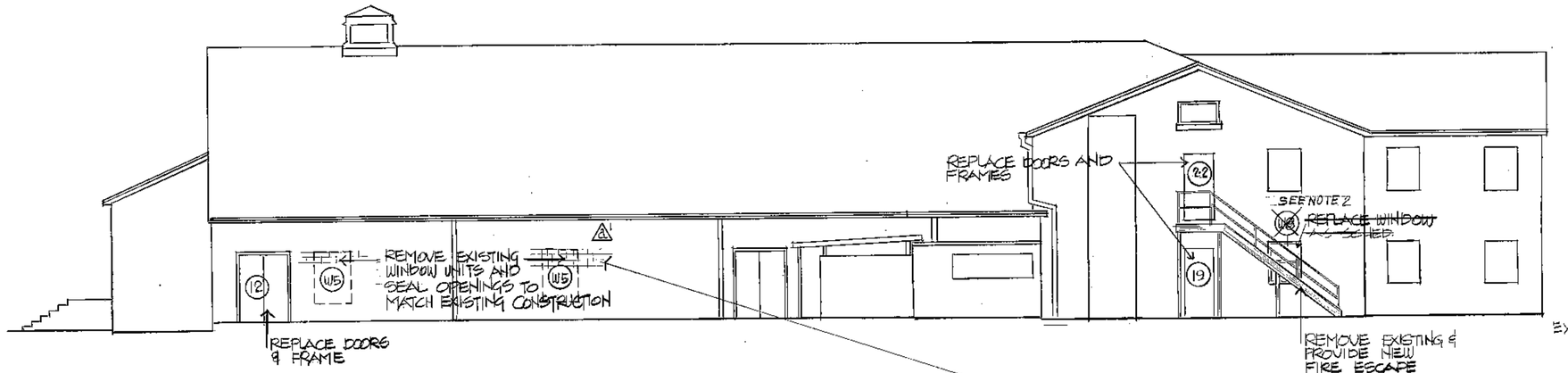
NORTH ELEVATION (1 / A2 | A5)

1/8" = 1'-0"



SOUTH ELEVATION (2 / A2 | A5)

1/8" = 1'-0"

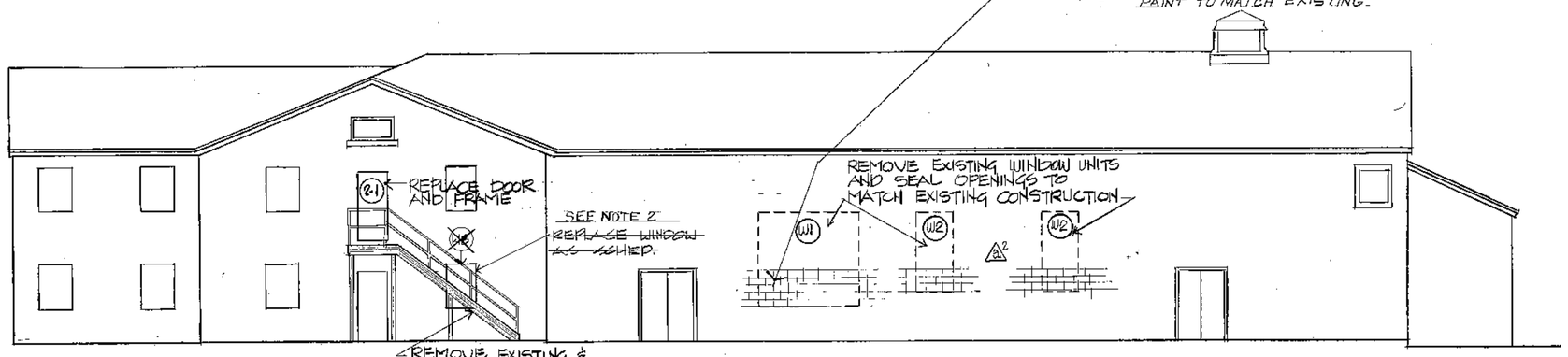


EAST ELEVATION (3 / A2 | A5)

1/8" = 1'-0"

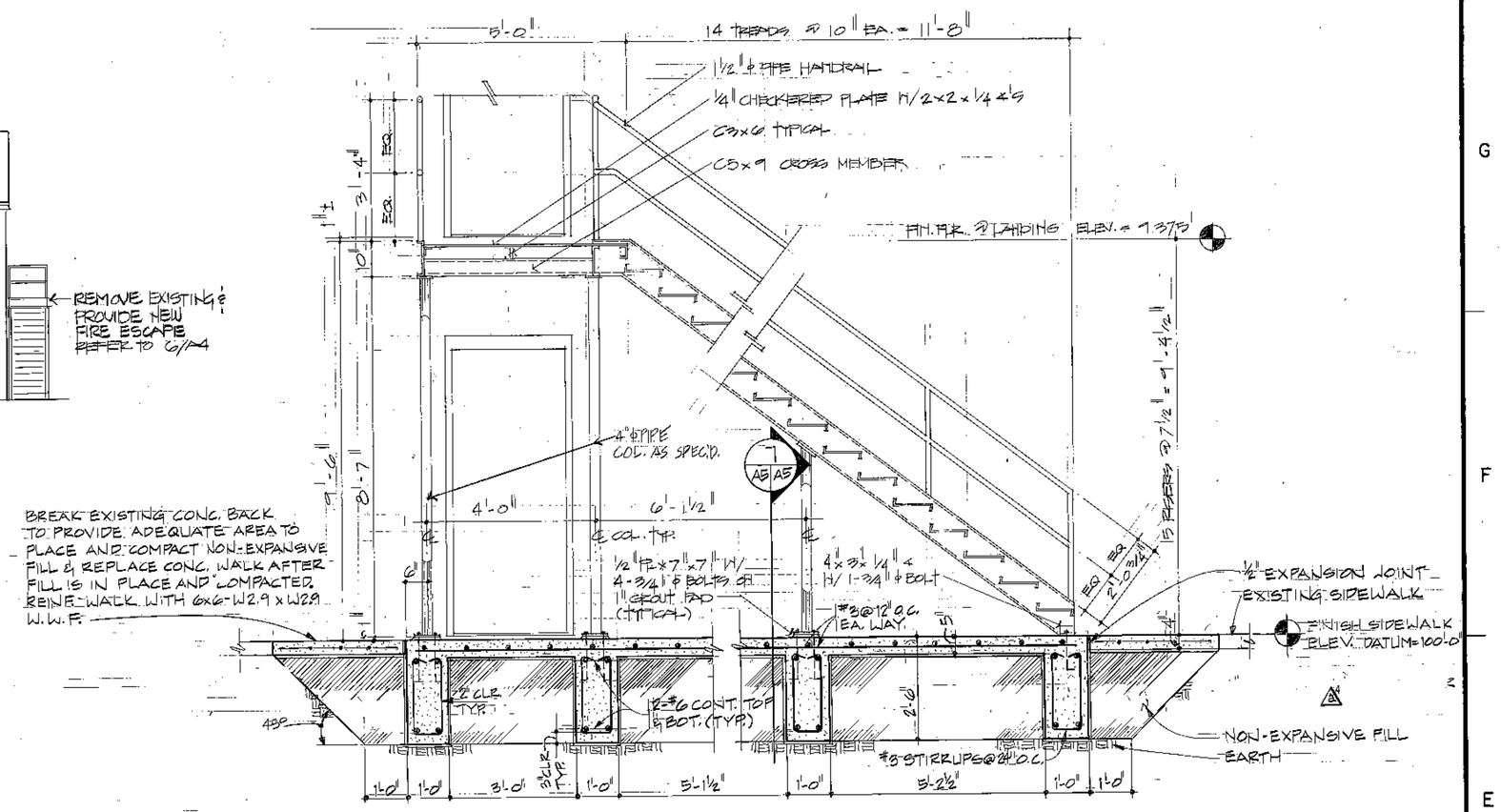
NOTE

1. NEW MASONRY CONSTRUCTION @ WINDOWS REMOVED SHALL BE TOOTHED IN TO PROVIDE CLEAN APPERANCE SEE 3/A5 & 4/A5 FOR LOCATIONS OF NEW EXTERIOR MASONRY WORK.
2. WINDOW ~~W-8 DELETED~~. FILL OPENING PAINT TO MATCH EXISTING.



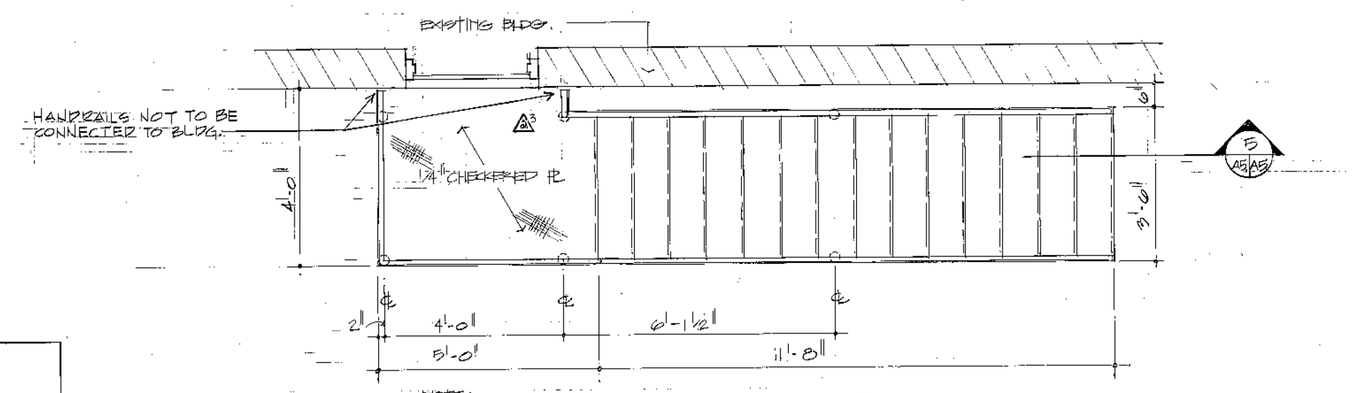
WEST ELEVATION (4 / A2 | A5)

1/8" = 1'-0"



SECTION @ STAIRS (5 / A5 | A5)

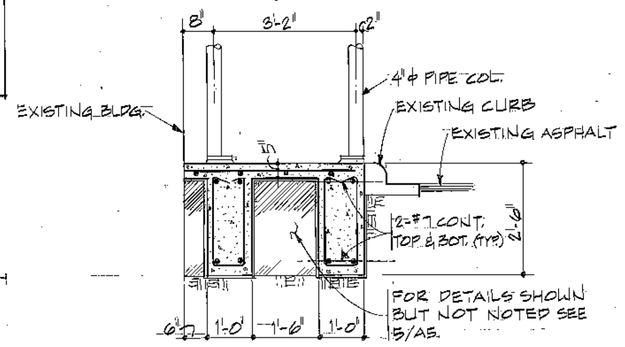
SC: ½" = 1'-0"



NOTE:
1. THE FIRE STAIRS TO BE FREE STANDING W/ 3/8" ROD DIAGONAL CROSS BRACING BETWEEN COLUMNS NORTH AND SOUTH OF EXIT DOORS.
2. FIELD WELD ALL CONNECTIONS

ENLARGED STAIR PLAN (6 / A2 | A5)

SC: ½" = 1'-0"



SECTION (7 / A5 | A5)

SC: ½" = 1'-0"

GRAPHIC SCALES

SCALE: ⅛" = 1'-0"

SCALE: ½" = 1'-0"

SYM.	DO. NO.	ACTION	DATE	DESCRIPTION OF REVISION
			22 MAR 82	[illegible]
4		AM# 0001	MAR 1981	TOOTH-IN NEW MASONRY TO EXISTING.
				~~TOOTH~~ FILL & HANDRAIL CORRECTION TO STAIR.

U.S. ARMY ENGINEER DISTRICT, FORT WORTH
CORPS OF ENGINEERS
FORT WORTH, TEXAS

DESIGNED BY: TEY
DRAWN BY: TEY, KEM
CHECKED BY: SHG
SUBMITTED BY:
ENGINEER:

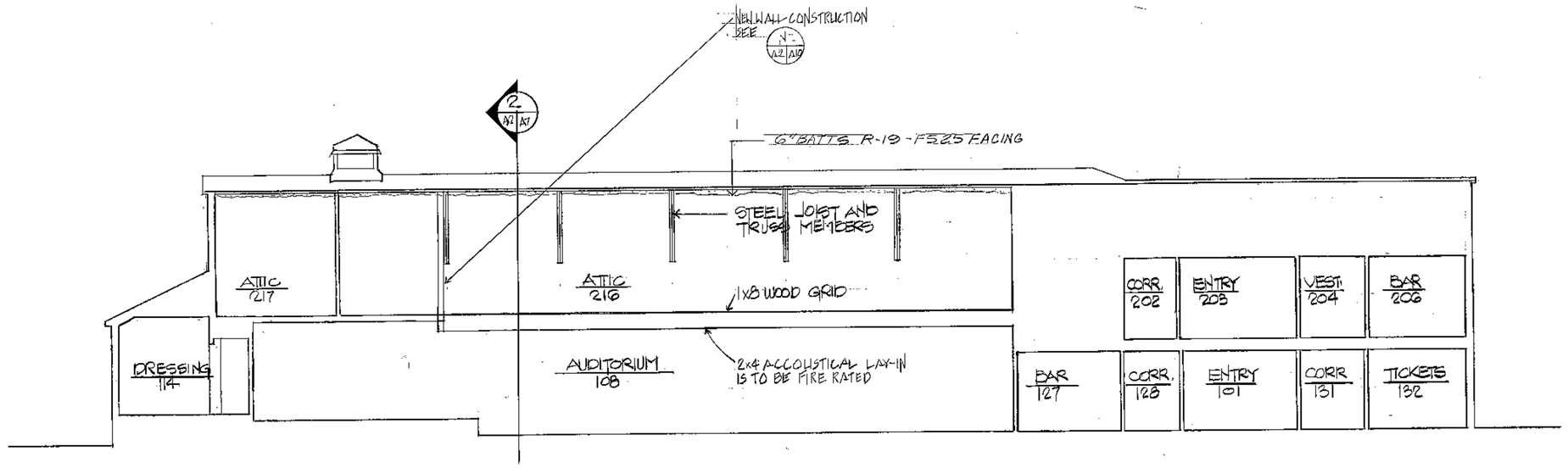
FORT SAM HOUSTON
SAN ANTONIO, TEXAS

UPGRADE BUILDING 2652
REPAIR BUILDING 2652

ELEVATIONS & DETAILS

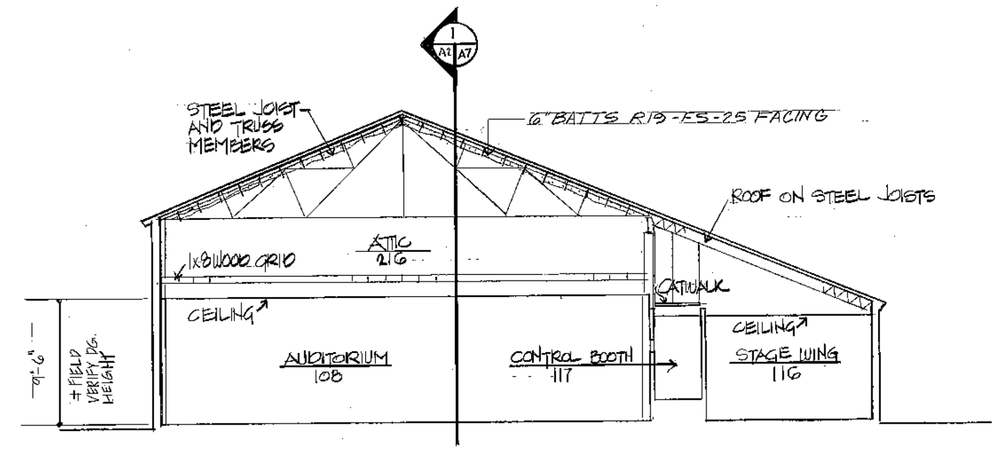
INV. NO. DACA62-81-B-0071 DATED: 2/13/81
CONTR. NO. DACA62-81-C-0080
DRAWING NUMBER: A5 OF 10
SHEET NO.
SEQUENCE NO. 6

CONTR. NO. DACA62-81-C-0080



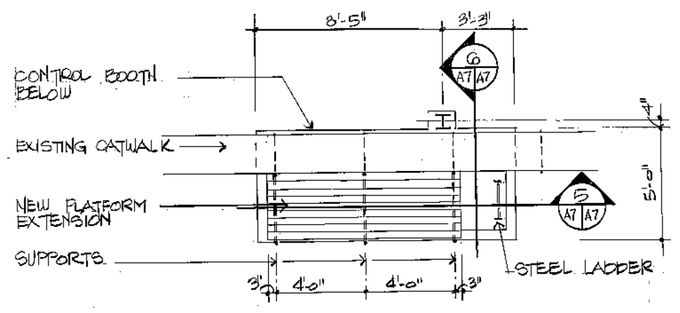
BUILDING SECTION 1 A2 A7

1/8" = 1'-0"



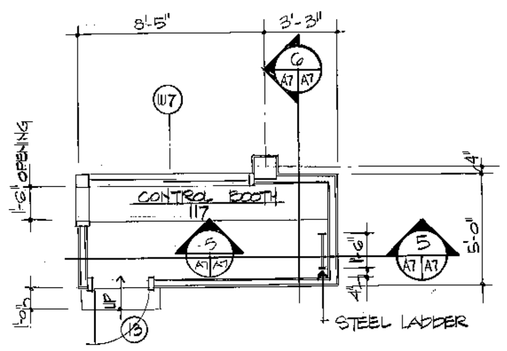
BUILDING SECTION 2 A2 A7

1/8" = 1'-0"



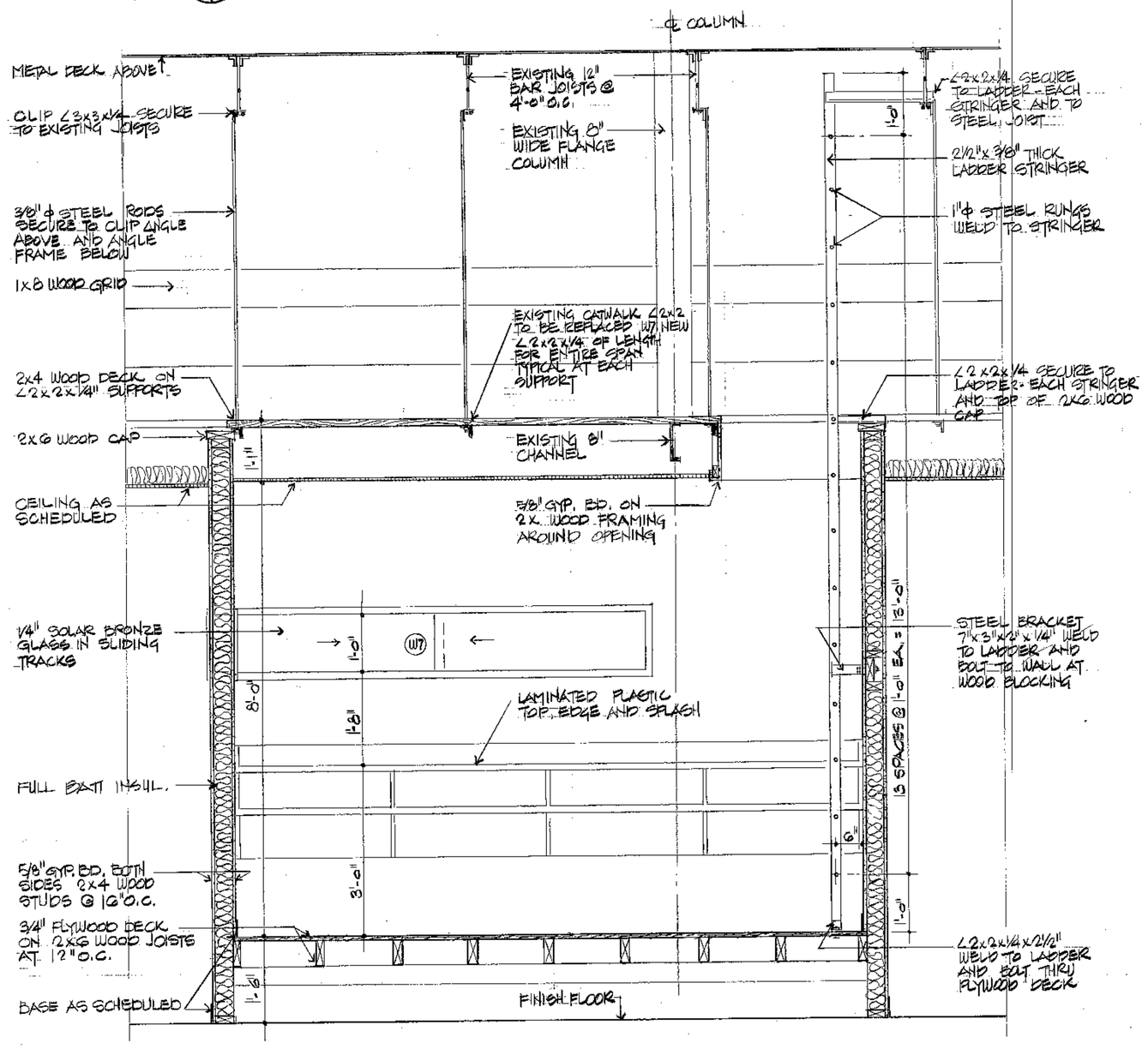
CONTROL BOOTH - UPPER PLATFORM 3 A2 A7

1/4" = 1'-0"



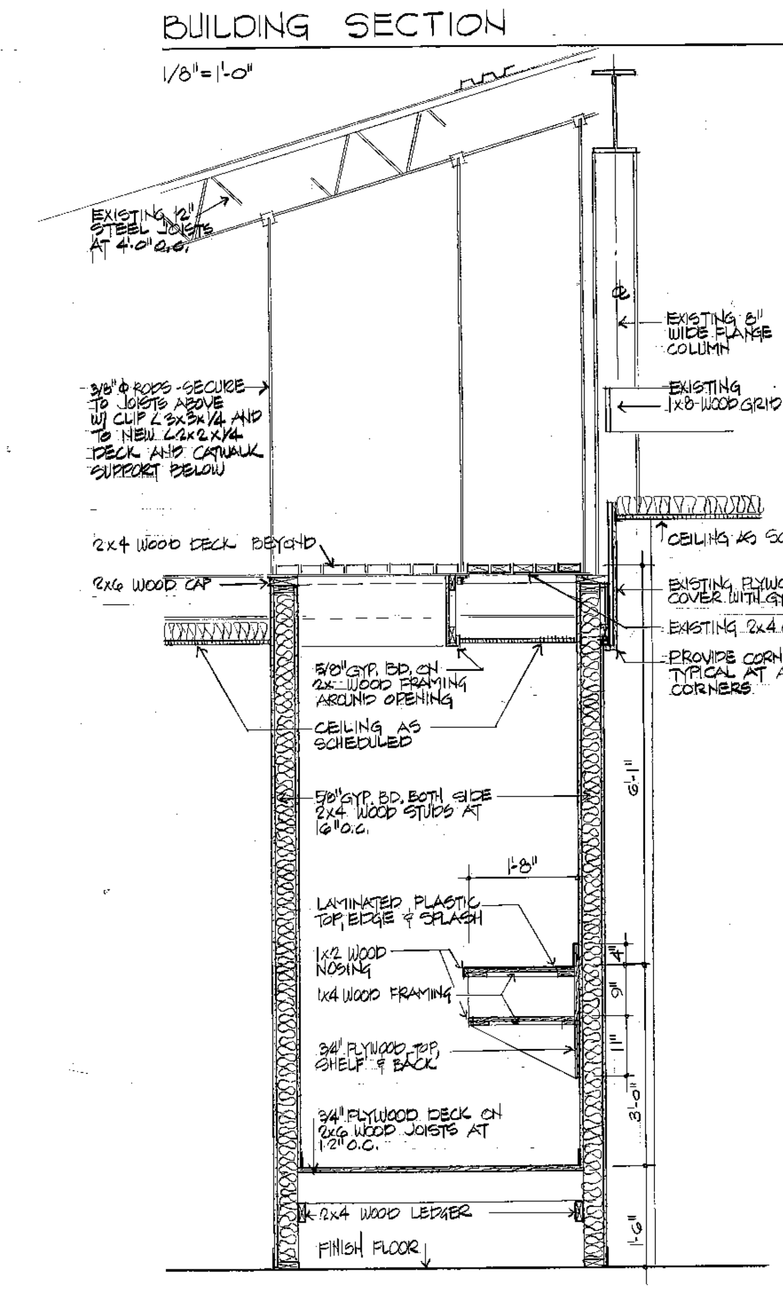
CONTROL BOOTH - FLOOR PLAN 4 A2 A7

1/4" = 1'-0"



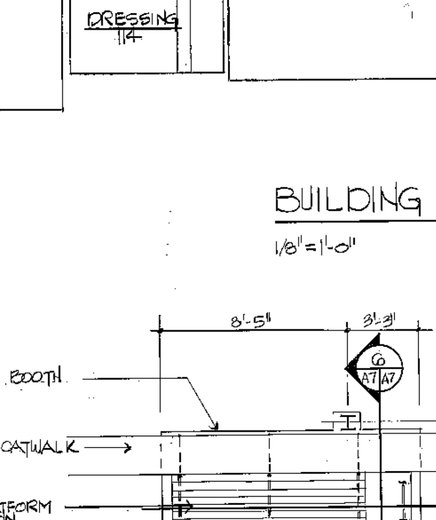
WALL SECTION 5 A7 A7

3/4" = 1'-0"



WALL SECTION 6 A7 A7

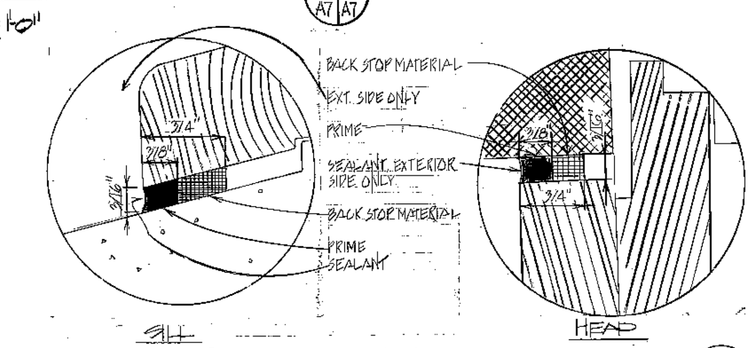
3/4" = 1'-0"



REMOVE COUNTERTOP WHERE INDICATED & APPLY PLASTIC LAMINATE @ CUT EDGES TO MATCH EXISTING TRIM. REFER TO SIMILAR DETAIL 5 A4/A7

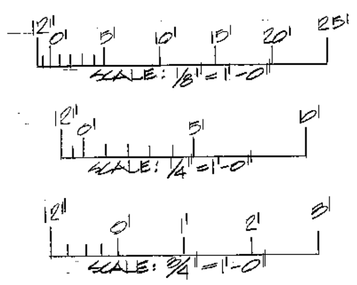
PHOTO @ RM. NO. 127 7 A4/A7

NO SCALE



JOINT TJC DETAILS 8 A4/A7

NO SCALE



SYM	DG. NO.	ACTION	DATE	DESCRIPTION OF REVISION
			22 MAR 82	"REVISED AS CHECKED"
	AM #0001		13 MAR 81	CORRECT TITLE REF. NOTE

KILLEBREW/RUCKER/ASSOC., INC.
600 PETROLEUM BLDG.
WICHITA FALLS, TEXAS

U.S. ARMY ENGINEER DISTRICT, FORT WORTH
CORPS OF ENGINEERS
FORT WORTH, TEXAS

DESIGNED BY: TEY
DRAWN BY: TEY
CHECKED BY: SKG

FORT SAM HOUSTON
SAN ANTONIO, TEXAS

UPGRADE BUILDING 2652
REPAIR BUILDING 2652

PARTIAL PLANS & SECTIONS

SUBMITTED BY:

ENGINEER:

INV. NO. DACA63-81-B-0071 DATED: 2/13/81

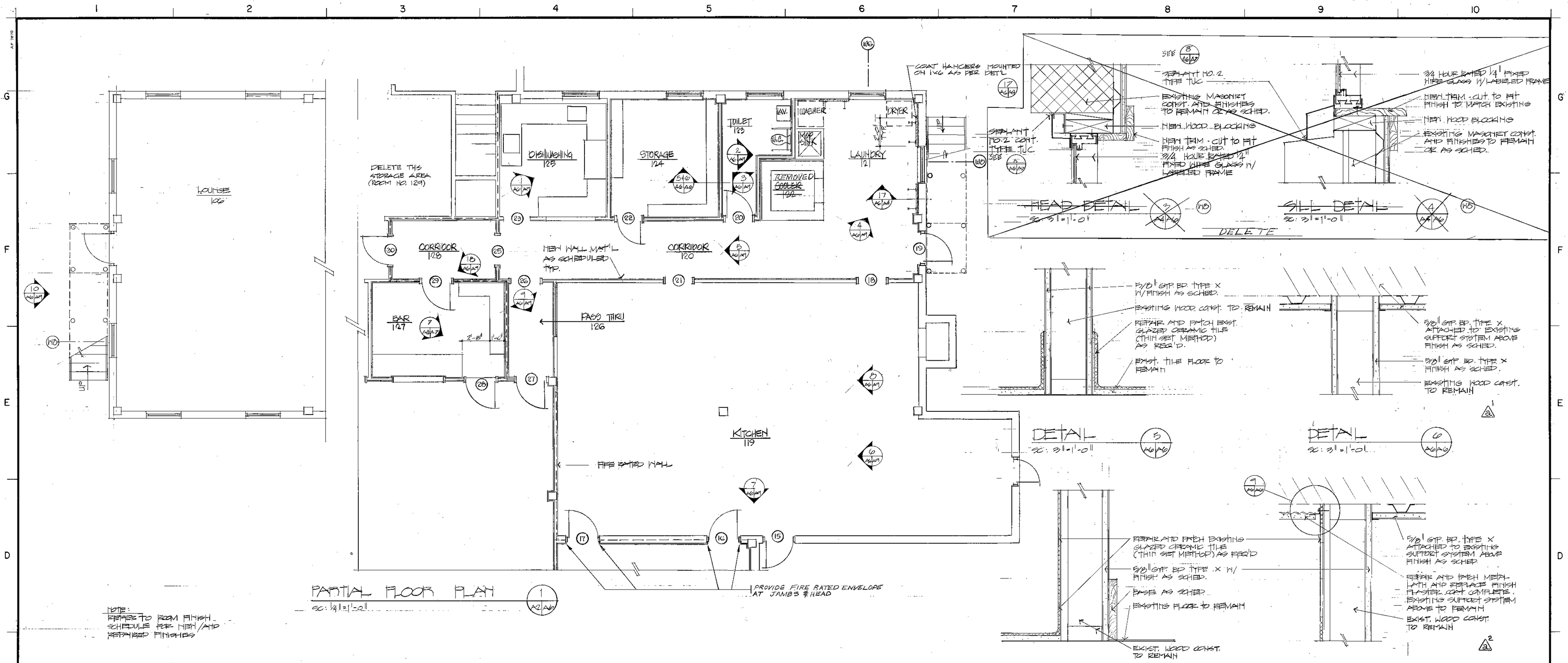
CONTR. NO. DACA63 81 C 0080

DRAWING NUMBER

SHEET NO. A7 OF 10

SEQUENCE NO. 8

CONTR. NO. DACA63 81 C 0080



SYM.	DO. NO.	ACTION	DATE	DESCRIPTION OF REVISION
			22 MAR 82	[illegible]
2	AM.#0001	13 MAR 81		DELETE INCORRECT REFERENCE NOTE

KILLEBREW/RUCKER/ASSOC., INC.
600 PETROLEUM BLDG.
WICHITA FALLS, TEXAS

U.S. ARMY ENGINEER DISTRICT, FORT WORTH
CORPS OF ENGINEERS
FORT WORTH, TEXAS

DESIGNED BY: TET
DRAWN BY: KEN, TET
CHECKED BY: SNG

FORT SAM HOUSTON
SAN ANTONIO, TEXAS

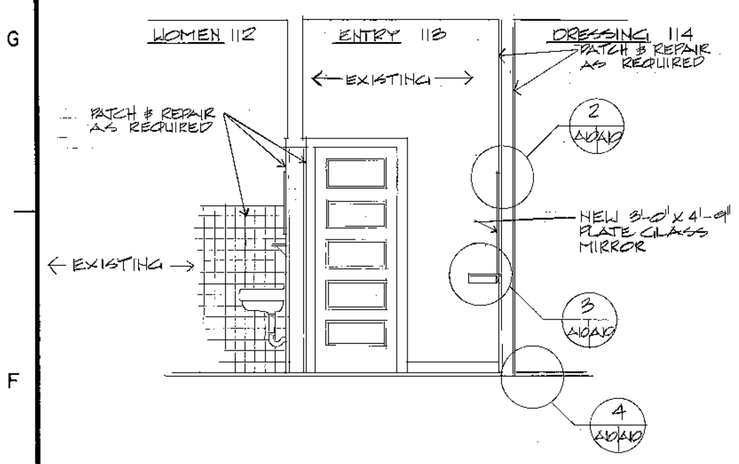
UPGRADE BUILDING 2652
REPAIR BUILDING 2652

ENLARGED PARTIAL PLANS & DETAILS

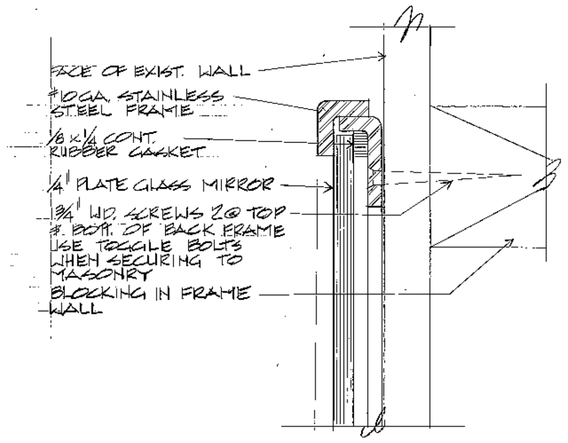
SUBMITTED BY:
ENGINEER:

INV. NO. DACA63-81-B-0071 DATED 2/13/81
CONTR. NO. DACA63-81-C-0080
DRAWING NUMBER
SHEET NO. A6 OF 10
SEQUENCE NO. 7

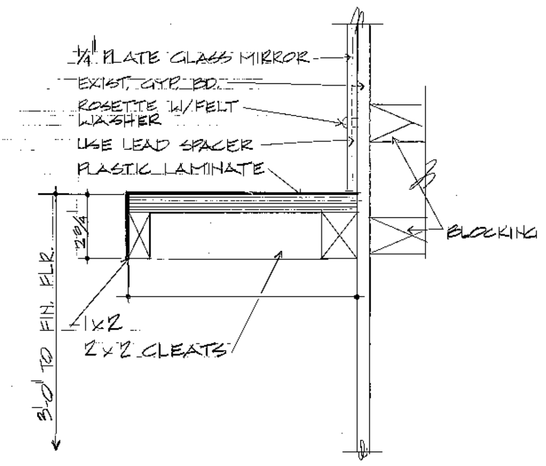
CONTR. NO. DACA63-81-C-0080



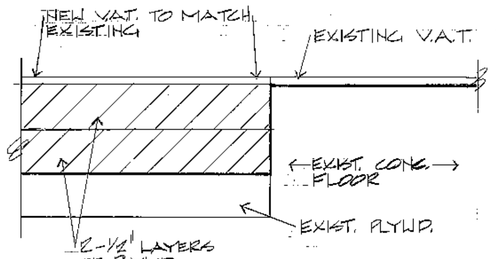
SECTION
SCALE 3/8" = 1'-0"
1 A9/A10



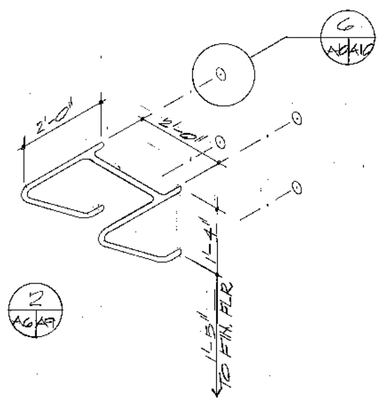
DETAIL @ MIRROR
FULL SCALE
2 A9/A10



DETAIL @ COUNTER
SCALE 3" = 1'-0"
3 A9/A10



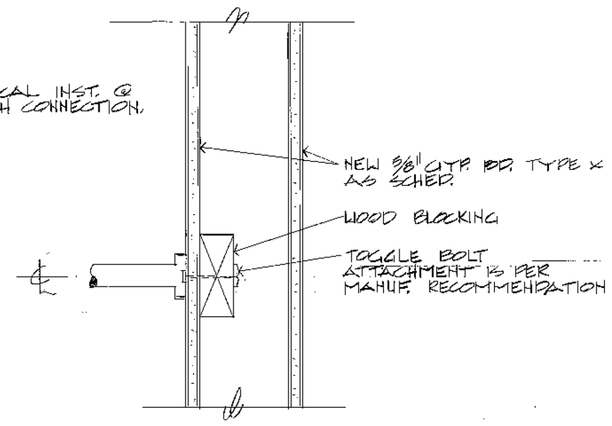
DETAIL
FULL SCALE
4 A9/A10



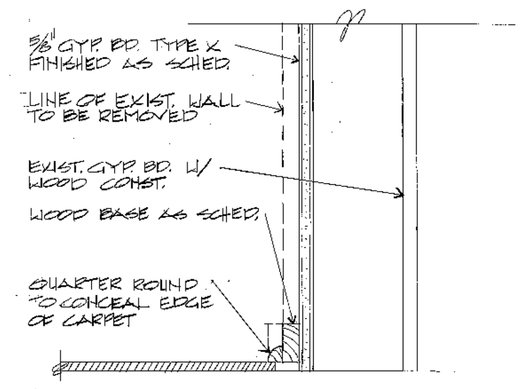
NOTE:
REFER TO PHOTO FOR GRAB BAR LOCATION
2 A6/A9

ISOMETRIC
SCALE 1/2" = 1'-0"
5 A9/A10

NOTE:
TYPICAL INST. @ EACH CONNECTION.

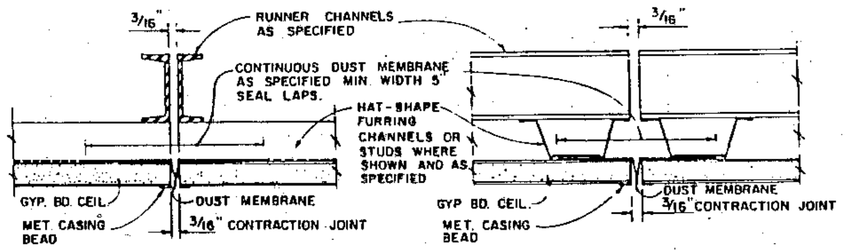


DETAIL
SCALE 3" = 1'-0"
6 A9/A10



EXISTING CARPET TO REMAIN, CONT. SHALL:
1. COVER AND PROTECT OR
2. REMOVE AND REINSTALL

DETAIL
SCALE 3" = 1'-0"
7 A10

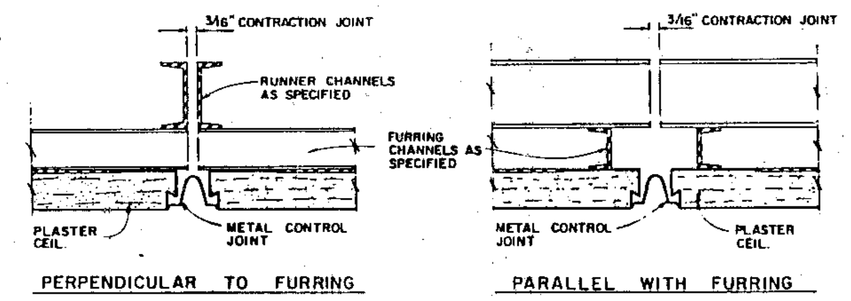


PERPENDICULAR TO FURRING

PARALLEL WITH FURRING

NOTE:
CONTROL JOINTS SHALL BE INSTALLED TO DIVIDE LONG NARROW AREAS OF GYPSUM WALLBOARD, SUCH AS OCCUR ON CORRIDOR CEILINGS, INTO LENGTHS OF NOT MORE THAN 40 FEET AND SHALL BE INSTALLED AT EXPANSION OR CONTRACTION JOINTS IN THE STRUCTURAL FRAMEWORK OVER WHICH GYPSUM BOARD IS REQUIRED. CONTROL JOINTS FOR WALLS SHALL BE INSTALLED AS DETAILED ABOVE WITH DUST MEMBRANE AND METAL CASING BEADS APPLIED TO TWO WALL STUDS OR FURRING MEMBERS SPACED 3/16" APART. ALL CEILING AND WALL JOINTS SHALL BE INSTALLED WHERE SHOWN ON THE ARCHITECTURAL WALL ELEVATIONS AND THE FLOOR AND REFLECTED CEILING PLANS.

GYPSUM BOARD



PERPENDICULAR TO FURRING

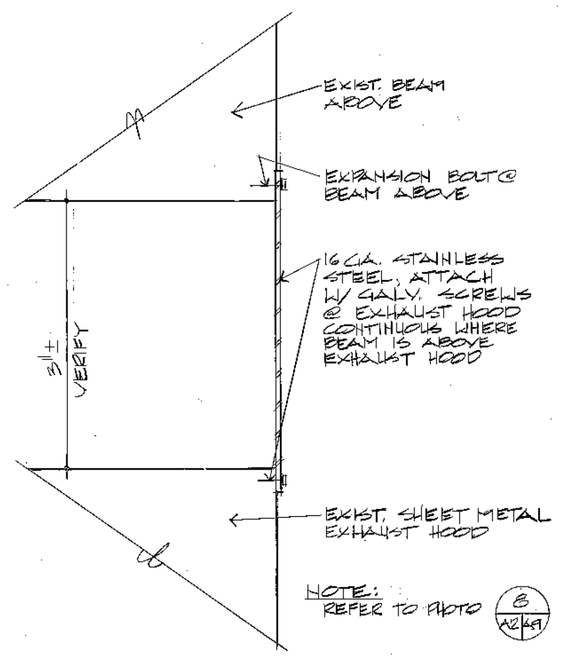
PARALLEL WITH FURRING

NOTE:
CONTROL OR EXPANSION JOINTS IN PLASTER WORK SHALL BE INSTALLED TO CONTINUE EXPANSION OR CONTROL JOINTS IN STRUCTURAL FRAME AND TO DIVIDE PLASTER AREAS INTO NOT MORE THAN 144 SQUARE FEET FOR PORTLAND CEMENT PLASTER AND 300 SQUARE FEET FOR GYPSUM OR LIME PLASTER. INTERVAL BETWEEN JOINTS SHALL BE REDUCED IN LONG NARROW AREAS SUCH AS CORRIDOR CEILINGS TO 12 FEET FOR PORTLAND CEMENT PLASTER AND 30 FEET FOR GYPSUM OR LIME PLASTER. UNRESTRAINED CEILING DESIGN SHALL BE INSTALLED WHEN TOTAL CEILING AREA EXCEEDS INTERVAL LIMITATIONS AS SPECIFIED ABOVE. CONTROL JOINTS FOR WALLS SHALL BE INSTALLED AS DETAILED ABOVE WITH METAL CONTROL JOINT APPLIED TO TWO WALL STUDS OR FURRING MEMBERS SPACED 3/16" APART. ALL CEILING AND WALL JOINTS SHALL BE INSTALLED WHERE SHOWN ON THE ARCHITECTURAL WALL ELEVATIONS AND THE FLOOR AND REFLECTED CEILING PLANS

PLASTER

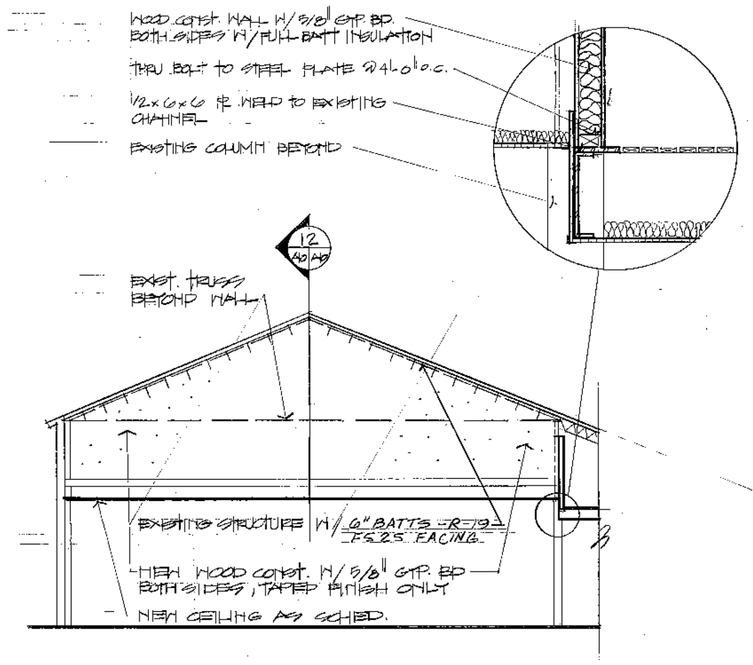
CEILING CONTROL JOINT DET'L
NO SCALE
8 A6/A10

CEILING CONTROL JOINT DET'L
NO SCALE
9 A6/A10

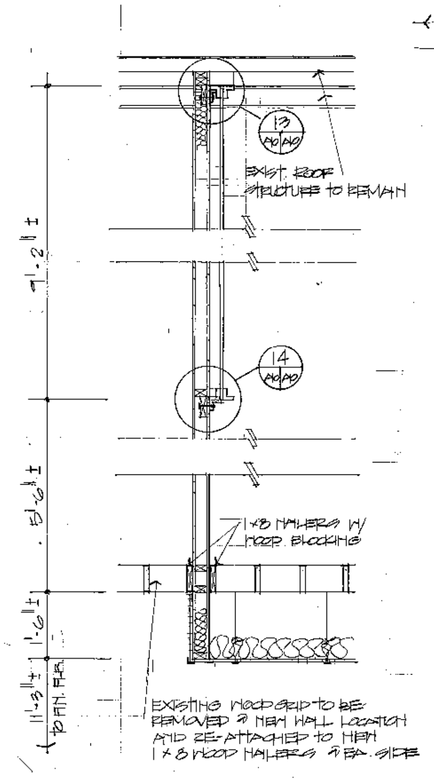


NOTE:
REFER TO PHOTO 8 A2/A9

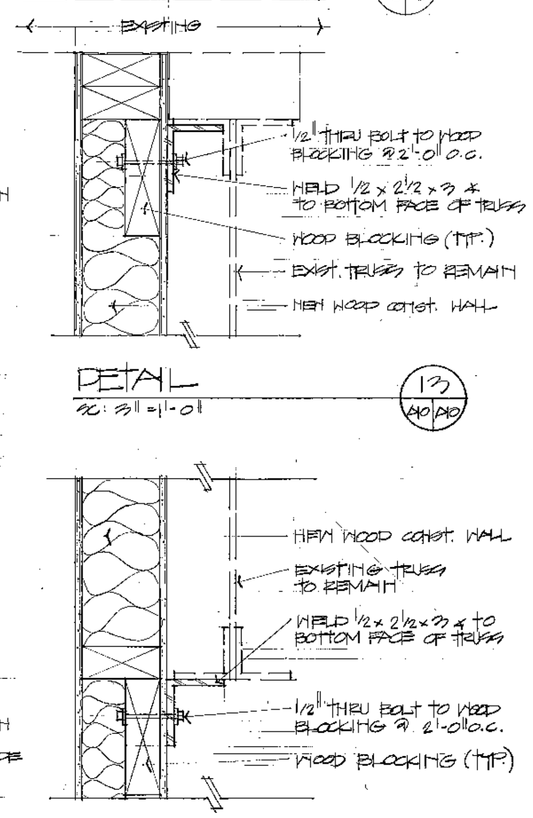
DETAIL
FULL SCALE
10 A9/A10



SECTION
SC: 1/4" = 1'-0"
11 A2/A10



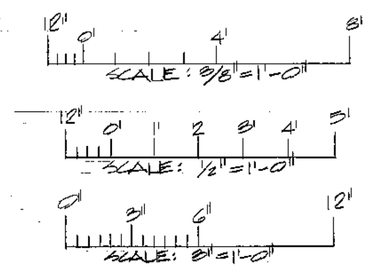
SECTION
SC: 1/2" = 1'-0"
12 A10/A10



DETAIL
SC: 3" = 1'-0"
13 A10/A10

DETAIL
SC: 3" = 1'-0"
14 A10/A10

GRAPHIC SCALES:



SYM.	DO. NO.	ACTION	DATE	DESCRIPTION OF REVISION
			12 MAR 82	"AS BUILT"

KILLEBREW/RUCKER/ASSOC., INC.
600 PETROLEUM BLDG.
WICHITA FALLS, TEXAS

U.S. ARMY ENGINEER DISTRICT, FORT WORTH
CORPS OF ENGINEERS
FORT WORTH, TEXAS

DESIGNED BY: KEH
DRAWN BY: BAT
CHECKED BY: SHC
SUBMITTED BY:
ENGINEER:

FORT SAM HOUSTON
SAN ANTONIO, TEXAS

UPGRADE BUILDING 2652
REPAIR BUILDING 2652

DETAILS

INV. NO. DACA63-81-B-0071 DATED: 2/17/81
CONTR. NO. DACA63-81-C-0080
DRAWING NUMBER
SHEET NO. A10 OF 10
SEQUENCE NO. 11

CONTR. NO. DACA63-81-C-0080

STATE OF TEXAS
JAMES N. KILLEBREW
9482
REGISTERED
ARCHITECT

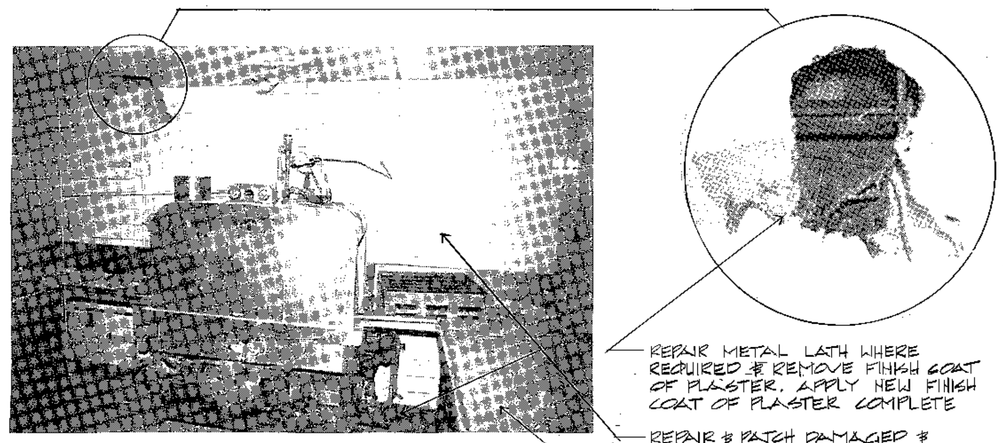


PHOTO @ RM. NO. 125 — 1 A6/A9
NO SCALE

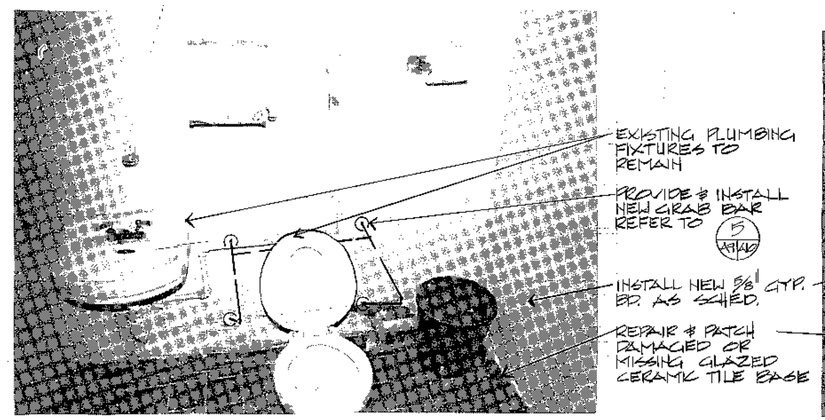
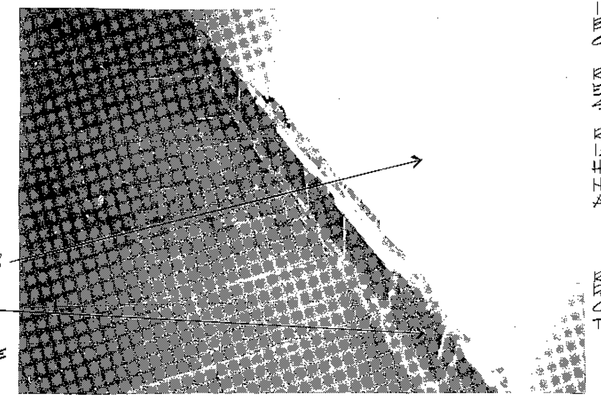


PHOTO @ RM. NO. 123 — 2 A6/A9
NO SCALE



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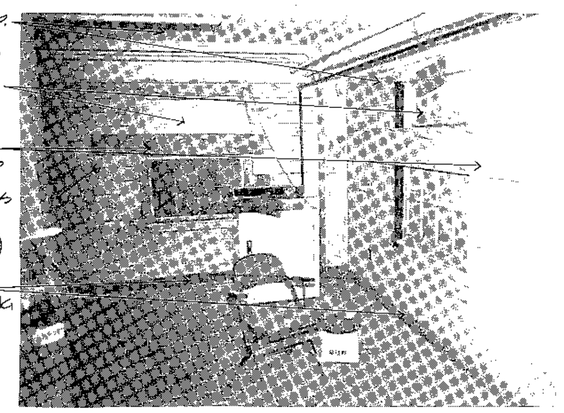
NO SCALE



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NO SCALE

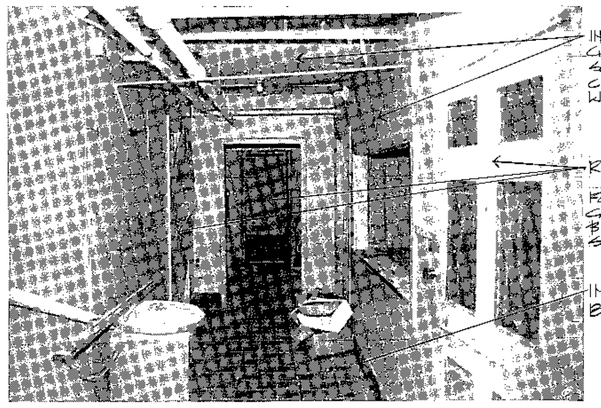

PHOTO @ RM. NO. 120 — 5 A6/A9
NO SCALE

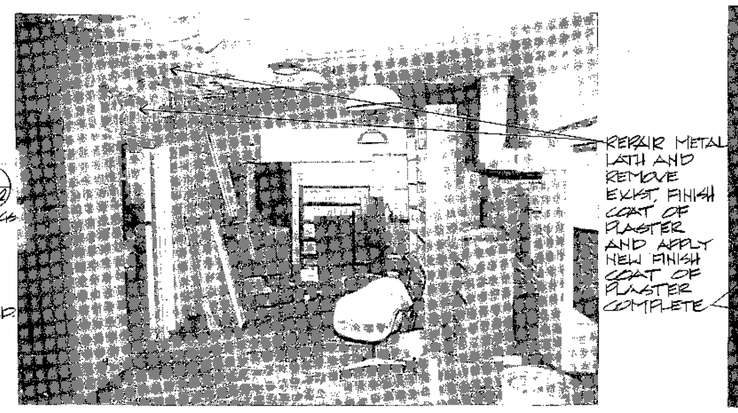

PHOTO @ RM. NO. 119 — 6 A6/A9
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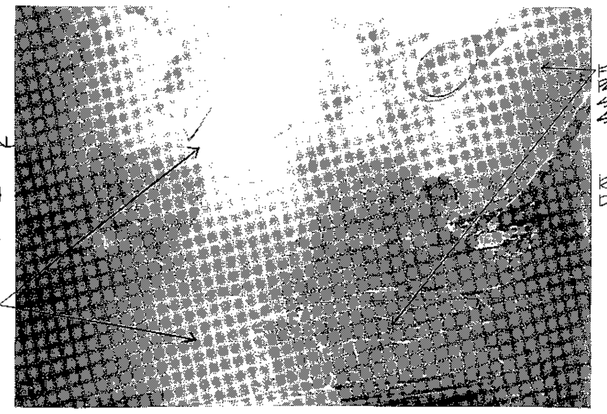

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NO SCALE

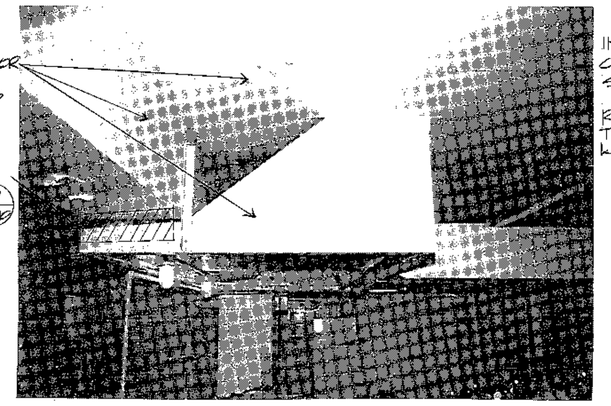


PHOTO @ RM. NO. 119 — 8 A6/A9
NO SCALE

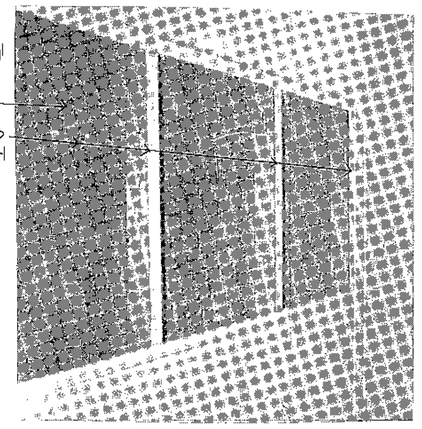
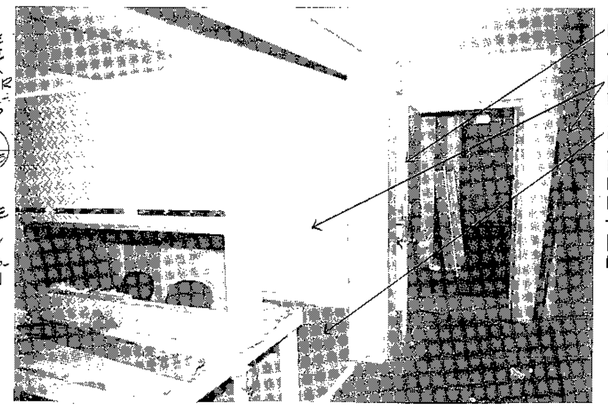


PHOTO @ RM. NO. 126 — 9 A6/A9
NO SCALE

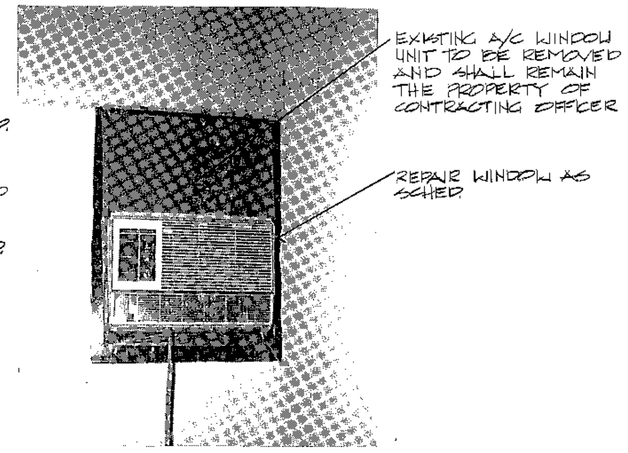
10

PHOTO @ WEST EXTERIOR — 10 A6/A9
NO SCALE



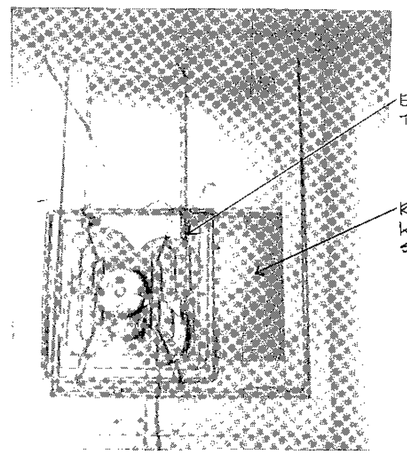
11

PHOTO @ RM. NO. 114 — 11 A6/A9
NO SCALE



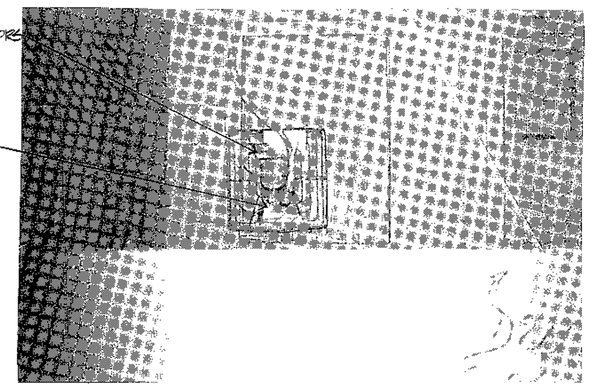
12

PHOTO @ RM. NO. 114 — 12 A6/A9
NO SCALE



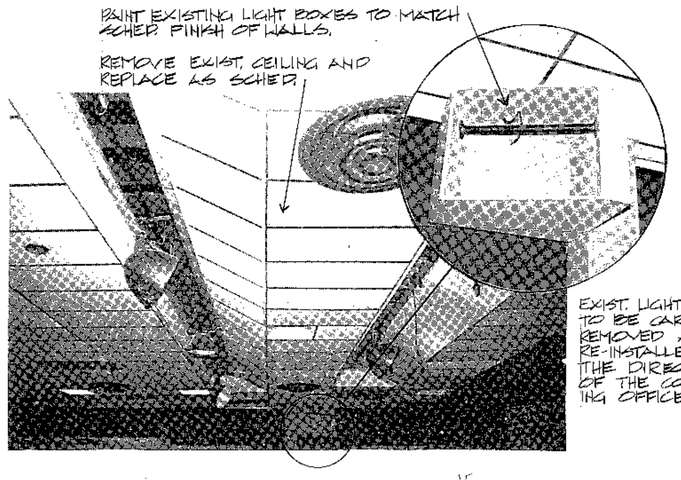
13

PHOTO @ RM. NO. 113 — 13 A6/A9
NO SCALE



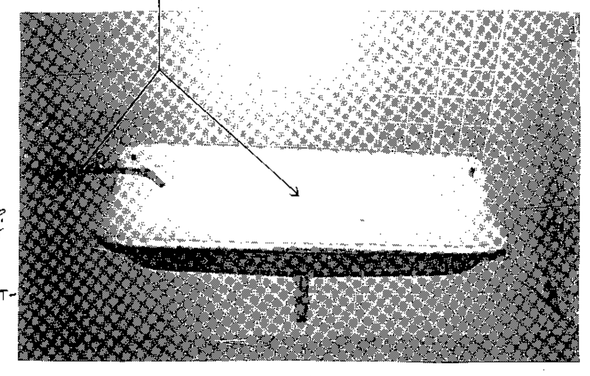
14

PHOTO @ RM. NO. 111 — 14 A6/A9
NO SCALE



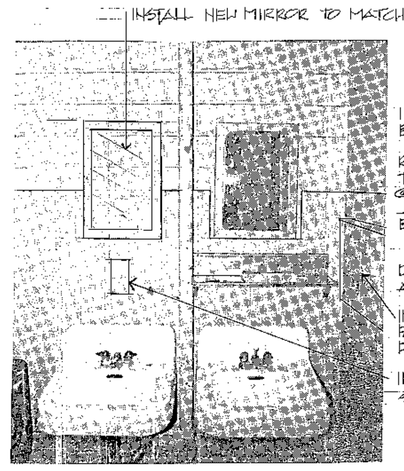
15

PHOTO @ RM. NO. 108 — 15 A6/A9
NO SCALE



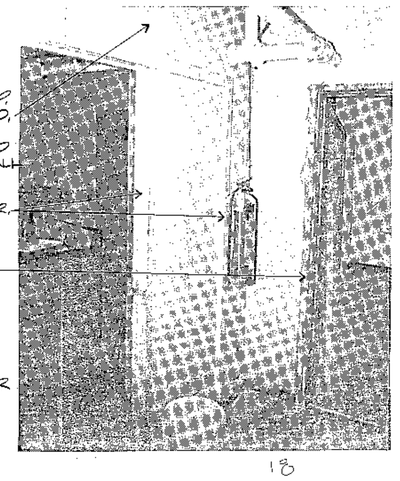
16

PHOTO @ RM. NO. 111 — 16 A6/A9
NO SCALE



17

PHOTO @ RM. NO. 112 — 17 A6/A9
NO SCALE



18

PHOTO @ RM. NO. 128 — 18 A6/A9
NO SCALE

SYM	DO. NO.	ACTION	DATE	DESCRIPTION OF REVISION
			22 MAR 82	

KILLEBREW/RUCKER/ASSOC., INC.
600 PETROLEUM BLDG.
WICHITA FALLS, TEXAS

U.S. ARMY ENGINEER DISTRICT, FORT WORTH
CORPS OF ENGINEERS
FORT WORTH, TEXAS

DESIGNED BY: KEW
DRAWN BY: PJT
CHECKED BY: SHG
SUBMITTED BY:
ENGINEER:

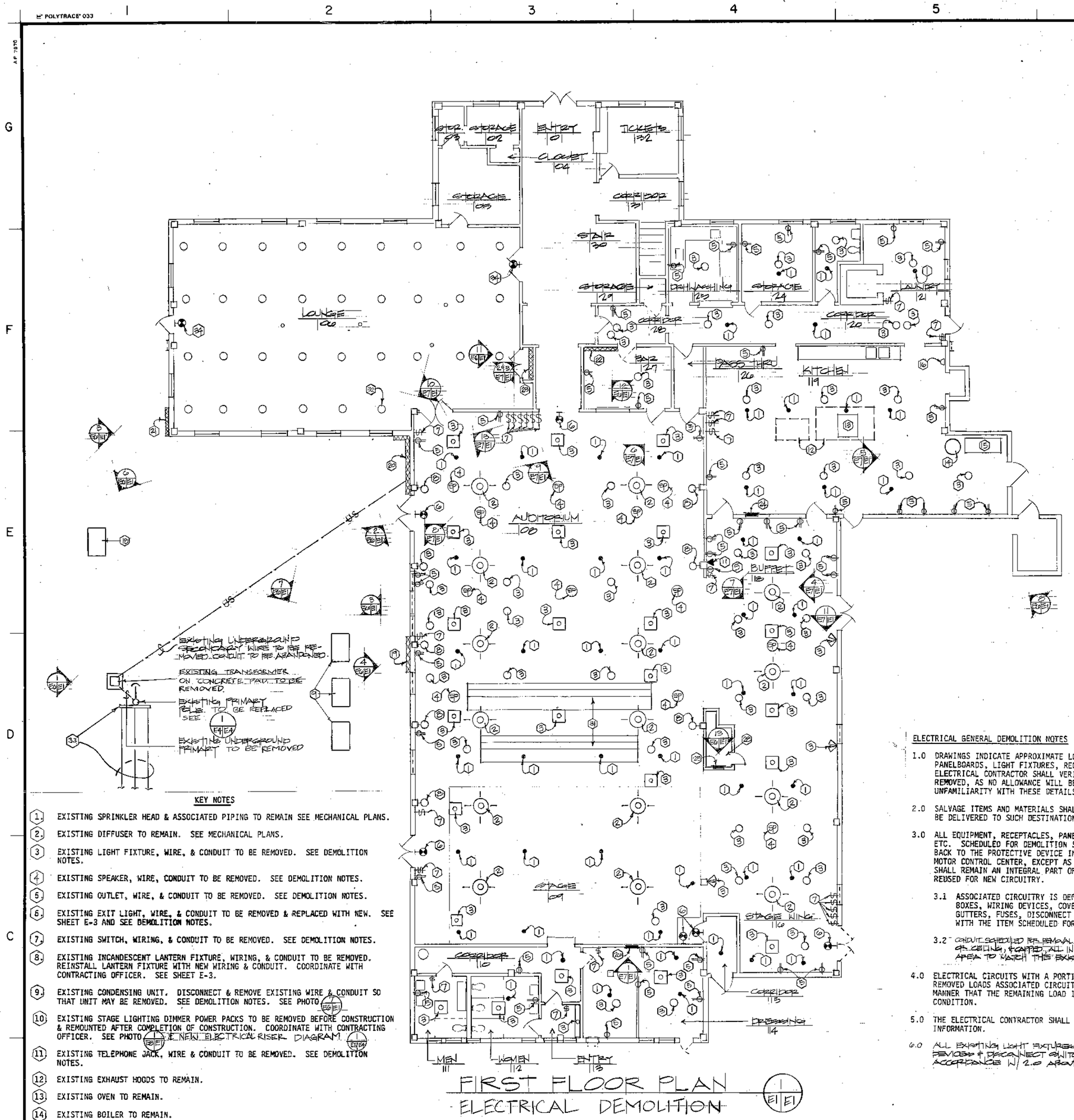
FORT SAM HOUSTON
SAN ANTONIO, TEXAS

UPGRADE BUILDING 2652
REPAIR BUILDING 2652

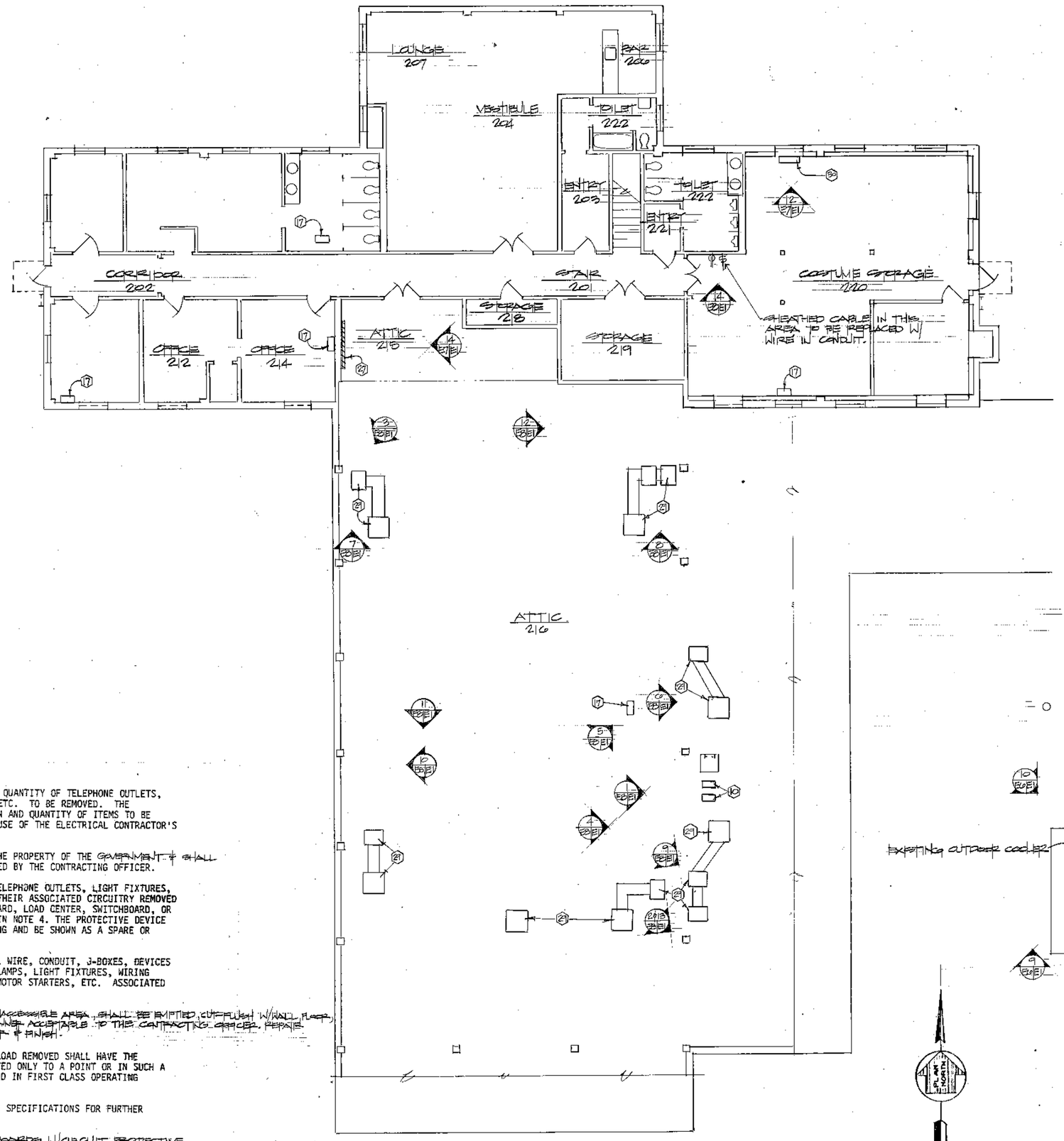
PHOTOS

INV. NO. DACA63-81-B-0071 DATED: 2/13/81
CONTR. NO. DACA63-81-C-0080
DRAWING NUMBER A9 SHEET NO. OF 10
SEQUENCE NO. 10

CONTR. NO. DACA63-81-C-0080



FIRST FLOOR PLAN
ELECTRICAL DEMOLITION



ATTIC FLOOR PLAN
ELECTRICAL DEMOLITION

GRAPHIC SCALES

KEY NOTES

1. EXISTING SPRINKLER HEAD & ASSOCIATED PIPING TO REMAIN SEE MECHANICAL PLANS.
2. EXISTING DIFFUSER TO REMAIN. SEE MECHANICAL PLANS.
3. EXISTING LIGHT FIXTURE, WIRE, & CONDUIT TO BE REMOVED. SEE DEMOLITION NOTES.
4. EXISTING SPEAKER, WIRE, CONDUIT TO BE REMOVED. SEE DEMOLITION NOTES.
5. EXISTING OUTLET, WIRE, & CONDUIT TO BE REMOVED. SEE DEMOLITION NOTES.
6. EXISTING EXIT LIGHT, WIRE, & CONDUIT TO BE REMOVED & REPLACED WITH NEW. SEE SHEET E-3 AND SEE DEMOLITION NOTES.
7. EXISTING SWITCH, WIRING, & CONDUIT TO BE REMOVED. SEE DEMOLITION NOTES.
8. EXISTING INCANDESCENT LANTERN FIXTURE, WIRING, & CONDUIT TO BE REMOVED. REINSTALL LANTERN FIXTURE WITH NEW WIRING & CONDUIT. COORDINATE WITH CONTRACTING OFFICER. SEE SHEET E-3.
9. EXISTING CONDENSING UNIT. DISCONNECT & REMOVE EXISTING WIRE & CONDUIT SO THAT UNIT MAY BE REMOVED. SEE DEMOLITION NOTES. SEE PHOTO
10. EXISTING STAGE LIGHTING DIMMER POWER PACKS TO BE REMOVED BEFORE CONSTRUCTION & REMOUNTED AFTER COMPLETION OF CONSTRUCTION. COORDINATE WITH CONTRACTING OFFICER. SEE PHOTO & NEW ELECTRICAL RISER DIAGRAM
11. EXISTING TELEPHONE JACK, WIRE & CONDUIT TO BE REMOVED. SEE DEMOLITION NOTES.
12. EXISTING EXHAUST HOODS TO REMAIN.
13. EXISTING OVEN TO REMAIN.
14. EXISTING BOILER TO REMAIN.
15. EXISTING STORAGE TANK TO REMAIN.
16. EXISTING DUMBWAITER TO BE REPAIRED & PLACED IN PROPER OPERATING CONDITION.
17. EXISTING UNIT HEATER. DISCONNECT & REMOVE EXISTING WIRE & CONDUIT SO THAT UNIT MAY BE REMOVED. SEE DEMOLITION NOTES.
18. EXISTING CONDENSING UNIT WATER PUMP. DISCONNECT & REMOVE EXISTING WIRE & CONDUIT SO THAT UNIT MAY BE REMOVED. SEE DEMOLITION NOTES. SEE PHOTO
19. EXISTING ELECTRICAL GEAR. SEE RISER DIAGRAM & PHOTO FOR DEMOLITION INFORMATION.
20. EXISTING ELECTRICAL GEAR. SEE RISER DIAGRAM & PHOTO FOR DEMOLITION INFORMATION.
21. EXISTING ELECTRICAL GEAR. SEE RISER DIAGRAM & PHOTO FOR DEMOLITION INFORMATION.
22. EXISTING ELECTRICAL GEAR. SEE RISER DIAGRAM & PHOTO FOR DEMOLITION INFORMATION.
23. EXISTING ELECTRICAL GEAR. SEE RISER DIAGRAM & PHOTO & FOR DEMOLITION INFORMATION.
24. EXISTING PANEL TO BE REMOVED & REPLACED WITH NEW PANEL. SEE SHEET E-4 & NEW RISER DIAGRAM

KEY NOTES (CONTINUED)

25. EXISTING PANEL TO BE REMOVED & REPLACED WITH NEW PANEL. SEE SHEET E-4 & NEW RISER DIAGRAM . FOR MORE DEMOLITION INFORMATION SEE PHOTO
26. EXISTING PANEL TO BE REMOVED & REPLACED WITH NEW PANEL. SEE SHEET E-4 & NEW RISER DIAGRAM . FOR MORE DEMOLITION INFORMATION SEE PHOTO
27. EXISTING RHEASTAT DIMMERS, WIRING, & CONDUIT TO BE REMOVED. SEE PHOTO FOR MORE DEMOLITION INFORMATION.
28. EXISTING CONTROL ROOM TO BE REMOVED & REPLACED BY OTHERS. FOR NEW CONTROL ROOM LIGHTING & POWER SEE SHEETS E-3 & E-4.
29. EXISTING MECHANICAL EQUIPMENT. EXISTING POWER & CONTROL WIRING & CONDUIT TO BE REPLACED WITH POWER & CONTROL WIRING & CONDUIT & RECONNECTED TO NEW PANEL 'F'. SEE SHEET E-4 & NEW RISER DIAGRAM

KEY NOTES (CONTINUED)

30. EXISTING WALL A/C UNIT. DISCONNECT & REMOVE EXISTING WIRE & CONDUIT SO THAT UNIT MAY BE REMOVED. SEE DEMOLITION NOTES. SEE PHOTO
31. EXISTING CEILING RUNNERS FOR STAGE TRACK LIGHTING TO REMAIN.
32. TYPICAL EXISTING INCANDESCENT FIXTURE IN LOUNGE TO REMAIN. DISCONNECT EXISTING CIRCUITING & RECONNECT TO NEW PANEL 'A' LOCATED IN BAR.
33. THE EXISTING POLE HARDWARE SHALL BE REMOVED ON THE NEW TRANSFORMER POLE ASSEMBLY. THE OVERHEAD PRIMARY CONDUCTORS ARE TO BE REUSED. SEE FOR NEW POLE DETAIL.
34. EXISTING EXIT LIGHT TO BE REMOVED & REPLACED WITH NEW. SEE KEY NOTE #4 ON SHEET E-3.

ELECTRICAL GENERAL DEMOLITION NOTES

1.0 DRAWINGS INDICATE APPROXIMATE LOCATION AND QUANTITY OF TELEPHONE OUTLETS, PANELBOARDS, LIGHT FIXTURES, RECEPTACLES, ETC. TO BE REMOVED. THE ELECTRICAL CONTRACTOR SHALL VERIFY LOCATION AND QUANTITY OF ITEMS TO BE REMOVED, AS NO ALLOWANCE WILL BE MADE BECAUSE OF THE ELECTRICAL CONTRACTOR'S UNFAMILIARITY WITH THESE DETAILS.

2.0 SALVAGE ITEMS AND MATERIALS SHALL REMAIN THE PROPERTY OF THE GOVERNMENT & SHALL BE DELIVERED TO SUCH DESTINATION AS DIRECTED BY THE CONTRACTING OFFICER.

3.0 ALL EQUIPMENT, RECEPTACLES, PANELBOARDS, TELEPHONE OUTLETS, LIGHT FIXTURES, ETC. SCHEDULED FOR DEMOLITION SHALL HAVE THEIR ASSOCIATED CIRCUITRY REMOVED BACK TO THE PROTECTIVE DEVICE IN A PANELBOARD, LOAD CENTER, SWITCHBOARD, OR MOTOR CONTROL CENTER, EXCEPT AS MENTIONED IN NOTE 4. THE PROTECTIVE DEVICE SHALL REMAIN AN INTEGRAL PART OF THE HOUSING AND BE SHOWN AS A SPARE OR REUSED FOR NEW CIRCUITRY.

3.1 ASSOCIATED CIRCUITRY IS DEFINED AS ALL WIRE, CONDUIT, J-BOXES, DEVICES BOXES, WIRING DEVICES, COVER PLATES, LAMPS, LIGHT FIXTURES, WIRING GUTTERS, FUSES, DISCONNECT SWITCHES, MOTOR STARTERS, ETC. ASSOCIATED WITH THE ITEM SCHEDULED FOR REMOVAL.

3.2 CONDUIT SCHEDULED FOR REMOVAL, BUT IN AN INACCESSIBLE AREA, SHALL BE EMPTIED, CUT FLUSH W/WALL FLOOR OR CEILING, CAPPED ALL IN A SAFE MANNER ACCEPTABLE TO THE CONTRACTING OFFICER. REPAIR AREA TO MATCH THE EXISTING COLOR & FINISH.

4.0 ELECTRICAL CIRCUITS WITH A PORTION OF THE LOAD REMOVED SHALL HAVE THE REMOVED LOADS ASSOCIATED CIRCUITRY TERMINATED ONLY TO A POINT OR IN SUCH A MANNER THAT THE REMAINING LOAD IS ACTIVE AND IN FIRST CLASS OPERATING CONDITION.

5.0 THE ELECTRICAL CONTRACTOR SHALL CONSULT THE SPECIFICATIONS FOR FURTHER INFORMATION.

6.0 ALL EXISTING LIGHT FIXTURES & PANELBOARDS W/CIRCUIT PROTECTIVE DEVICES & DISCONNECT SWITCHES SHALL BE SALVAGED IN ACCORDANCE W/ 2.0 ABOVE.

VOIDED

SYM	DWG. NO.	ACTION	DATE	DESCRIPTION OF REVISION
			22MAR82	"DRAWING OF RECORD" AS BUILT

KILLEBREW/RUCKER/ASSOC., INC.
600 PETROLEUM BLDG.
WICHITA FALLS, TEXAS

U.S. ARMY ENGINEER DISTRICT, FORT WORTH
CORPS OF ENGINEERS
FORT WORTH, TEXAS

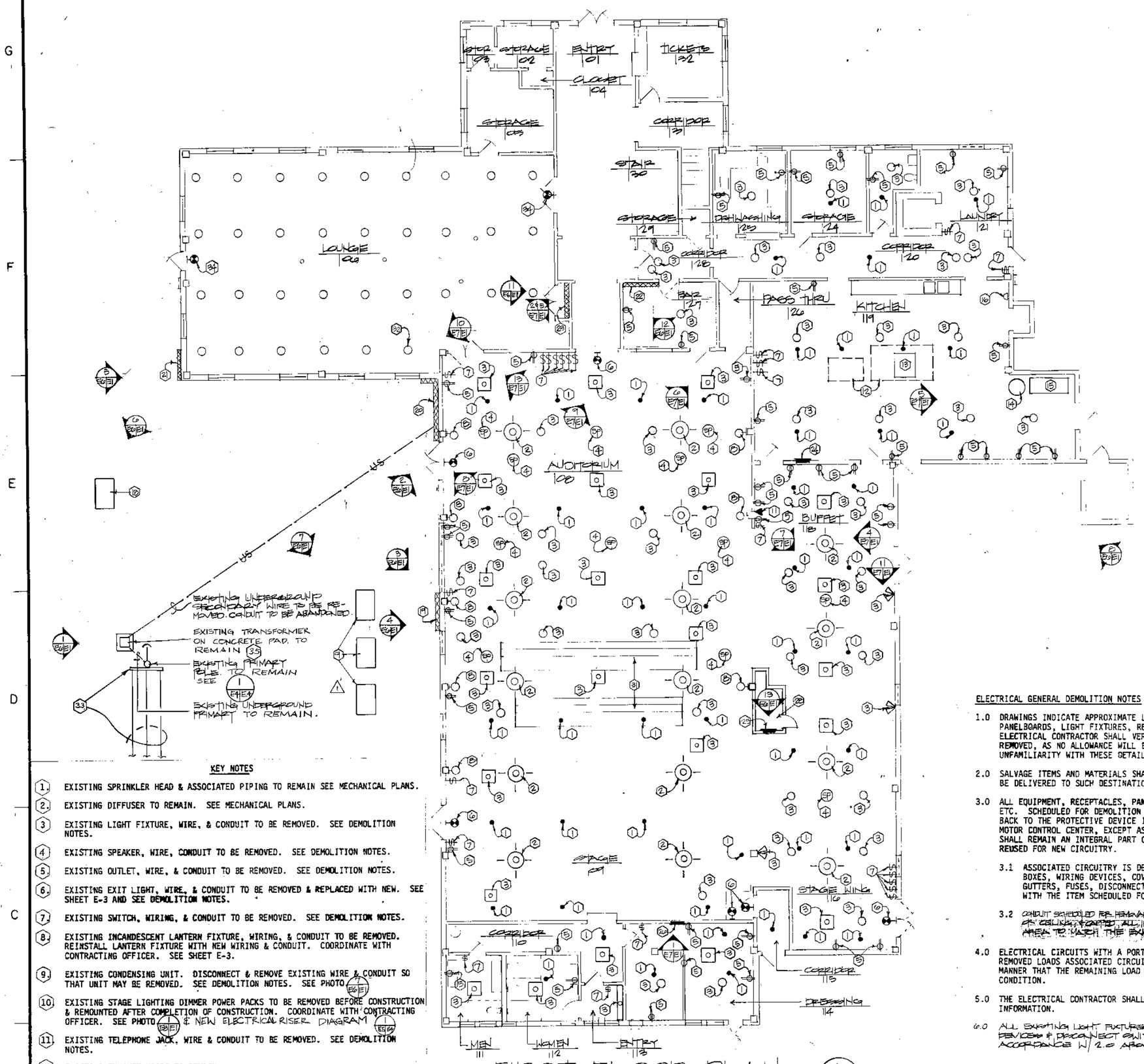
DESIGNED BY: EMB
DRAWN BY: EMB
CHECKED BY: CRR
SUBMITTED BY:
ENGINEER:

FORT SAM HOUSTON
SAN ANTONIO, TEXAS

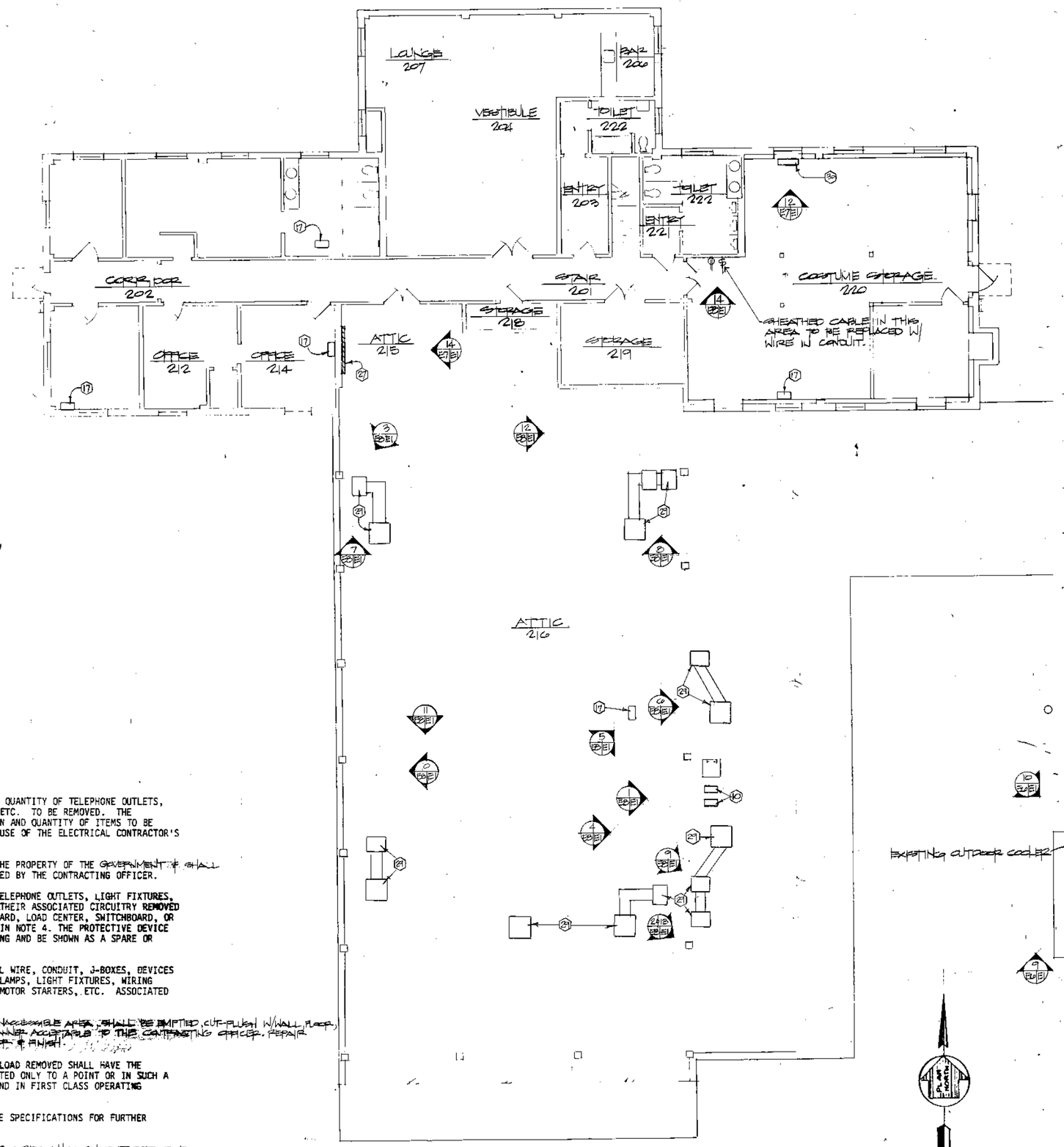
UPGRADE BUILDING 2252
REPAIR BUILDING 2652

ELECTRICAL DEMOLITION

INV. NO. DACA63-81-B-0071 DATED: 2/13/81
CONTR. NO. DACA63-81-C-0080
DRAWING NUMBER E1 OF 8
SEQUENCE NO. 20
CONTR. NO. DACA63-81-C-0080



FIRST FLOOR PLAN
ELECTRICAL DEMOLITION



ATTIC FLOOR PLAN
ELECTRICAL DEMOLITION

GRAPHIC SCALES

KEY NOTES

1. EXISTING SPRINKLER HEAD & ASSOCIATED PIPING TO REMAIN SEE MECHANICAL PLANS.
2. EXISTING DIFFUSER TO REMAIN. SEE MECHANICAL PLANS.
3. EXISTING LIGHT FIXTURE, WIRE, & CONDUIT TO BE REMOVED. SEE DEMOLITION NOTES.
4. EXISTING SPEAKER, WIRE, CONDUIT TO BE REMOVED. SEE DEMOLITION NOTES.
5. EXISTING OUTLET, WIRE, & CONDUIT TO BE REMOVED. SEE DEMOLITION NOTES.
6. EXISTING EXIT LIGHT, WIRE, & CONDUIT TO BE REMOVED & REPLACED WITH NEW. SEE SHEET E-3 AND SEE DEMOLITION NOTES.
7. EXISTING SWITCH, WIRING, & CONDUIT TO BE REMOVED. SEE DEMOLITION NOTES.
8. EXISTING INCANDESCENT LANTERN FIXTURE, WIRING, & CONDUIT TO BE REMOVED. REINSTALL LANTERN FIXTURE WITH NEW WIRING & CONDUIT. COORDINATE WITH CONTRACTING OFFICER. SEE SHEET E-3.
9. EXISTING CONDENSING UNIT. DISCONNECT & REMOVE EXISTING WIRE & CONDUIT SO THAT UNIT MAY BE REMOVED. SEE DEMOLITION NOTES. SEE PHOTO (6/E4/E1)
10. EXISTING STAGE LIGHTING DIMMER POWER PACKS TO BE REMOVED BEFORE CONSTRUCTION & REMOUNTED AFTER COMPLETION OF CONSTRUCTION. COORDINATE WITH CONTRACTING OFFICER. SEE PHOTO (8/E4/E1) & NEW ELECTRICAL RISER DIAGRAM (1/E5/E4)
11. EXISTING TELEPHONE JACK, WIRE & CONDUIT TO BE REMOVED. SEE DEMOLITION NOTES.
12. EXISTING EXHAUST HOODS TO REMAIN.
13. EXISTING OVEN TO REMAIN.
14. EXISTING BOILER TO REMAIN.
15. EXISTING STORAGE TANK TO REMAIN.
16. EXISTING DUMBWAITER TO BE REPAIRED & PLACED IN PROPER OPERATING CONDITION.
17. EXISTING UNIT HEATER. DISCONNECT & REMOVE EXISTING WIRE & CONDUIT SO THAT UNIT MAY BE REMOVED. SEE DEMOLITION NOTES.
18. EXISTING CONDENSING UNIT WATER PUMP. DISCONNECT & REMOVE EXISTING WIRE & CONDUIT SO THAT UNIT MAY BE REMOVED. SEE DEMOLITION NOTES. SEE PHOTO (6/E4/E1)
19. EXISTING ELECTRICAL GEAR. SEE RISER DIAGRAM (5/E2/E1) & PHOTO (4/E4/E1) FOR DEMOLITION INFORMATION.
20. EXISTING ELECTRICAL GEAR. SEE RISER DIAGRAM (4/E2/E1) & PHOTO (2/E4/E1) FOR DEMOLITION INFORMATION.
21. EXISTING ELECTRICAL GEAR. SEE RISER DIAGRAM (1/E2/E1) & PHOTO (5/E4/E1) FOR DEMOLITION INFORMATION.
22. EXISTING ELECTRICAL GEAR. SEE RISER DIAGRAM (3/E2/E1) & PHOTO (12/E4/E1) FOR DEMOLITION INFORMATION.
23. EXISTING ELECTRICAL GEAR. SEE RISER DIAGRAM (2/E2/E1) & PHOTO (11/E4/E1), (2/E7/E1), & (3/E7/E1) FOR DEMOLITION INFORMATION.
24. EXISTING PANEL TO BE REMOVED & REPLACED WITH NEW PANEL. SEE SHEET E-4 & NEW RISER DIAGRAM (1/E5/E4)

ELECTRICAL GENERAL DEMOLITION NOTES

1.0 DRAWINGS INDICATE APPROXIMATE LOCATION AND QUANTITY OF TELEPHONE OUTLETS, PANELBOARDS, LIGHT FIXTURES, RECEPTACLES, ETC. TO BE REMOVED. THE ELECTRICAL CONTRACTOR SHALL VERIFY LOCATION AND QUANTITY OF ITEMS TO BE REMOVED, AS NO ALLOWANCE WILL BE MADE BECAUSE OF THE ELECTRICAL CONTRACTOR'S UNFAMILIARITY WITH THESE DETAILS.

2.0 SALVAGE ITEMS AND MATERIALS SHALL REMAIN THE PROPERTY OF THE GOVERNMENT & SHALL BE DELIVERED TO SUCH DESTINATION AS DIRECTED BY THE CONTRACTING OFFICER.

3.0 ALL EQUIPMENT, RECEPTACLES, PANELBOARDS, TELEPHONE OUTLETS, LIGHT FIXTURES, ETC. SCHEDULED FOR DEMOLITION SHALL HAVE THEIR ASSOCIATED CIRCUITRY REMOVED BACK TO THE PROTECTIVE DEVICE IN A PANELBOARD, LOAD CENTER, SWITCHBOARD, OR MOTOR CONTROL CENTER, EXCEPT AS MENTIONED IN NOTE 4. THE PROTECTIVE DEVICE SHALL REMAIN AN INTEGRAL PART OF THE HOUSING AND BE SHOWN AS A SPARE OR REUSED FOR NEW CIRCUITRY.

3.1 ASSOCIATED CIRCUITRY IS DEFINED AS ALL WIRE, CONDUIT, J-BOXES, DEVICES BOXES, WIRING DEVICES, COVER PLATES, LAMPS, LIGHT FIXTURES, WIRING GUTTERS, FUSES, DISCONNECT SWITCHES, MOTOR STARTERS, ETC. ASSOCIATED WITH THE ITEM SCHEDULED FOR REMOVAL.

3.2 CONDUIT SCHEDULED FOR REMOVAL, BUT IN AN INACCESSIBLE AREA, SHALL BE EMPTIED, CUT-PLUGGED W/WALL PLATE, OR CEILING PLATE. ALL IN A SAFE MANNER ACCEPTABLE TO THE CONTRACTING OFFICER. REPAIR AREA TO MATCH THE EXISTING COLOR & FINISH.

4.0 ELECTRICAL CIRCUITS WITH A PORTION OF THE LOAD REMOVED SHALL HAVE THE REMOVED LOADS ASSOCIATED CIRCUITRY TERMINATED ONLY TO A POINT OR IN SUCH A MANNER THAT THE REMAINING LOAD IS ACTIVE AND IN FIRST CLASS OPERATING CONDITION.

5.0 THE ELECTRICAL CONTRACTOR SHALL CONSULT THE SPECIFICATIONS FOR FURTHER INFORMATION.

6.0 ALL EXISTING LIGHT FIXTURES & PANELBOARDS W/CIRCUIT PROTECTIVE DEVICES & DISCONNECT SWITCHES SHALL BE SALVAGED IN ACCORDANCE W/ 2.0 ABOVE.

KEY NOTES (CONTINUED)

25. EXISTING PANEL TO BE REMOVED & REPLACED WITH NEW PANEL. SEE SHEET E-4 & NEW RISER DIAGRAM (1/E5/E4). FOR MORE DEMOLITION INFORMATION SEE PHOTO (13/E4/E1).
26. EXISTING PANEL TO BE REMOVED & REPLACED WITH NEW PANEL. SEE SHEET E-4 & NEW RISER DIAGRAM (1/E5/E4). FOR MORE DEMOLITION INFORMATION SEE PHOTO (1/E7/E1).
27. EXISTING RHEOSTAT DIMMERS, WIRING, & CONDUIT TO BE REMOVED. SEE PHOTO (14/E7/E1) FOR MORE DEMOLITION INFORMATION.
28. EXISTING CONTROL ROOM TO BE REMOVED & REPLACED BY OTHERS. FOR NEW CONTROL ROOM LIGHTING & POWER SEE SHEETS E-3 & E-4.
29. EXISTING MECHANICAL EQUIPMENT. EXISTING POWER & CONTROL WIRING & CONDUIT TO BE REPLACED WITH POWER & CONTROL WIRING & CONDUIT & RECONNECTED TO NEW PANEL 'F'. SEE SHEET E-4 & NEW RISER DIAGRAM (1/E5/E4).

KEY NOTES (CONTINUED)

30. EXISTING WALL A/C UNIT. DISCONNECT & REMOVE EXISTING WIRE & CONDUIT SO THAT UNIT MAY BE REMOVED. SEE DEMOLITION NOTES. SEE PHOTO (12/E7/E1)
31. EXISTING CEILING RUNNERS FOR STAGE TRACK LIGHTING TO REMAIN.
32. TYPICAL EXISTING INCANDESCENT FIXTURE IN LOUNGE TO REMAIN. DISCONNECT EXISTING CIRCUITING & RECONNECT TO NEW PANEL 'A' LOCATED IN BAR.
33. THE EXISTING POLE HARDWARE SHALL REMAIN. THE OVERHEAD PRIMARY CONDUCTORS SHALL REMAIN. SEE (33) LOCATION. SEE (1/E4/E4) FOR NEW TRANSFORMER.
34. EXISTING EXIT LIGHT TO BE REMOVED & REPLACED WITH NEW. SEE KEY NOTE #44 SHEET E-3.
35. EXISTING TRANSFORMER SERVES BLDG 2652 AND ADJACENT BLDGS. EXISTING UNDERGROUND SERVICE TO BLDG 2652 SHALL BE TERMINATED AS INDICATED ON PLANS BUT EXISTING UNDERGROUND SERVICES TO ADJACENT BLDGS SHALL REMAIN.

SYM	DR. NO.	ACTION	DATE	DESCRIPTION OF REVISION
			22 MAR 82	"AS BUILT"
1	F4935	D #50001	20 NOV 81	Revise Notes

KILLEBREW/RUCKER/ASSOC., INC.
600 PETROLEUM BLDG.
WICHITA FALLS, TEXAS

U.S. ARMY ENGINEER DISTRICT, FORT WORTH
CORPS OF ENGINEERS
FORT WORTH, TEXAS

DESIGNED BY: EMP
DRAWN BY: EMP
CHECKED BY: CRP
SUBMITTED BY:
ENGINEER:

FORT SAM HOUSTON
SAN ANTONIO, TEXAS

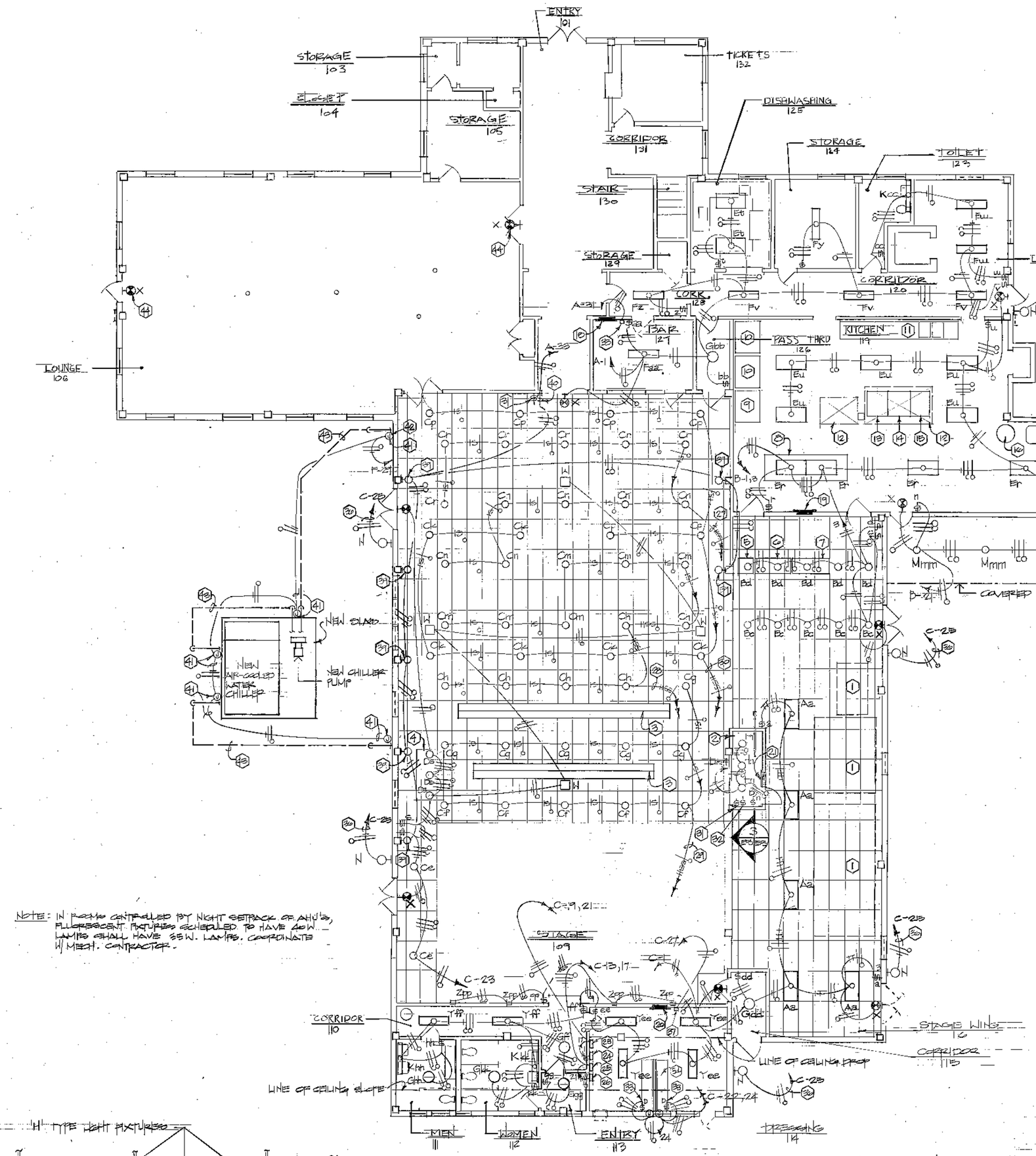
UPGRADE BUILDING 2652
REPAIR BUILDING 2652

ELECTRICAL DEMOLITION

INV. NO. DACA63-81-B-0071 DATED: 2/13/81
CONTR. NO. DACA63-81-C-0080
DRAWING NUMBER: E1.1 of 8
SHEET NO.
SEQUENCE NO. 20A

CONTR. NO. DACA63-81-C-0080

SUPERSEDES SHEET NO E1 of 8



FIRST FLOOR - ELECTRICAL RENOVATION (1 E3 E3)

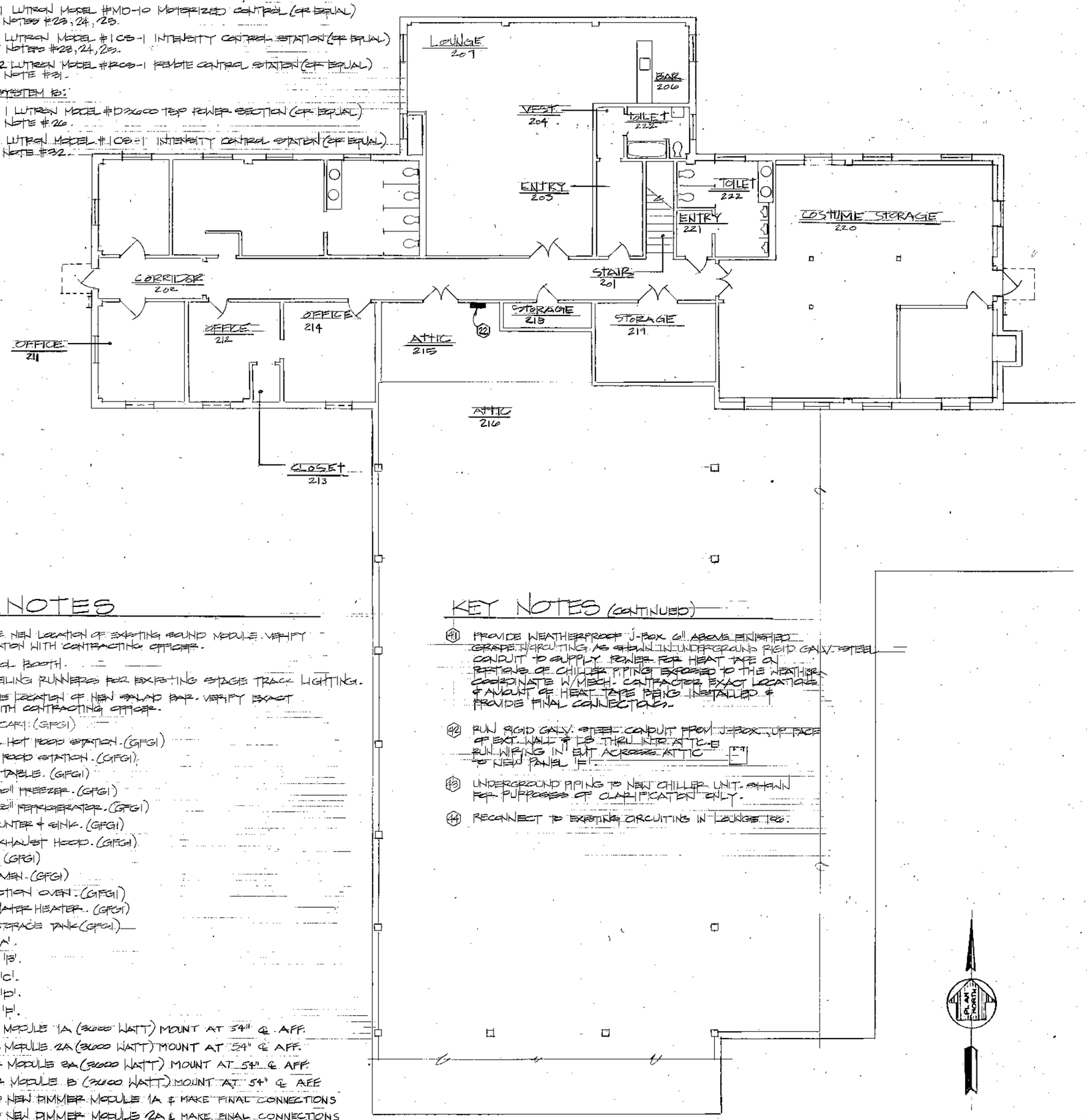
SCALE 1/8" = 1'-0"

DIMMER NOTES

1. DIMMING SYSTEM A:
   A. PROVIDE 3 LUTRON MODEL #D3000 TRP POWER SECTIONS (OR EQUAL) SEE KEY NOTES #23, 24, 25.
   B. PROVIDE 1 LUTRON MODEL #MD-10 MOTORIZED CONTROL (OR EQUAL) SEE KEY NOTES #23, 24, 25.
   C. PROVIDE 1 LUTRON MODEL #1CS-1 INTENSITY CONTROL STATION (OR EQUAL) SEE KEY NOTES #22, 24, 25.
   D. PROVIDE 2 LUTRON MODEL #RCS-1 REMOTE CONTROL STATION (OR EQUAL) SEE KEY NOTE #21.
2. DIMMING SYSTEM B:
   A. PROVIDE 1 LUTRON MODEL #D3600 TRP POWER SECTION (OR EQUAL) SEE KEY NOTE #26.
   B. PROVIDE 1 LUTRON MODEL #1CS-1 INTENSITY CONTROL STATION (OR EQUAL) SEE KEY NOTE #32.

KEY NOTES

1. APPROXIMATE NEW LOCATION OF EXISTING SOUND MODULE. VERIFY EXACT LOCATION WITH CONTRACTING OFFICER.
2. NEW CONTROL BOOTH.
3. EXISTING CEILING RUNNERS FOR EXISTING STAGE TRACK LIGHTING.
4. APPROXIMATE LOCATION OF NEW SALAD BAR. VERIFY EXACT LOCATION WITH CONTRACTING OFFICER.
5. NEW ENTRE CART. (GFGI)
6. NEW 4-WELL HOT FOOD STATION. (GFGI)
7. NEW COLD FOOD STATION. (GFGI)
8. NEW WORK TABLE. (GFGI)
9. NEW 30"x60" FREEZER. (GFGI)
10. NEW 30"x60" REFRIGERATOR. (GFGI)
11. EXISTING COUNTER & SINK. (GFGI)
12. EXISTING EXHAUST HOOD. (GFGI)
13. NEW STOVE. (GFGI)
14. EXISTING OVEN. (GFGI)
15. NEW CONVECTION OVEN. (GFGI)
16. EXISTING WATER HEATER. (GFGI)
17. EXISTING STORAGE TANK (GFGI)
18. NEW PANEL 'A'.
19. NEW PANEL 'B'.
20. NEW PANEL 'C'.
21. NEW PANEL 'D'.
22. NEW PANEL 'F'.
23. NEW DIMMER MODULE 1A (3000 WATT) MOUNT AT 54" @ AFF.
24. NEW DIMMER MODULE 2A (3000 WATT) MOUNT AT 54" @ AFF.
25. NEW DIMMER MODULE 3A (3000 WATT) MOUNT AT 54" @ AFF.
26. NEW DIMMER MODULE B (3000 WATT) MOUNT AT 54" @ AFF.
27. HOMERUN TO NEW DIMMER MODULE 1A & MAKE FINAL CONNECTIONS
28. HOMERUN TO NEW DIMMER MODULE 2A & MAKE FINAL CONNECTIONS
29. HOMERUN TO NEW DIMMER MODULE 3A & MAKE FINAL CONNECTIONS
30. HOMERUN TO NEW DIMMER MODULE B & MAKE FINAL CONNECTIONS
31. NEW DIMMER CONTROL FOR NEW DIMMER MODULES 1A, 2A, & 3A. PROVIDE & INSTALL CONTROL WIRING IN CONDUIT AS REQ'D BETWEEN DIMMER CONTROL & MODULE. (2 LOCATIONS AS SHOWN ON PLANS. MOTORIZED CONTROLS)
32. NEW DIMMER CONTROL FOR NEW DIMMER MODULE B. PROVIDE & INSTALL CONTROL WIRING IN CONDUIT AS REQUIRED BETWEEN DIMMER CONTROL & MODULE.
33. SURFACE METAL RACEWAY (WIREMOLD 2100 OR EQUAL) SHALL HAVE A SEPARATE CIRCUIT & DIMMER FOR EACH SIDE OF DRESSING TABLE. MOUNT DIMMER 6" ABOVE TOP OF DRESSING TABLE IN WALL IN LOCATION SHOWN ON (1 E3 E3)
34. EXISTING DRESSING TABLE. PROVIDE NEW LIGHTING & POWER AS SHOWN IN PHOTO (4 E3 E3)
35. FOR CLARIFICATION PURPOSES, THE MAKE-UP LIGHTS HAVE BEEN LABELED 'E' TYPE FIXTURES IN THE LIGHTING FIXTURE SCHEDULE ON SHEET E-5.
36. HOMERUN THROUGH NEW TIMECLOCK FLUSH-MOUNTED IN WALL NEXT TO PANEL 'C' IN LOCATION SHOWN.
37. NEW TIMECLOCK FLUSH-MOUNTED IN WALL IN LOCATION SHOWN & ON THE OWN CIRCUIT (C-27).
38. NEW LOCATION OF EXISTING TIMECLOCK: RECONNECT EXISTING OUTSIDE LIGHTING FROM PREVIOUS TIMECLOCK LOCATION & PROVIDE WIRING, & CONDUIT AS REQUIRED.
39. EXISTING LANTERN FIXTURE REMOUNTED IN EXISTING LOCATION. PROVIDE & INSTALL NEW WIRING IN CONDUIT AS SHOWN.
40. NEW SWITCH FOR EXISTING LANTERN FIXTURES. PROVIDE & INSTALL NEW WIRING IN CONDUIT AS SHOWN.

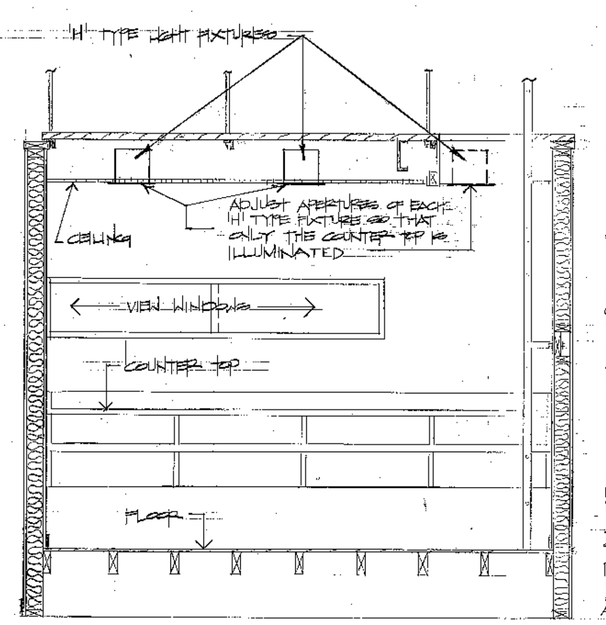


SECOND FLOOR - ELECTRICAL RENOVATION (2 E3 E3)

SCALE 1/8" = 1'-0"

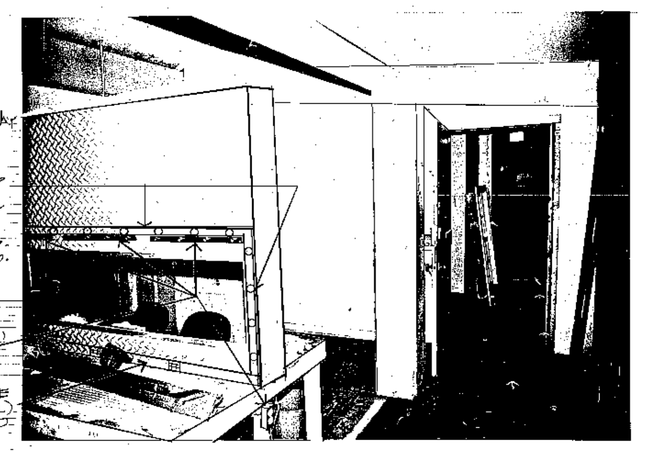
KEY NOTES (CONTINUED)

41. PROVIDE WEATHERPROOF J-BOX 6" ABOVE FINISHED GRADE NEXT TO NEW CHILLER UNIT. PROVIDE GROUTING AS SHOWN IN UNDERGROUND RIGID GALV. STEEL CONDUIT TO SUPPLY POWER FOR HEAT TAPE ON PORTIONS OF CHILLER PIPING EXPOSED TO THE WEATHER. COORDINATE W/ MECH. CONTRACTOR EXACT LOCATIONS & AMOUNT OF HEAT TAPE BEING INSTALLED & PROVIDE FINAL CONNECTIONS.
42. RUN RIGID GALV. STEEL CONDUIT FROM J-BOX ON FACE OF EXT. WALL @ DB THRU INTO ATTIC & RUN WIRING IN EXT ACROSS ATTIC TO NEW PANEL 'F'.
43. UNDERGROUND PIPING TO NEW CHILLER UNIT SHOWN FOR PURPOSES OF CLARIFICATION ONLY.
44. RECONNECT TO EXISTING CIRCUITING IN LOUNGE 106.



SECTION (3)

SCALE: 1/2" = 1'-0" E3 E3



ON BOTH SIDES OF TABLE PROVIDE SURFACE METAL RACEWAY (WIREMOLD 2100 OR EQUAL) W/ (WIREMOLD 2120 F OR EQUAL) WHITE PORCELAIN KEYLESS SOCKETS EQUALLY SPACED AS SHOWN. PROVIDE ADAPTERS TO SOCKETS AS NECESSARY TO ACCOMMODATE 60 WATT G-40 INCANDESCENT LAMPS. SEE KEY NOTES #33 & #34.

ON BOTH SIDES OF EXISTING DRESSING TABLE MAKE-UP LIGHTS & OUTLET WIRE & CONDUIT TO BE REMOVED. SEE DEMO. NOTES SHEET E-1.

PROVIDE ON BOTH SIDES OF TABLE NEW G-40 WIREMOLD (OR EQUAL) W/ 3 OUTLETS EQUALLY SPACED ALONG LENGTH OF TABLE

PHOTO @ DRESSING TABLE (4 E3 E3)

NO SCALE

VOIDED

REGISTERED ARCHITECT STATE OF TEXAS 1067

James R. Killebrew

STATE OF TEXAS JAMES R. KILLEBREW 9482 REGISTERED PROFESSIONAL ENGINEER

SYM.	NO.	ACTION	DATE	DESCRIPTION OF REVISION
			22 MAR 82	"DRAWING OF RECORD AS BUILT"

KILLEBREW/RUCKER/ASSOC., INC.
600 PETROLEUM BLDG.
WICHITA FALLS, TEXAS

U.S. ARMY ENGINEER DISTRICT, FORT WORTH
CORPS OF ENGINEERS
FORT WORTH, TEXAS

DESIGNED BY: PWP
DRAWN BY: PWP
CHECKED BY: RPO
SUBMITTED BY:
ENGINEER:

FORT SAM HOUSTON
SAN ANTONIO, TEXAS

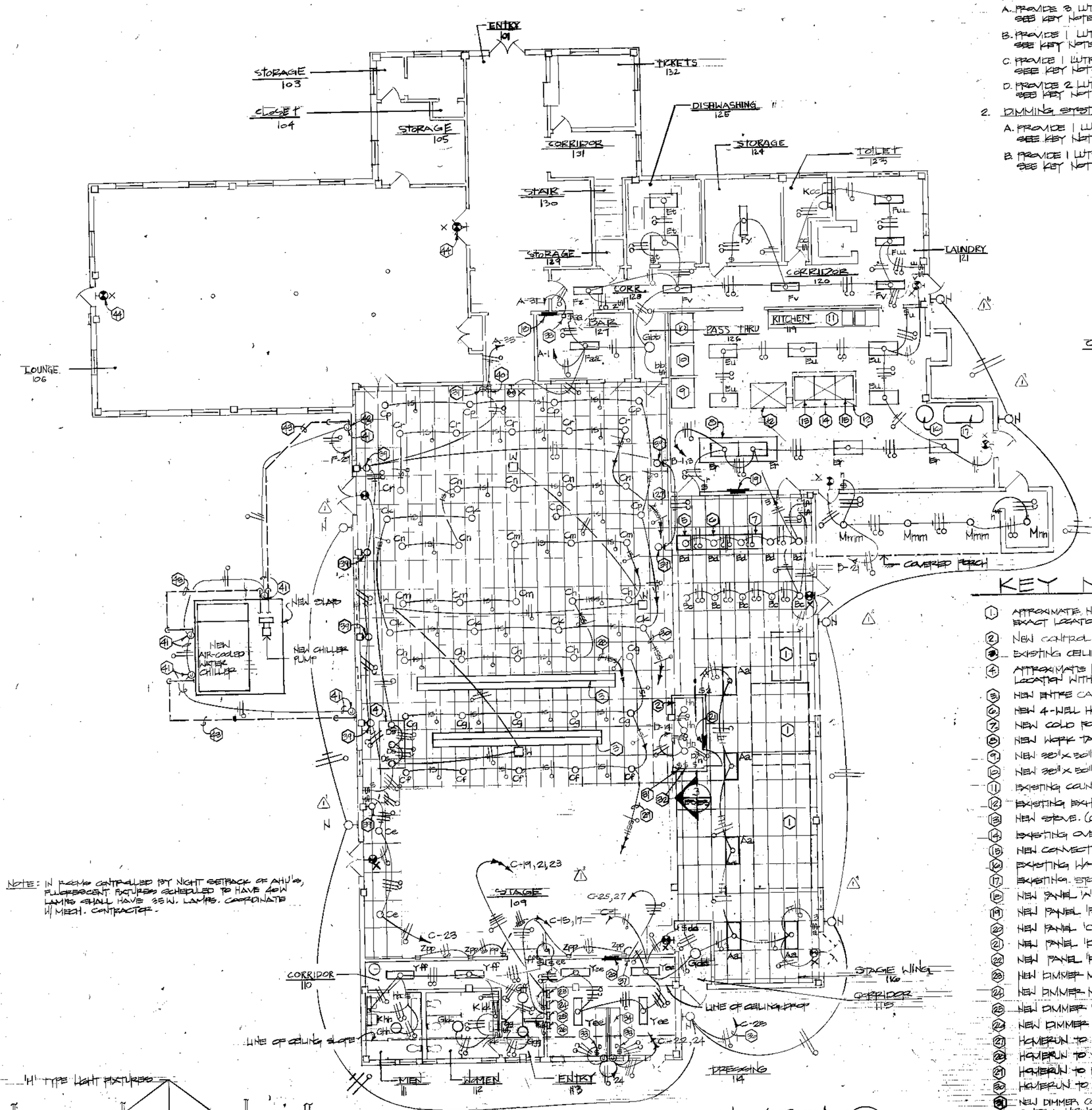
UPGRADE BUILDING 2652
REPAIR BUILDING 2652

LIGHTING - RENOVATED FLOOR PLAN

INV. NO. DACA63-81-B-0071 DATED: 2/3/81
CONTR. NO. DACA63-81-C-0080
DRAWING NUMBER
SHEET NO. E3 OF 8
SEQUENCE NO. 22
CONTR. NO. DACA63-81-C-0080

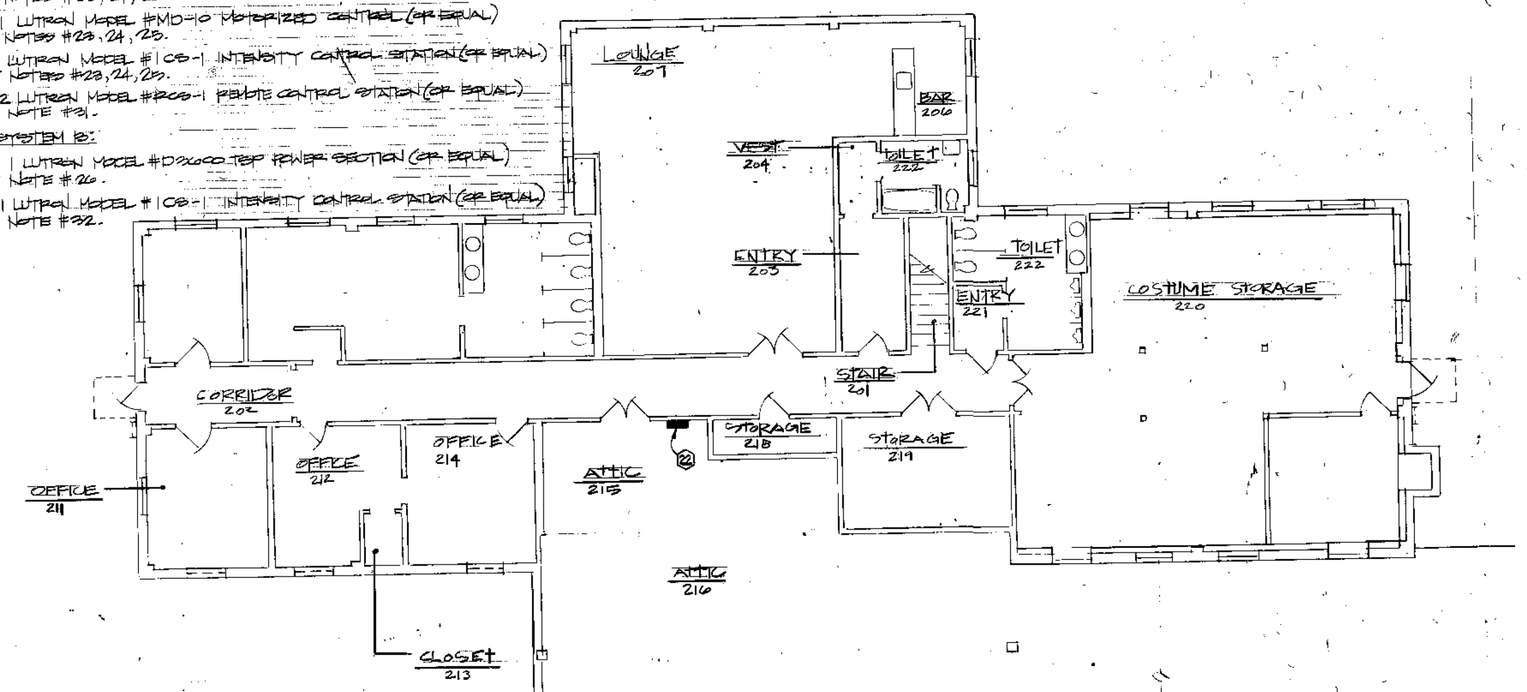
DIMMER NOTES

1. DIMMING SYSTEM A:
   A. PROVIDE 3 LUTRON MODEL #D3000 TRP POWER SECTIONS (OR EQUAL) SEE KEY NOTES #23, 24, 25.
   B. PROVIDE 1 LUTRON MODEL #MD-10 MOTORIZED CONTROL (OR EQUAL) SEE KEY NOTES #23, 24, 25.
   C. PROVIDE 1 LUTRON MODEL #1CS-1 INTENSITY CONTROL STATION (OR EQUAL) SEE KEY NOTES #23, 24, 25.
   D. PROVIDE 2 LUTRON MODEL #RCS-1 PRESET CONTROL STATION (OR EQUAL) SEE KEY NOTE #31.
2. DIMMING SYSTEM B:
   A. PROVIDE 1 LUTRON MODEL #D3000 TRP POWER SECTION (OR EQUAL) SEE KEY NOTE #26.
   B. PROVIDE 1 LUTRON MODEL #1CS-1 INTENSITY CONTROL STATION (OR EQUAL) SEE KEY NOTE #32.



NOTE: IN ROOMS CONTROLLED BY NIGHT SETBACK OF AHU'S, FLUORESCENT FIXTURES SCHEDULED TO HAVE 60W LAMPS SHALL HAVE 55W. LAMPS. COORDINATE W/ MECH. CONTRACTOR.

FIRST FLOOR - ELECTRICAL RENOVATION (1 / E3/E3)
SCALE 1/8" = 1'-0"



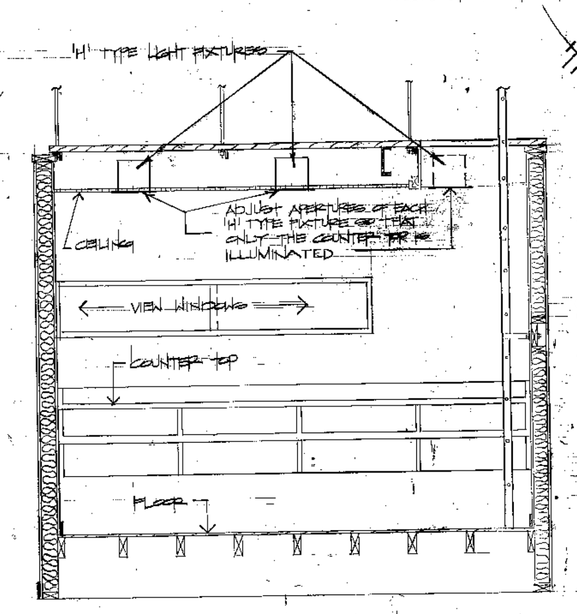
SECOND FLOOR - ELECTRICAL RENOVATION (2 / E3/E3)
SCALE 1/8" = 1'-0"

KEY NOTES

1. APPROXIMATE NEW LOCATION OF EXISTING SOUND MODULE. VERIFY EXACT LOCATION WITH CONTRACTING OFFICER.
2. NEW CONTROL BOOTH
3. EXISTING CEILING RUNNERS FOR EXISTING STAGE TRACK LIGHTING.
4. APPROXIMATE LOCATION OF NEW SALAD BAR. VERIFY EXACT LOCATION WITH CONTRACTING OFFICER.
5. NEW ENTREE CART (GFGI)
6. NEW 4-WELL HOT FOOD STATION. (GFGI)
7. NEW COLD FOOD STATION. (GFGI)
8. NEW WORK TABLE. (GFGI)
9. NEW 36" x 30" FREEZER. (GFGI)
10. NEW 36" x 30" REFRIGERATOR. (GFGI)
11. EXISTING COUNTER & SINK. (GFGI)
12. EXISTING EXHAUST HOOD. (GFGI)
13. NEW STOVE. (GFGI)
14. EXISTING OVEN. (GFGI)
15. NEW CONVECTION OVEN. (GFGI)
16. EXISTING WATER HEATER. (GFGI)
17. EXISTING STORAGE TANK (GFGI)
18. NEW PANEL 'N'.
19. NEW PANEL 'B'.
20. NEW PANEL 'C'.
21. NEW PANEL 'D'.
22. NEW PANEL 'F'.
23. NEW DIMMER MODULE 1A (3000 WATT) MOUNT AT 54" @ AFF.
24. NEW DIMMER MODULE 2A (3000 WATT) MOUNT AT 54" @ AFF.
25. NEW DIMMER MODULE 3A (3000 WATT) MOUNT AT 54" @ AFF.
26. NEW DIMMER MODULE B (3000 WATT) MOUNT AT 54" @ AFF.
27. HOMERUN TO NEW DIMMER MODULE 1A & MAKE FINAL CONNECTIONS
28. HOMERUN TO NEW DIMMER MODULE 2A & MAKE FINAL CONNECTIONS
29. HOMERUN TO NEW DIMMER MODULE 3A & MAKE FINAL CONNECTIONS
30. HOMERUN TO NEW DIMMER MODULE B & MAKE FINAL CONNECTIONS
31. NEW DIMMER CONTROL FOR NEW DIMMER MODULES 1A, 2A, & 3A. PROVIDE & INSTALL CONTROL WIRING IN CONDUIT AS REQ'D BETWEEN DIMMER CONTROL & MODULE. (2 LOCATIONS AS SHOWN ON PLANS. MOTORIZED CONTROLS)
32. NEW DIMMER CONTROL FOR NEW DIMMER MODULE B. PROVIDE & INSTALL CONTROL WIRING IN CONDUIT AS REQUIRED BETWEEN DIMMER CONTROL & MODULE.
33. SURFACE METAL RACEWAY (SLIMLINE 2100 OR EQUAL) SHALL HAVE A SEPARATE CIRCUIT & DIMMER FOR EACH SIDE OF DRESSING TABLE. MOUNT DIMMER 6" ABOVE TOP OF DRESSING TABLE IN WALL IN LOCATION SHOWN ON (1 / E3/E3)
34. EXISTING DRESSING TABLE. PROVIDE NEW LIGHTING & POWER AS SHOWN IN PHOTO (4 / E3/E3)
35. FOR CLARIFICATION PURPOSES, THE MAKE-UP LIGHTS HAVE BEEN LABELED 'R' TYPE FIXTURES IN THE LIGHTING FIXTURE SCHEDULE ON SHEET E-3.
36. HOMERUN THROUGH NEW TIMECLOCK FLUSH-MOUNTED IN WALL NEXT TO PANEL 'C' IN LOCATION SHOWN.
37. NEW TIMECLOCK FLUSH-MOUNTED IN WALL IN LOCATION SHOWN ON THE OWN CIRCUIT (C-27).
38. NEW LOCATION OF EXISTING TIMECLOCK. RECONNECT EXISTING OUTSIDE LIGHTING FROM PREVIOUS TIMECLOCK LOCATION & PROVIDE J-BOXES, WIRING & CONDUIT AS REQUIRED.
39. EXISTING LANTERN FIXTURE REMOUNTED IN EXISTING LOCATION. PROVIDE & INSTALL NEW WIRING IN CONDUIT AS SHOWN.
40. NEW SWITCH FOR EXISTING LANTERN FIXTURES. PROVIDE & INSTALL NEW WIRING IN CONDUIT AS SHOWN.

KEY NOTES (CONTINUED)

41. PROVIDE WEATHERPROOF J-BOX 6" ABOVE FINISHED GRADE W/ RECEPTACLE AS SHOWN IN UNDERGROUND RIGID GALV. STEEL CONDUIT TO SUPPLY POWER FOR HEAT TAPE ON PORTIONS OF CHILLER PIPING EXPOSED TO THE WEATHER. COORDINATE W/ MECH. CONTRACTOR EXACT LOCATIONS & AMOUNT OF HEAT TAPE BEING INSTALLED & PROVIDE FINAL CONNECTION.
42. RUN RIGID GALV. STEEL CONDUIT FROM J-BOX UP FACE OF EXT. WALL & 1'-0" THRU INTO ATTIC. RUN WIRING IN EXT. ACROSS ATTIC TO NEW PANEL 'H'.
43. UNDERGROUND PIPING TO NEW CHILLER UNIT. SHOWN FOR PURPOSES OF CLARIFICATION ONLY.
44. RECONNECT TO EXISTING CIRCUITING IN LOUNGE 106.



SECTION (3 / E3/E3)
SCALE 1/2" = 1'-0"

ON BOTH SIDES OF TABLE PROVIDE SURFACE METAL RACEWAY (SLIMLINE 2100 OR EQUAL) W/ (SLIMLINE 2100 P OR EQUAL) WHITE ENGLISH KEYLESS SOCKETS EQUALLY SPACED AS SHOWN. PROVIDE ADAPTERS TO SOCKETS AS NECESSARY TO ACCOMMODATE 60 WATT G16½ INCANDESCENT LAMPS. SEE KEY NOTES #30 & #33.

ON BOTH SIDES OF EXISTING DRESSING TABLE VANITY LIGHTS & OUTLETS. LINES & CIRCUIT TO BE RELOCATED. SEE REV. NOTES SHEET E-1.

PROVIDE ON BOTH SIDES OF TABLE NEW (3-2000 WIRE-MOLD OR EQUAL) W/ 3 OUTLETS EQUALLY SPACED ALONG LENGTH OF TABLE

PHOTO @ DRESSING TABLE (4 / E3/E3)
NO SCALE

SYM	NO.	ACTION	DATE	DESCRIPTION OF REVISION
			22 MAR 82	[illegible]
1	E4935	COR P00001	20 NOV 81	REVISE CIRCUIT

KELLEBREW/RUCKER/ASSOC., INC.
800 PETROLEUM BLDG.
WICHITA FALLS, TEXAS

U.S. ARMY ENGINEER DISTRICT, FORT WORTH
CORPS OF ENGINEERS
FORT WORTH, TEXAS

DESIGNED BY: DWP
DRAWN BY: DWP
CHECKED BY: RPO
SUBMITTED BY:
ENGINEER:

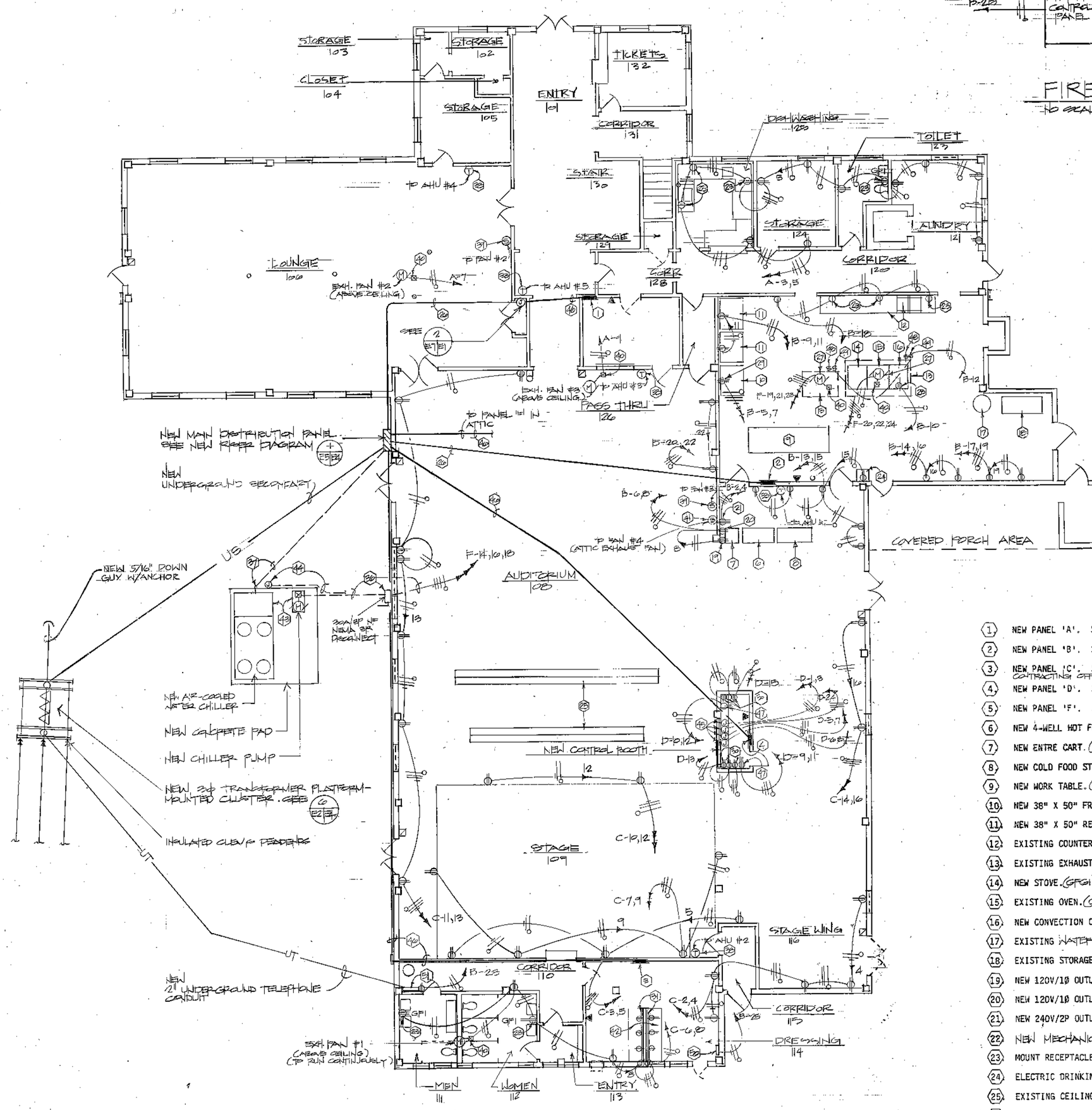
FORT SAM HOUSTON
SAN ANTONIO, TEXAS

UPGRADE BUILDING 2652
REPAIR BUILDING 2652

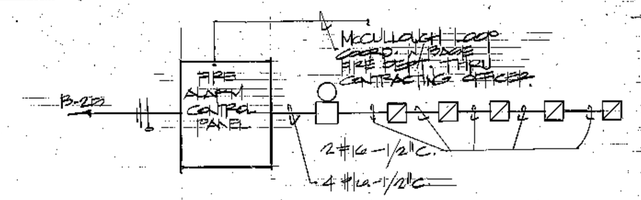
LIGHTING - RENOVATED FLOOR PLAN

INV. NO. DACA 63-81-B-0071 DATED 2/3/81
CONTR. NO.
DRAWING NUMBER
SHEET 2 OF 22

REGISTERED ARCHITECT STATE OF TEXAS
JAMES R. KELLEBREW 9482 REGISTERED PROFESSIONAL ENGINEER STATE OF TEXAS



FIRST FLOOR

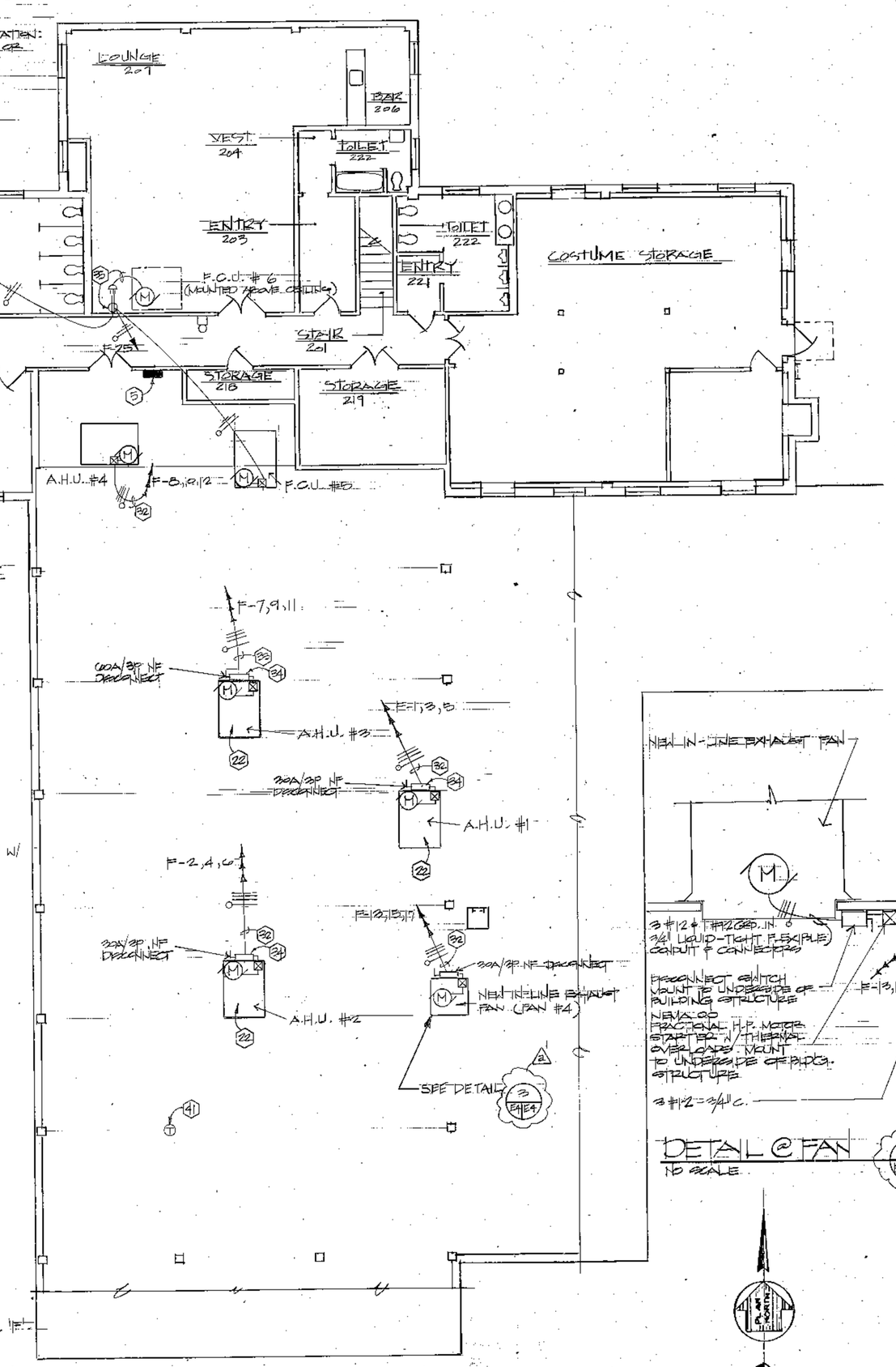


FIRE ALARM RISER DIAGRAM

NO SCALE

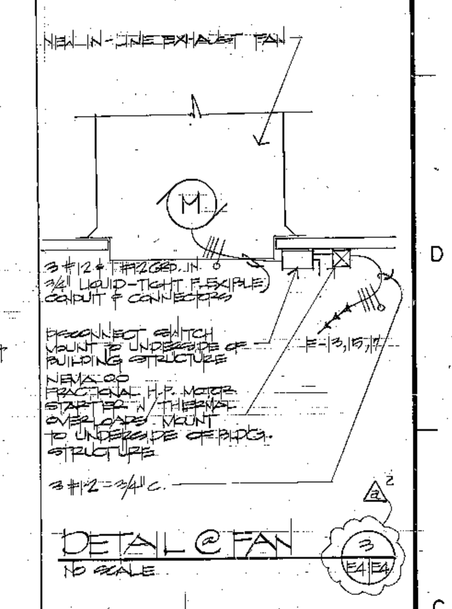
EQUIPMENT NOTES

1. FIRE ALARM CONTROL PANEL: EDWARDS #EQ1000 OR EQUAL
2. HORN/LIGHT COMBINATION: UNITED #720 24V/6PP FLANGE OR EQUAL.
3. FIRE ALARM PULL-STATION: EDWARDS #270-EPS OR EQUAL



ATTIC

NOTE:
COORDINATE W/CONTRACTING OFFICER THE PATCHING OF WALLS TO CONCEAL CONDUIT IN ROOMS W/WALLS NOT SCHEDULED FOR REFURBISHING.



DETAIL @ FAN

NO SCALE

GRAPHIC SCALES

KEY NOTES

1. NEW PANEL 'A'. SEE SHEET E-5 FOR PANEL SCHEDULE & NEW RISER DIAGRAM.
2. NEW PANEL 'B'. SEE SHEET E-5 FOR PANEL SCHEDULE & NEW RISER DIAGRAM.
3. NEW PANEL 'C'. SEE SHEET E-5 FOR PANEL SCHEDULE & NEW RISER DIAGRAM. COORDINATE INSTALLATION W/ CONTRACTING OFFICER TO MAINTAIN INTEGRITY OF FIRE WALL.
4. NEW PANEL 'D'. SEE SHEET E-5 FOR PANEL SCHEDULE & NEW RISER DIAGRAM.
5. NEW PANEL 'F'. SEE SHEET E-5 FOR PANEL SCHEDULE & NEW RISER DIAGRAM.
6. NEW 4-WELL HOT FOOD STATION. (GFGI)
7. NEW ENTRE CART. (GFGI)
8. NEW COLD FOOD STATION. (GFGI)
9. NEW WORK TABLE. (GFGI)
10. NEW 38" X 50" FREEZER. (GFGI)
11. NEW 38" X 50" REFRIGERATOR. (GFGI)
12. EXISTING COUNTER AND SINK. (GFGI)
13. EXISTING EXHAUST HOOD. (GFGI)
14. NEW STOVE. (GFGI)
15. EXISTING OVEN. (GFGI)
16. NEW CONVECTION OVEN. (GFGI)
17. EXISTING WATER HEATER (GFGI)
18. EXISTING STORAGE TANK. (GFGI)
19. NEW 120V/1Ø OUTLET FOR ENTRE CART. MOUNT 12" A.F.F.
20. NEW 120V/1Ø OUTLET FOR COFFEE MAKER. MOUNT 12" A.F.F.
21. NEW 240V/2P OUTLET FOR 4 WELL HOT FOOD STATION. MOUNT 12" A.F.F. OUTLETS ARE 20A.
22. NEW MECHANICAL EQUIPMENT. PROVIDE CIRCUITING IN CONDUIT AS SHOWN TO NEW PANEL 'F'.
23. MOUNT RECEPTACLE HORIZONTALLY 6" ABOVE LINE OF TOP OF SINGS.
24. ELECTRIC DRINKING FOUNTAIN.
25. EXISTING CEILING TRACK RUNNERS FOR STAGE LIGHTING.
26. FEEDER CONDUIT. EXISTING TO REMAIN. SEE NEW RISER DIAGRAM (E5/E4)
27. EXISTING EXHAUST HOOD FANS TO BE RECIRCULATED AS SHOWN.
28. NEW 120V/1Ø OUTLET FOR CONVECTION OVEN. STUB DOWN FROM CEILING & SUPPORT FROM FLOOR.
29. NEW 240V/2P OUTLET FOR NEW 38" x 50" FREEZER. MOUNT 12" A.F.F. OUTLETS ARE 20A.
30. MOUNT RECEPTACLE 12" ABOVE TOP OF LINE OF COUNTER.
31. NEW WIREMOLD. SEE DRESSING TABLE DETAIL (E3/E3)
32. 3 #12 & 1 #12 GROUND IN 3/4" C.
33. 3 #10 & 1 #12 GROUND IN 3/4" C.
34. MOUNT DISCONNECT SWITCH ON SIDE OF AHU. DISCONNECT SWITCH SHALL UNDER NO CIRCUMSTANCES BE MOUNTED ON THE A.H.U. OR ACCESS PANEL OR IN ANY MANNER THAT HINDERS PROPER ACCESS TO THE UNIT.
35. MOUNT OUTLET ABOVE CEILING ON ACCESSIBLE WALL OR COMPARABLE SUPPORT NO MORE THAN 3'-0" FROM UNIT. PROVIDE CORD & CAP & FINAL CONNECTIONS.
36. 3 #10 & 1 #12 IN 3/4" C. RUN FROM NEW CHILLER PUMP UNDERGROUND TO NEW DISCONNECT SWITCH MOUNTED ON EXTERIOR WALL. RUN FROM SWITCH UP SIDE OF EXTERIOR WALL, UP THRU INTO ATTIC & RUN TO 20A/2P CIRCUIT BREAKER IN NEW PANEL 'F'.
37. UNDERGROUND SERVICE FROM NEW MDP TO NEW CHILLER. STUB UP & UP INTO UNIT. SEE NEW RISER DIAGRAM SHEET E-5 FOR WIRE & CONDUIT SIZES.
38. PROVIDE CONTROL WIRING IN CONDUIT BETWEEN THERMOSTAT & UNIT. COORDINATE W/MECH. FOR NUMBER OF WIRES & CONNECTIONS TO UNIT.

RENOVATED FLOOR PLAN - POWER

SCALE: 1/8" = 1'-0"

KEY NOTES (CONTINUED)

39. PROVIDE & INSTALL NEW CONDUIT & WIRING TO RECONNECT EXISTING STAGE LIGHT DIMMER MODULES. SEE NEW RISER DIAGRAM & NEW PANEL SCHEDULES 'D'.
40. NEW PANELBOARD FEEDER CONDUIT & WIRING. SEE NEW RISER DIAGRAM
41. NEW 20A/240V/1Ø OUTLET.
42. EXISTING SWITCH TO EXISTING EXHAUST FAN.
43. EXISTING SWITCH TO EXISTING EXHAUST HOOD LIGHT. PROVIDE NEW CIRCUITING AS SHOWN. CLEAN & RELAMP FIXTURE.
44. NEW FIRE ALARM CONTROL PANEL. MOUNT 84" AFF. SINGLE ZONE SINGLE CIRCUIT PANEL. SEE FIRE ALARM RISER DIAGRAM
45. NEW 18" x 24" x 6" TELEPHONE CABINET. SQUARE D #12246B BOX OR EQUAL; SQUARE D #1224 TF FRONT OR EQUAL; & SQUARE D #1224 BACKING OR EQUAL.

KEY NOTES (CONTINUED)

46. PROVIDE CONTROL WIRING IN CONDUIT BETWEEN EXHAUST INLET SWITCH & FAN. COORDINATE W/MECH. FOR NUMBER OF WIRES & CONNECTIONS TO FAN.
47. PROVIDE EXHAUST FAN W/A FRACTIONAL H.P. MANUAL STARTER W/THERMAL OVERLOAD RELAY, RATED 240 VAC 1 H.P.
48. PROVIDE CONTROL WIRING IN CONDUIT BETWEEN WINTER/SUMMER SWITCH, THERMOSTAT IN ATTIC, & ATTIC EXHAUST FAN ON ROOF. COORDINATE W/MECH. FOR NUMBER OF WIRES, CONNECTIONS TO FAN, & EXACT LOCATION OF THERMOSTAT IN ATTIC.
49. EXISTING DRESSING TABLE TO REMAIN. PROVIDE NEW G-2000 WIREMOLD (OR EQUAL) W/2 OUTLETS BRANCH, SPACED AS SHOWN. REF FURTHER INFORMATION SEE PHOTO AT DRESSING TABLE (4/E3/E3)
50. FOR HEAT STRIP CIRCUITING SEE (E3/E3)
51. CONTROL WIRING IN 3/4" RIGID GALV. STEEL CONDUIT. STUB UP TO WEATHERPROOF J-BOX & RUN IN 3/4" LIQUID TIGHT FLEX. CONDUIT TO UNIT. RUN IN RIGID GALV. CONDUIT FROM J-BOX UNDERGROUND UP FACE OF EXT. WALL, UP THRU WALL & RUN IN EMT ACROSS ATTIC TO NEW PANEL 'F' (CIRCUIT #27)

VOIDED

James R. Ketchum

STATE OF TEXAS JAMES R. KETCHUM 9432 REGISTERED PROFESSIONAL ENGINEER

Am 0001 13 March 81

SYM.	DO. NO.	ACTION	DATE	DESCRIPTION OF REVISION
			22MAR82	CHANGE OF WORK AS BUILT
A	AM# 0001		13 MAR. 81	DETAIL @ FAN CHANGED FROM (2/E4/E4) TO (3/E4/E4)

KILLEBREW/RUCKER/ASSOC., INC.
600 PETROLEUM BLDG.
WICHITA FALLS, TEXAS

U.S. ARMY ENGINEER DISTRICT, FORT WORTH
CORPS OF ENGINEERS
FORT WORTH, TEXAS

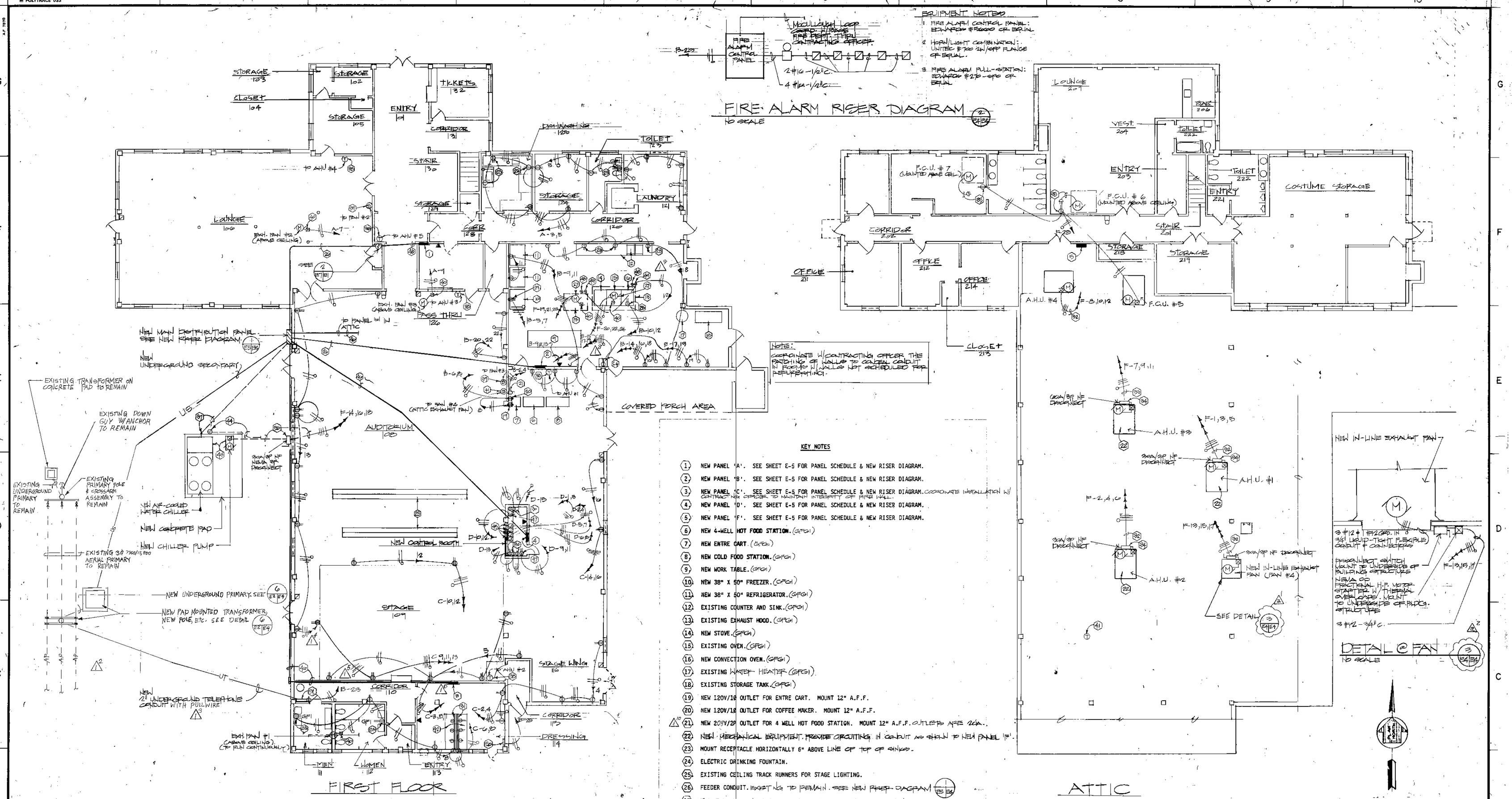
DESIGNED BY: DMD
DRAWN BY: DMD
CHECKED BY: CRD
SUBMITTED BY:
ENGINEER:

FORT SAM HOUSTON
SAN ANTONIO, TEXAS

UPGRADE BUILDING 2652
REPAIR BUILDING 2652
POWER & FIRE ALARM SYSTEM-
RENOVATED FLOOR PLAN

INV. NO. DACA63-81-B-0071 DATED: 2/13/81
CONTR. NO. DACA63-81-C-0080
DRAWING NUMBER
SHEET NO. E4 OF 8
SEQUENCE NO. 27

CONTR. NO. DACA63-81-C-0080



NOTE: COORDINATE W/ CONTRACTING OFFICER THE PATCHING OF WALLS TO CONCEAL CONDUIT IN ROOMS W/ WALLS NOT SCHEDULED FOR REFURBISHING.

EQUIPMENT NOTES

1. FIRE ALARM CONTROL PANEL: EDWARDS #F6000 OR EQUAL
2. HORN/LIGHT COMBINATION: UNITED #730 24V/60PP FLANGE OR EQUAL.
3. FIRE ALARM PULL-STATION: EDWARDS #270-OPS OR EQUAL

KEY NOTES

1. NEW PANEL 'A'. SEE SHEET E-5 FOR PANEL SCHEDULE & NEW RISER DIAGRAM.
2. NEW PANEL 'B'. SEE SHEET E-5 FOR PANEL SCHEDULE & NEW RISER DIAGRAM.
3. NEW PANEL 'C'. SEE SHEET E-5 FOR PANEL SCHEDULE & NEW RISER DIAGRAM. COORDINATE INSTALLATION W/ CONTRACTING OFFICER TO MAINTAIN INTEGRITY OF FIRE WALL.
4. NEW PANEL 'D'. SEE SHEET E-5 FOR PANEL SCHEDULE & NEW RISER DIAGRAM.
5. NEW PANEL 'F'. SEE SHEET E-5 FOR PANEL SCHEDULE & NEW RISER DIAGRAM.
6. NEW 4-WELL HOT FOOD STATION. (GFGI)
7. NEW ENTRE CART. (GFGI)
8. NEW COLD FOOD STATION. (GFGI)
9. NEW WORK TABLE. (GFGI)
10. NEW 38" X 50" FREEZER. (GFGI)
11. NEW 38" X 50" REFRIGERATOR. (GFGI)
12. EXISTING COUNTER AND SINK. (GFGI)
13. EXISTING EXHAUST HOOD. (GFGI)
14. NEW STOVE. (GFGI)
15. EXISTING OVEN. (GFGI)
16. NEW CONVECTION OVEN. (GFGI)
17. EXISTING WATER HEATER (GFGI)
18. EXISTING STORAGE TANK. (GFGI)
19. NEW 120V/1φ OUTLET FOR ENTRE CART. MOUNT 12" A.F.F.
20. NEW 120V/1φ OUTLET FOR COFFEE MAKER. MOUNT 12" A.F.F.
21. NEW 208V/2P OUTLET FOR 4 WELL HOT FOOD STATION. MOUNT 12" A.F.F. OUTLETS ARE 20A.
22. NEW MECHANICAL EQUIPMENT. PROVIDE CIRCUITING IN CONDUIT AS SHOWN TO NEW PANEL 'F'.
23. MOUNT RECEPTACLE HORIZONTALLY 6" ABOVE LINE OF TOP OF SINKS.
24. ELECTRIC DRINKING FOUNTAIN.
25. EXISTING CEILING TRACK RUNNERS FOR STAGE LIGHTING.
26. FEEDER CONDUIT. EXISTING TO REMAIN. SEE NEW RISER DIAGRAM
27. EXISTING EXHAUST HOOD FANS TO BE RECIRCUITED AS SHOWN.
28. NEW 120V/1φ OUTLET FOR CONVECTION OVEN. STUB DOWN FROM CEILING & SUPPORT FROM FLOOR.
29. NEW 208V/2P OUTLET FOR NEW 38"x 50" FREEZER. MOUNT 12" A.F.F. OUTLETS ARE 20A.
30. MOUNT RECEPTACLE 12" ABOVE TOP OF LINE OF COUNTER.
31. NEW WIREMOLD. SEE DRESSING TABLE DETAIL
32. 3 #12 + 1 #12 GROUND IN 3/4" C.
33. 3 #10 & 1 #12 GROUND IN 3/4" C.
34. MOUNT DISCONNECT SWITCH ON SIDE OF A.H.U. DISCONNECT SWITCH SHALL UNDER NO CIRCUMSTANCES BE MOUNTED ON THE A.H.U. + ACCESS PANEL OR IN ANY MANNER THAT HINDERS FRONT ACCESS TO THE UNIT.
35. MOUNT OUTLET ABOVE CEILING ON ACCESSIBLE NUT OR COMPARABLE SUPPORT NO MORE THAN 3'-6" FROM UNIT. PROVIDE CORD + CAP + FINAL CONNECTION.
36. 3 #12 + 1 #12 IN 3/4" C. RUN FROM NEW CHILLER PUMP UNDERGROUND TO NEW DISCONNECT SWITCH MOUNTED ON EXTERIOR WALL. RUN FROM SWITCH UP PIPE OF EXTERIOR WALL UP THRU INTO ATTIC + RUN TO 20A/2P CIRCUIT BREAKER IN NEW PANEL 'F'.
37. UNDERGROUND SERVICE FROM NEW MDP TO NEW CHILLER. STUB UP + UP INTO UNIT. SEE NEW RISER DIAGRAM SHEET E-5 FOR WIRE + CONDUIT SIZES.
38. PROVIDE CONTROL WIRING IN CONDUIT BETWEEN THERMOSTAT + UNIT. COORDINATE W/ MECH. FOR NUMBER OF WIRES + CONNECTIONS TO UNIT.

KEY NOTES (CONTINUED)

39. PROVIDE + INSTALL NEW CONDUIT + WIRING TO RECONNECT EXISTING STAGE LIGHT DIMMER MODULES. SEE NEW RISER DIAGRAM + NEW PANEL SCHEDULE 'D'.
40. NEW PANELBOARD FEEDER CONDUIT + WIRING. SEE NEW RISER DIAGRAM
41. NEW 20A/208V/1φ OUTLET.
42. EXISTING SWITCH TO EXISTING EXHAUST FAN.
43. EXISTING SWITCH TO EXISTING EXHAUST HOOD LIGHT. PROVIDE NEW CIRCUITING AS SHOWN. CLEAN + RELAMP FIXTURE.
50. NEW FIRE ALARM CONTROL PANEL. MOUNT 84" AFF. SINGLE ZONE SINGLE CIRCUIT PANEL. SEE FIRE ALARM RISER DIAGRAM
51. NEW 12"x 24"x 60" TELEPHONE CABINET. SQUARE D #12224LCB BOX OR EQUAL. SQUARE D #12224 TF FRONT OR EQUAL; + SQUARE D #12224 BACKING OR EQUAL.

KEY NOTES (CONTINUED)

53. PROVIDE CONTROL WIRING IN CONDUIT BETWEEN EXHAUST TIMER SWITCH + FAN. COORDINATE W/MECH. FOR NUMBER OF WIRES + CONNECTIONS TO FAN.
54. PROVIDE EXHAUST FAN W/A FRACTIONAL H.P. MANUAL STARTER W/THERMAL OVERLOAD RELAY, RATED 208 VAC 1 H.P.
55. PROVIDE CONTROL WIRING IN CONDUIT BETWEEN WINTER/SUMMER SWITCH, THERMOSTAT IN ATTIC, + ATTIC EXHAUST FAN ON ROOF. COORDINATE W/MECH. FOR NUMBER OF WIRES, CONNECTIONS TO FAN, + EXACT LOCATION OF THERMOSTAT IN ATTIC.
56. EXISTING DRESSING TABLE TO REMAIN. PROVIDE NEW (3)-2000 WIREMOLD (6 GANG) W/3 OUTLETS BEHIND, SPACED AS SHOWN. FOR FURTHER INFORMATION SEE PHOTO AT DRESSING TABLE
59. FOR HEAT REPAIR CIRCUITING SEE
60. CONTROL WIRING IN 3/4" RIGID GALV. STEEL CONDUIT. STUB UP TO WEATHERPROOF J-BOX + RUN IN 3/4" RIGID THWF FLEX. CONDUIT TO UNIT. PAINT RIGID GALV. CONDUIT FROM J-BOX UNDERGROUND UP FACE OF EXT. WALL UP THRU WALL + RUN IN EMT ACROSS ATTIC TO NEW PANEL 'F'.

SYM.	IDO. NO.	ACTION	DATE	DESCRIPTION OF REVISION
			22MAR82	
1	F4935	10th POD001	20 NOV 81	REVISE NOTES AND DRAWINGS
2		AMP 0001	13 MAR. 81	DETAIL @ FAN CHANGED FROM 3/E5 TO 3/E4

KILLEBREW/RUCKER/ASSOC., INC.
600 PETROLEUM BLDG.
WICHITA FALLS, TEXAS

U.S. ARMY ENGINEER DISTRICT, FORT WORTH
CORPS OF ENGINEERS
FORT WORTH, TEXAS

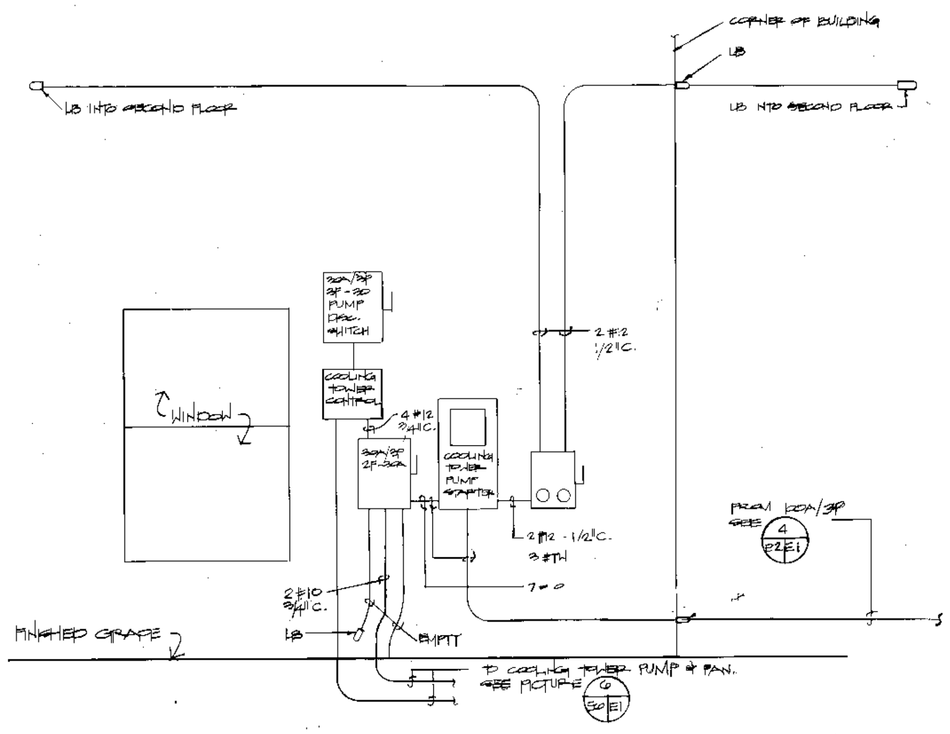
DESIGNED BY: DMD
DRAWN BY: DMD
CHECKED BY: CHP
SUBMITTED BY:
ENGINEER:

FORT SAM HOUSTON
SAN ANTONIO, TEXAS

UPGRADE BUILDING 2652
REPAIR BUILDING 2652
POWER & FIRE ALARM SYSTEM-
RENOVATED FLOOR PLAN

INV. NO. DACA63-81-B-0071 DATED: 2/13/81
CONTR. NO. DACA63-81-C-0080
DRAWING NUMBER
SHEET NO. E4 OF 8
SEQUENCE NO. 27A

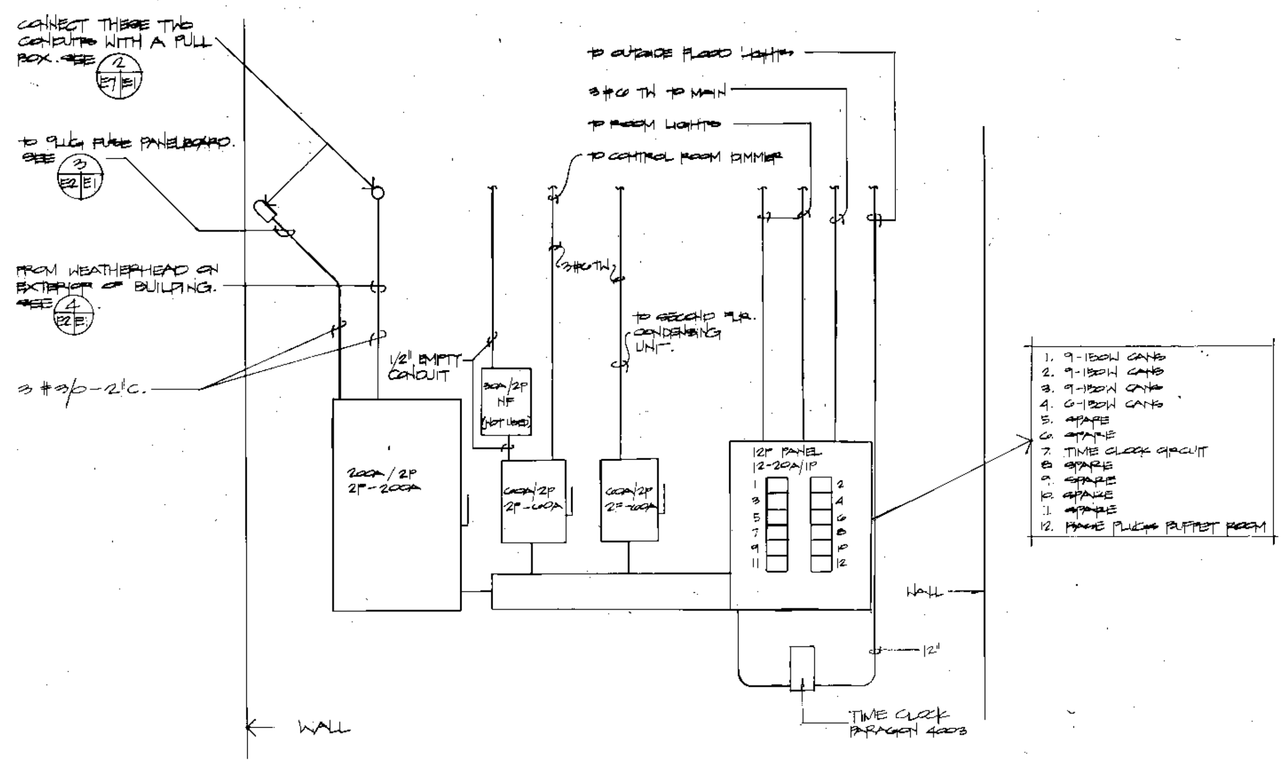
SUPERSEDES SHT. E4 OF 8.



EXISTING WEST WING ELECTRICAL RISER SERVED FROM 100A/3P

NO SCALE (1 E2|E1)

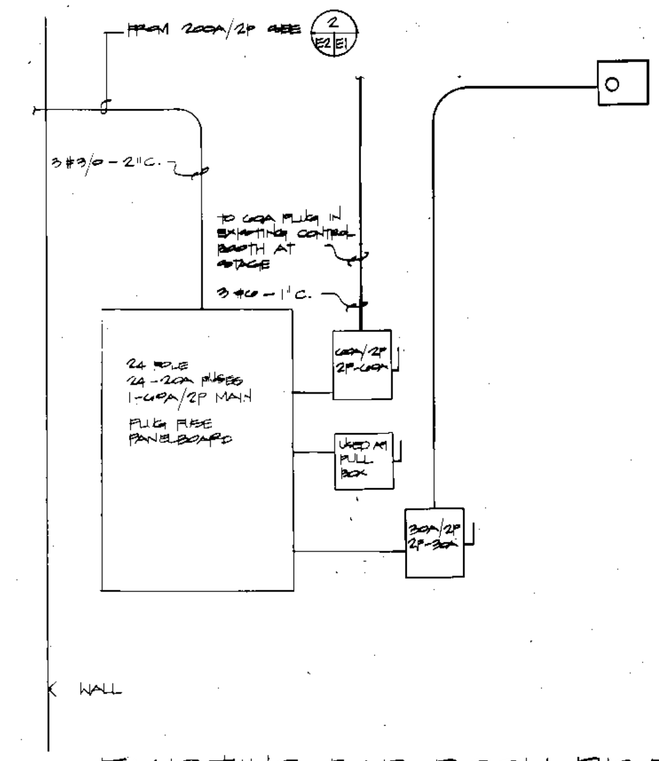
1. ALL ELECTRICAL GEAR, CONDUIT, WIRE, ETC. SHOWN IS TO BE REMOVED.
2. SEE PICTURE 5 E2|E1.



EXISTING SUB-DISTRIBUTION CENTER #1 RISER

NO SCALE (2 E2|E1)

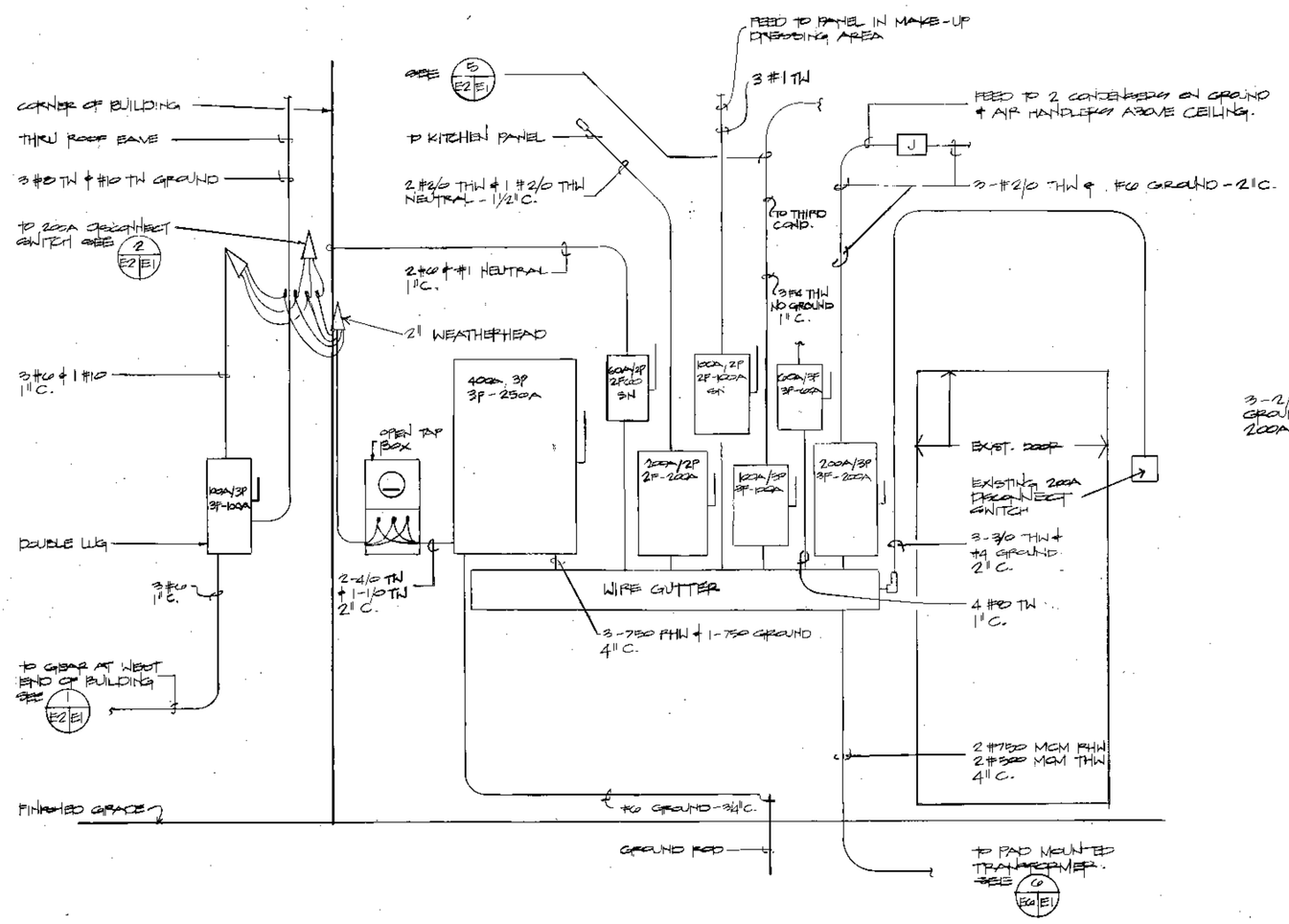
1. ALL ELECTRICAL GEAR, CONDUIT, WIRE, ETC. SHOWN IS TO BE REMOVED UNLESS OTHERWISE NOTED.
2. SEE PICTURE 11 E2|E1.



EXISTING BAR ROOM RISER

NO SCALE (3 E2|E1)

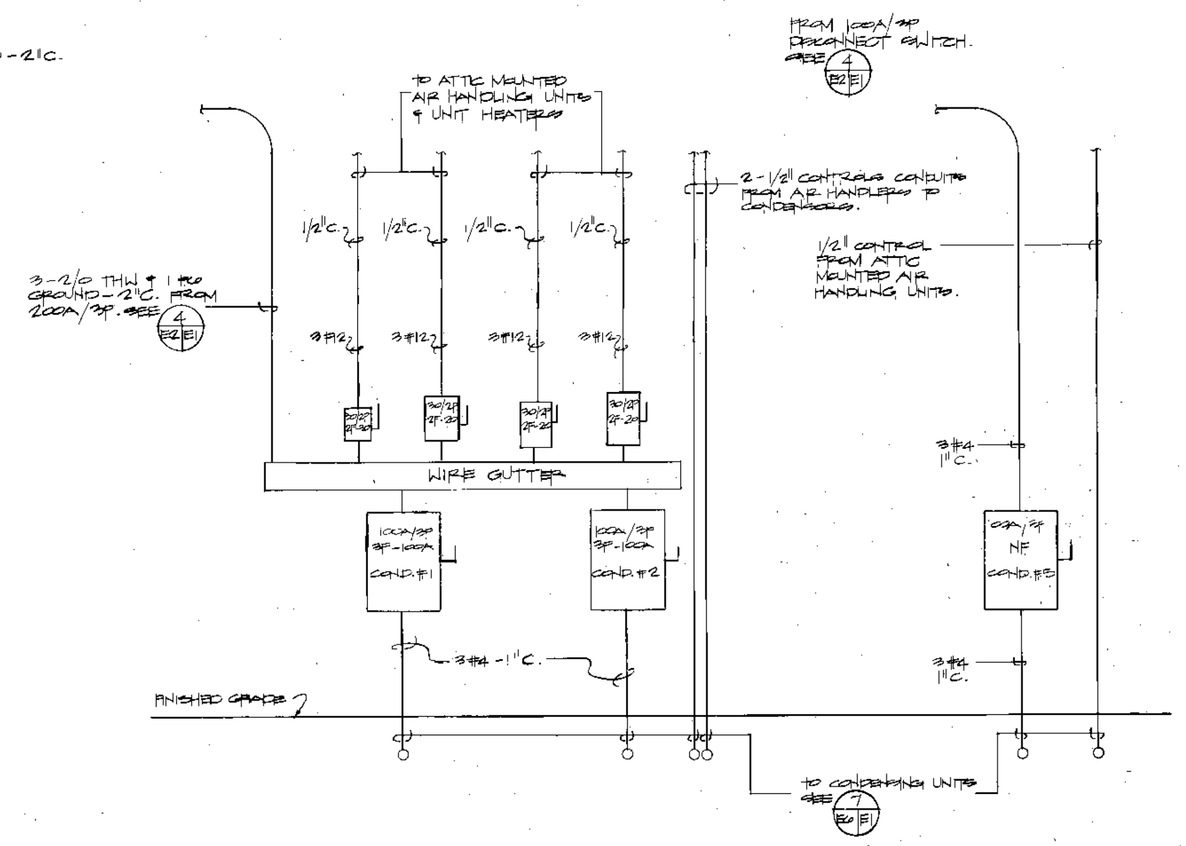
1. THIS PLUG FUSE PANELBOARD IS TO BE REPLACED WITH NEW PANEL IN. DISCONNECT SWITCHES ARE TO BE REMOVED. SEE NEW RISER. 3 E2|E1
2. SEE PICTURE 12 E2|E1



EXISTING MAIN ELECTRICAL RISER

NO SCALE (4 E2|E1)

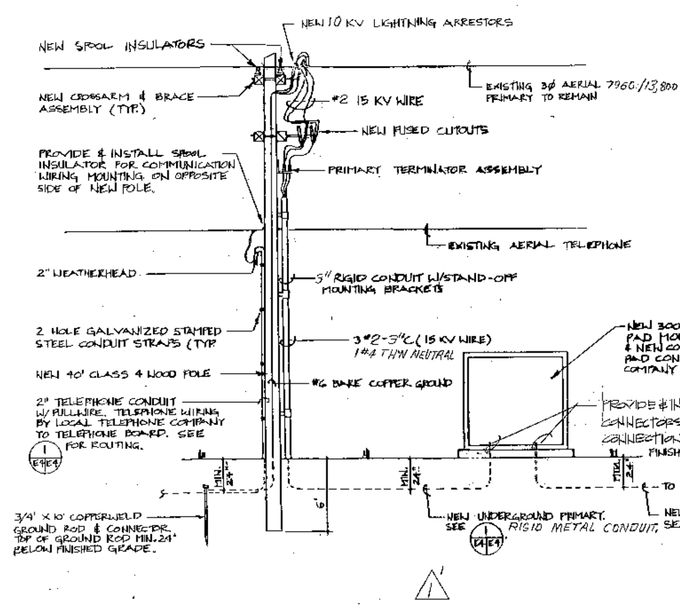
1. ALL ELECTRICAL GEAR, CONDUIT, WIRE, ETC. SHOWN IN THIS DIAGRAM SHALL BE REMOVED UNLESS OTHERWISE NOTED.
2. SEE PICTURE 2 E2|E1.



EXISTING CONDENSING UNITS RISER DIAGRAM

NO SCALE (5 E2|E1)

1. ALL ELECTRICAL GEAR, CONDUIT, WIRE, ETC. SHOWN IS TO BE REMOVED.
2. SEE PICTURES 3 E2|E1 + 4 E2|E1.



NEW PAD MOUNTED TRANSFORMER DETAIL

NO SCALE (6 E2|E1)

NOTES:

1. SEE GENERAL DEMOLITION NOTES SHEET E-1.
2. ALL EXISTING ELECTRICAL ITEMS SHOWN ON THIS SHEET ARE TO BE REMOVED. FOR NEW RISER SEE SHEET E-3.

SYM.	CO. NO.	ACTION	DATE	DESCRIPTION OF REVISION
			22 MAR 82	"UPGRADE OF 2652 AS BUILT"
1	F4935	CO P00001	20 NOV 81	GENERAL REVISION

KILLEBREW/RUCKER & ASSOC., INC.
600 PETROLEUM BLDG.
WICHITA FALLS, TEXAS

U.S. ARMY ENGINEER DISTRICT, FORT WORTH
CORPS OF ENGINEERS
FORT WORTH, TEXAS

DESIGNED BY: GB
DRAWN BY: GB
CHECKED BY: CEE
SUBMITTED BY:
ENGINEER:

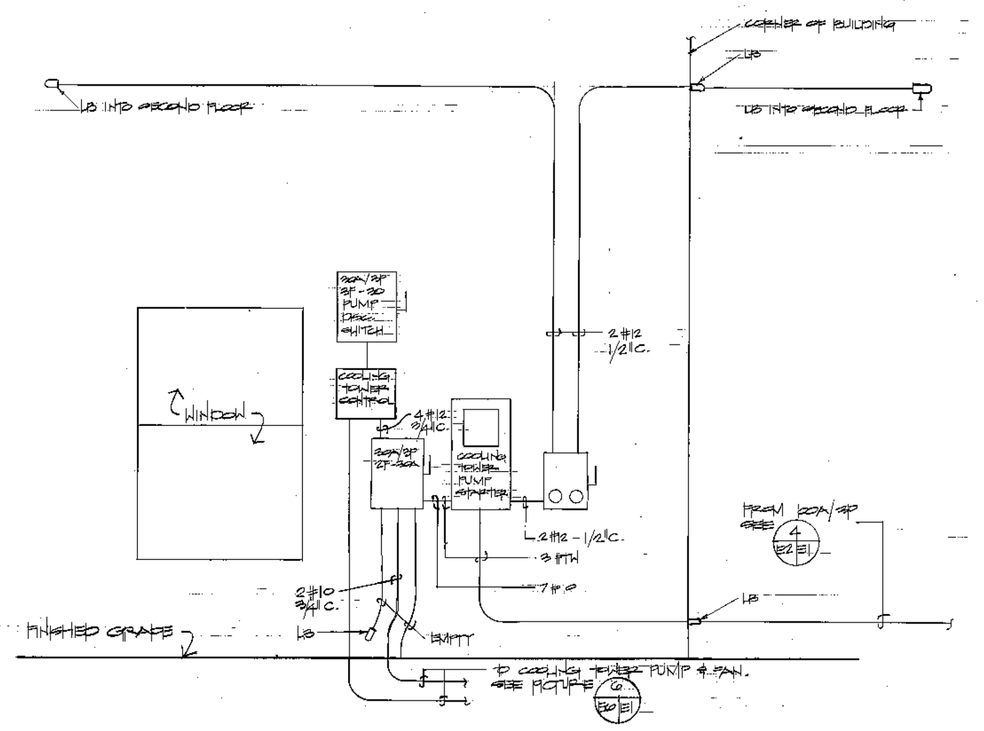
FORT SAM HOUSTON
SAN ANTONIO, TEXAS

UPGRADE BUILDING 2652
REPAIR BUILDING 2652
DEMOLITION - RISER DIAGRAM
& NEW POLE DETAIL

INV. NO. DACA63-81-B-0071 DATED: 2/13/81
CONTR. NO. DACA63 81 C-0080
DRAWING NUMBER
SHEET NO. E2.1 OF 8
SEQUENCE NO. 21

CONTR. NO. DACA63 81-C-0080

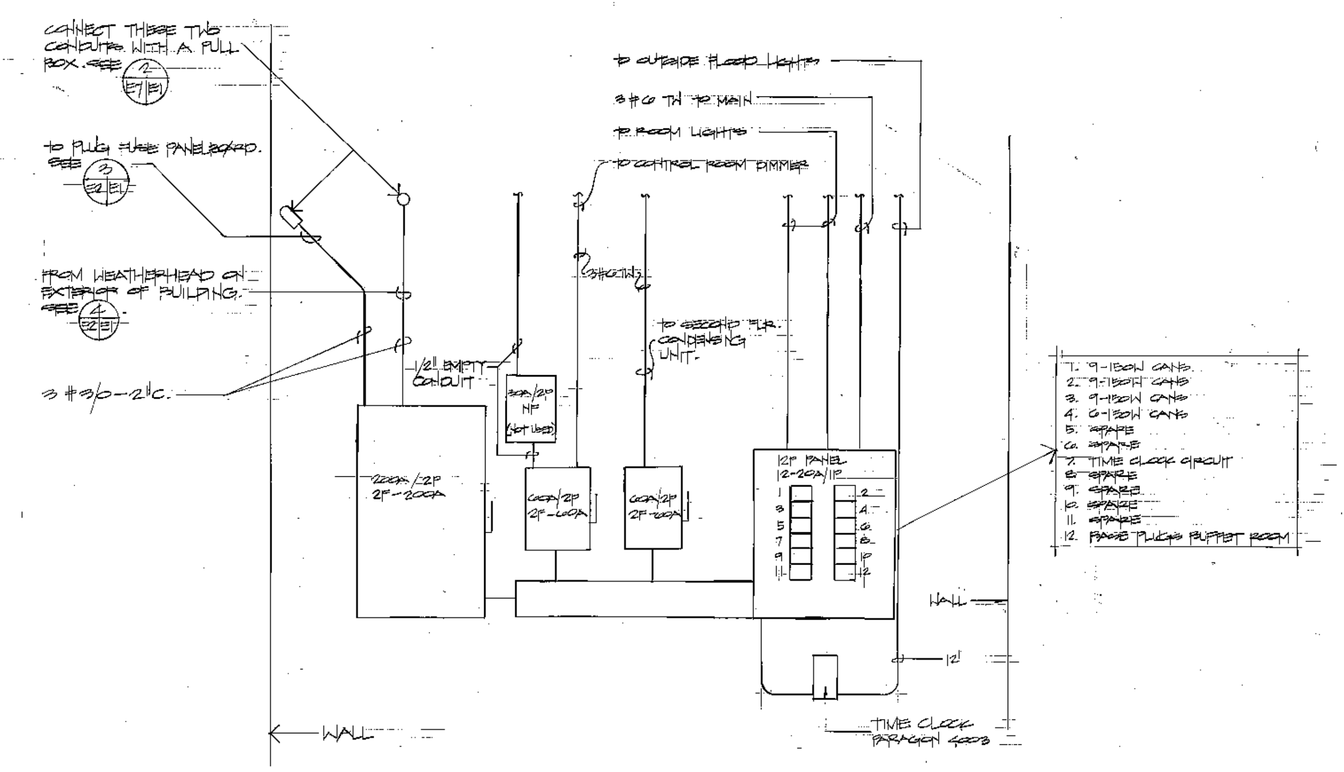
SUPERSEDES SHT 22 OF 8.



EXISTING WEST WING ELECTRICAL RISER SERVED FROM 100A/3P

NO SCALE

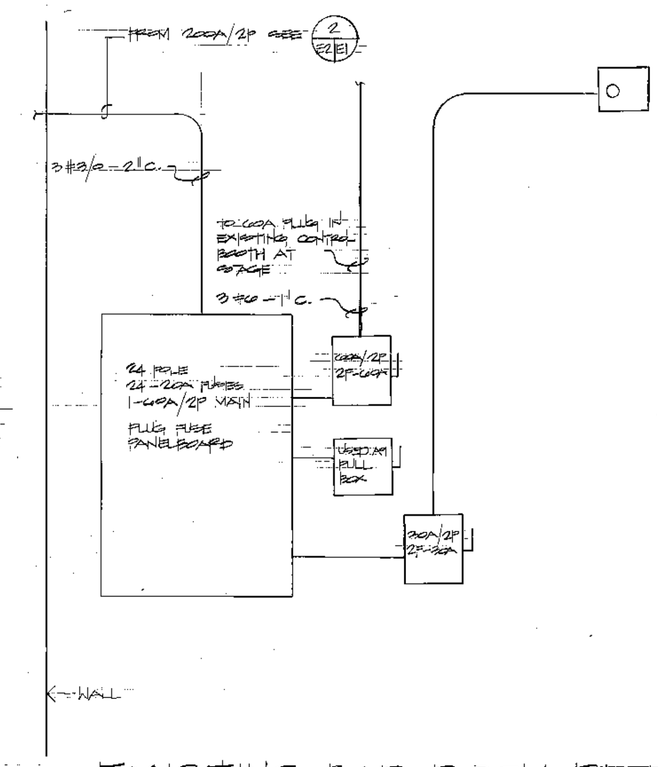
1. ALL ELECTRICAL GEAR, CONDUIT, WIRE, ETC. SHOWN IS TO BE REMOVED.
2. SEE PICTURE



EXISTING SUB-DISTRIBUTION CENTER #1 RISER

NO SCALE

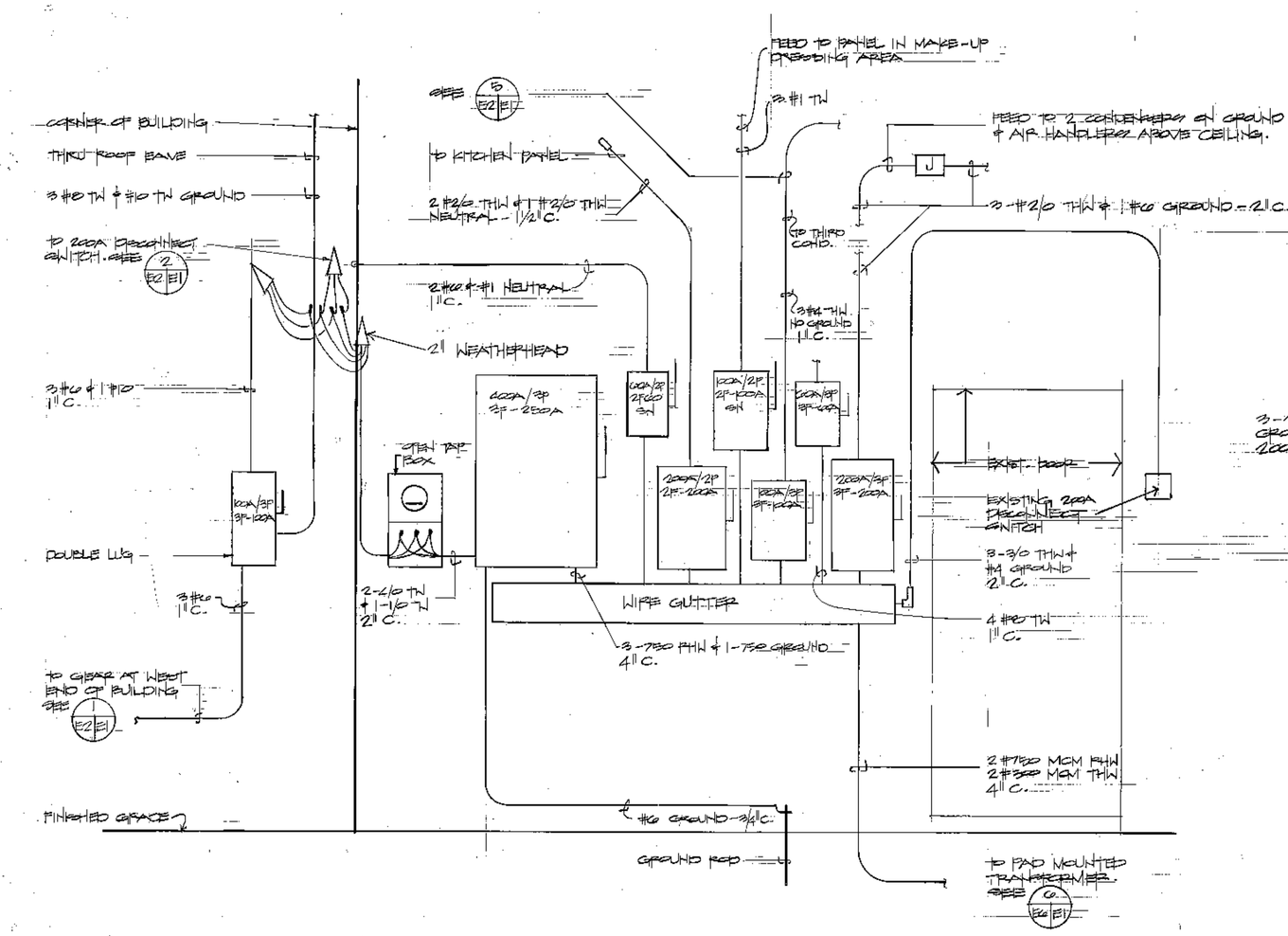
1. ALL ELECTRICAL GEAR, CONDUIT, WIRE, ETC. SHOWN IS TO BE REMOVED UNLESS OTHERWISE NOTED.
2. SEE PICTURE



EXISTING BAR ROOM RISER

NO SCALE

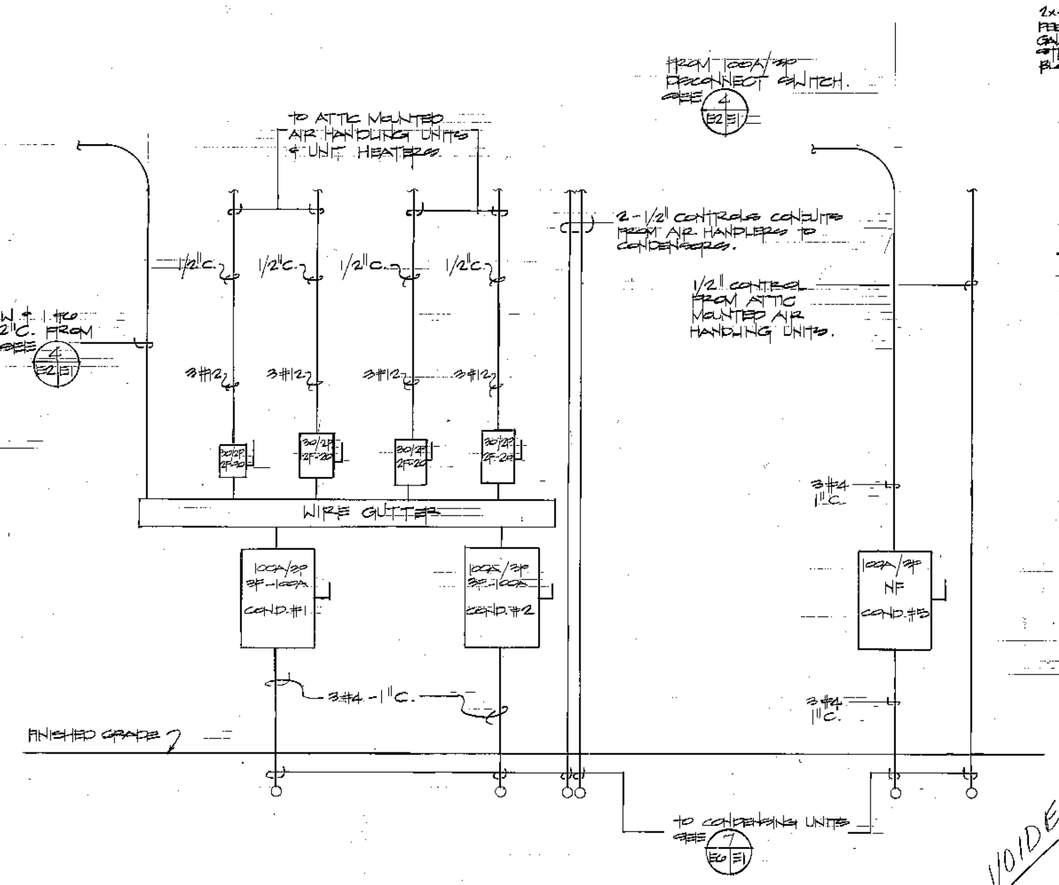
1. THE PLUG FUSE PANELBOARD IS TO BE REPLACED WITH NEW PANEL W/ DISCONNECT SWITCHES ARE TO BE REMOVED. SEE NEW PANEL
2. SEE PICTURE



EXISTING MAIN ELECTRICAL RISER

NO SCALE

1. ALL ELECTRICAL GEAR, CONDUIT, WIRE, ETC. SHOWN IN THE DIAGRAM SHALL BE REMOVED UNLESS OTHERWISE NOTED.
2. SEE PICTURE



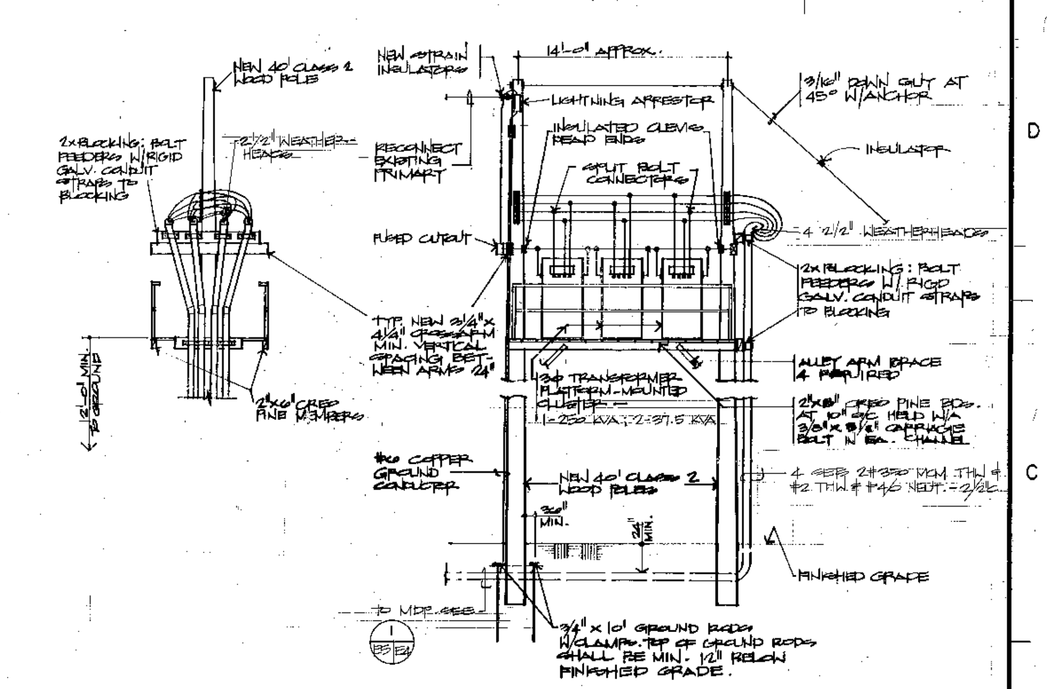
EXISTING CONDENSING UNITS RISER DIAGRAM

NO SCALE

1. ALL ELECTRICAL GEAR, CONDUIT, WIRE, ETC. SHOWN IS TO BE REMOVED.
2. SEE PICTURES

NOTES:

1. SEE GENERAL DEMOLITION NOTES SHEET E-1.
2. ALL EXISTING ELECTRICAL FEEDERS SHOWN ON THIS SHEET ARE TO BE REMOVED. FOR NEW RISER SEE SHEET E-3.



NEW PLATFORM-MOUNTED TRANSFORMER CLUSTER

NO SCALE

SYM.	DG. NO.	ACTION	DATE	DESCRIPTION OF REVISION
			22MAR82	DRAWING UP TO DATE AS BUILT

KILLEBREW/RUCKER/ASSOC., INC.
600 PETROLEUM BLDG.
WICHITA FALLS, TEXAS

U.S. ARMY ENGINEER DISTRICT, FORT WORTH
CORPS OF ENGINEERS
FORT WORTH, TEXAS

DESIGNED BY: CRP
DRAWN BY: LCE DJD
CHECKED BY: RRD
SUBMITTED BY:
ENGINEER:

FORT SAM HOUSTON
SAN ANTONIO, TEXAS

UPGRADE BUILDING 2652
REPAIR BUILDING 2652
DEMOLITION - RISER DIAGRAMS & NEW POLE DETAIL

INV. NO. DACA63 81-B-0071 DATED: 2/13/81
CONTR. NO. DACA63 81-C-0080
DRAWING NUMBER: E2 SHEET NO. 8 OF 8
SEQUENCE NO. 21A

VOIDED

CONTR. NO. DACA63 81-C-0080

225A/MLO; 120/208/3φ/4W FLUSH MOUNTED; GROUND BAR & LUG; 42 POLE

PANEL 'A' VOLTS 120/208 MAIN AMPS 175 PHASE 3 WIRES 4 MCB (MLO) 225 A.

CIRC NO.	FUSE SIZE	SWITCH SIZE	LOAD SERVED	PHASE LOAD V A A	B	C	LOAD SERVED	CIRC NO.	FUSE SIZE	SWITCH SIZE
1	20	1	* LIGHTING	1200 / 1200			EXISTING LOUNGE LTG.	2	20	1
3	20	1	6 OUTLETS		1200 / 1200		EXIST. LOUNGE LIGHTING	4	20	1
5	20	1	7 OUTLETS			1200 / 1200	EXIST. LOUNGE LIGHTING	6	20	1
7	20	1	EXHAUST FAN #2	600				8	20	1
9	20	1	EXHAUST FAN #3		600			10	20	1
11	20	1						12	20	1
13	20	1						14	20	1
15	20	1						16	20	1
17	20	1						18	20	1
19	20	1						20	20	1
21	20	1						22	20	1
23	20	1						24	20	1
25	20	1						26	20	1
27	20	1						28	20	1
29	20	1						30	20	1
31	20	1	EXISTING TIMECLOCK	1200				32	20	1
33	20	1	EXIST. LANTERN FIXTR.		1200			34	20	1
35	20	1	SPARE			1200 / 1200	SPARE	36	20	1
37	20	1	SPARE	1200			SPACE	38	—	—
39	20	1	SPARE		1200		SPACE	40	—	—
41	20	1	SPARE			1200	SPACE	42	—	—
			TOTAL	15/1800	15/1755	16/1935				

TOTAL CONNECTED LOAD 11.791 KVA
ESTIMATED DEMAND LOAD 11.791 KVA
DEMAND LINE AMPS 116.66

* FOR CORRIDOR 120, LAUNDRY 121, TOILET 123, STORAGE 124, DISHWASHING 125, PARS THRU 126, BAR 127, & CORRIDOR 128.

NOTE: THERE ARE 24 EXISTING CIRCUITS THAT ARE TO BE RECONNECTED TO NEW PANEL 'A'. TRACE OUT EACH OF THESE CIRCUITS, RECONNECT TO NEW 20A/1P CIRCUIT BREAKERS & PROPERLY FILL IN PANELBOARD INDEX.

400 MLO; 120/208/3φ/4W FLUSH MOUNTED; GROUND BAR & LUG; 24 POLE.

PANEL 'B' VOLTS 120/208 MAIN AMPS 300 PHASE 3 WIRES 4 MCB (MLO) 400 A.

CIRC NO.	FUSE SIZE	SWITCH SIZE	LOAD SERVED	PHASE LOAD V A A	B	C	LOAD SERVED	CIRC NO.	FUSE SIZE	SWITCH SIZE
1	60	2	EXISTING STAGE DIMMER	5520 / 5520			EXISTING STAGE DIMMER	2	60	2
3					5520 / 5520			4		
5	60	2	EXISTING STAGE DIMMER			5520 / 5520	EXISTING STAGE DIMMER	6	60	2
7				5520 / 5520				8		
9	60	2	220V φ		5520 / 5520		220V φ	10	60	2
11						5520 / 5520		12		
13	20	1	3 φ	300 / 600			3'11' FIXTURES	14	20	1
15	20	1	3 φ		1200 / 600			16	20	1
17	20	1	SPARE			1200 / 1200	SPARE	18	20	1
19	20	1	SPARE	1200			SPACE	20	—	—
21	20	1	SPARE		1200		SPACE	22	—	—
23	20	1	SPARE			1200	SPACE	24	—	—
25								26		
27								28		
29								30		
31								32		
33								34		
35								36		
37								38		
39								40		
41								42		
			TOTAL	24/1800	25/2000	25/2000				

TOTAL CONNECTED LOAD 14.94 KVA
ESTIMATED DEMAND LOAD 6.14 KVA
DEMAND LINE AMPS 188.17

PANEL 'MDP' VOLTS 120/208 MAIN AMPS 1200 PHASE 3 WIRES 4 MCB (MLO) 1200 A.

CIRC NO.	FUSE SIZE	SWITCH SIZE	LOAD SERVED	PHASE LOAD V A A	B	C	LOAD SERVED	CIRC NO.	FUSE SIZE	SWITCH SIZE
1				12/1223.3 / 21140				2		
3	225/3P	400A/3P	AIR COOLED WATER CHILLER		12/1223.3 / 21140			4	225/3P	400A/3P
5						12/1223.3 / 21140	PANEL 'E'	6		
7				13/2250 / 12/2214				8		
9	225/3P	400A/3P	PANEL 'A'		15/1800 / 22/2500		PANEL 'C'	10	225/3P	400A/3P
11						16/2500 / 27/2500		12		
13				11/1800 / 22/1800				14		
15	225/3P	400A/3P	PANEL 'B'		11/1800 / 22/1800		PANEL 'F'	16	225/3P	400A/3P
17						11/1800 / 22/1800		18		
19								20		
21								22		
23								24		
25								26		
27								28		
29								30		
31								32		
33								34		
35								36		
37								38		
39								40		
41								42		
			TOTAL	102/2553	101/1728	99/1614.3				

TOTAL CONNECTED LOAD 301.013 KVA
ESTIMATED DEMAND LOAD 278.202 KVA
DEMAND LINE AMPS 772.81

NOTE: METAL CONDUIT MAY SERVE AS EQUIP. GROUNDING CONDUCTOR.

LEGEND

- FLUORESCENT LIGHT FIXTURE. CAPITAL LETTER DENOTES TYPE IN FIXTURE SCHEDULE. SMALL LETTERS DENOTES SWITCH WITH SAME LETTER OPERATES THIS FIXTURE. CIRCLE IS OUTLET BOX.
- INCANDESCENT OR MERCURY VAPOR LIGHT FIXTURE. CAPITAL LETTER DENOTES TYPE IN FIXTURE SCHEDULE. SMALL LETTER DENOTES SWITCH WITH SAME LETTER OPERATES THIS FIXTURE.
- WP — DUPLEX RECEPTACLE. 15A., 125V., GROUNDED TYPE MOUNTED 12 INCHES ABOVE FLOOR UNLESS OTHERWISE NOTED. WP INDICATES WEATHERPROOF.
- —UT— UNDERGROUND TELEPHONE CONDUIT
- $a — SINGLE POLE WALL SWITCH MOUNT 4'-0" ABOVE FLOOR. SMALL LETTER DENOTES SWITCH OPERATES FIXTURE HAVING SAME LETTER.
- $a $3 $4 $a $3 — SAME AS SWITCH ABOVE WITH THE FOLLOWING EXCEPTIONS: 2 DENOTES 2 POLE, 3 DENOTES 3 WAY, 4 DENOTES 4 WAY, K DENOTES KEY OPERATED, 6 DENOTES OUTLIT LIGHT
- M 5 — MOTOR. NUMERAL INDICATES HORSEPOWER.
- MOTOR CONTROLLER.
- R — SAFETY SWITCH. UNFUSED UNLESS OTHERWISE NOTED. R INDICATES RAINTIGHT.
- PANELBOARD, LIGHTING
- CO — CONDUIT & CONDUCTORS CONCEALED IN WALL, CEILING OR FLOOR WITH NEUTRAL, HOT, SWITCH AND GROUND CONDUCTORS INDICATED RESPECTIVELY. NO CROSS LINES INDICATES 2 CONDUCTORS.
- CO — CONDUIT & CONDUCTORS RUN EXPOSED, WITH NEUTRAL, HOT, SWITCH AND GROUND CONDUCTORS INDICATED RESPECTIVELY. NO CROSS LINES INDICATES 2 CONDUCTORS.
- BRANCH CIRCUIT HOME RUN TO PANELBOARD WITH CIRCUIT NUMBERS SHOWN. NUMBERS OF ARROWS INDICATES NUMBER OF CIRCUITS.
- J — JUNCTION BOX.
- TELEPHONE JACK
- HORN/LIGHT COMBINATION (MOUNT MAX. 12" BELOW CEILING.
- —AP— EXISTING AERIAL PRIMARY
- FIRE ALARM PULL STATION (MOUNT 54" A.F.F.)
- EXISTING POLE TO REMAIN
- NEW POLE
- GFI — GROUND FAULT INTERRUPTER TYPE OUTLET
- MDP — MAIN DISTRIBUTION PANELBOARD
- GFCI — GOVERNMENT FURNISHED; GOVERNMENT INSTALLED
- —US— EXISTING UNDERGROUND SECONDARY
- PICTURE DESIGNATION - ARROW DESIGNATES DIRECTION PICTURE WAS TAKEN
- KEY NOTE DESIGNATION
- —US— NEW UNDERGROUND SECONDARY

100A MLO; 120/208/3φ/4W FLUSH MOUNTED; GROUND BAR & LUG; 30 POLE

PANEL 'B' VOLTS 120/208 MAIN AMPS 100 PHASE 3 WIRES 4 MCB (MLO) 100 A.

CIRC NO.	FUSE SIZE	SWITCH SIZE	LOAD SERVED	PHASE LOAD V A A	B	C	LOAD SERVED	CIRC NO.	FUSE SIZE	SWITCH SIZE
1	20	1	KITCHEN LIGHTING	1200 / 1200			4 WELL HOT FOOD CONTROL	2	20	2
3	20	1	BUFFET LIGHTING		1200 / 1200			4		
5	20	2	FREEZER			720 / 714	COFFEE MAKER	6	20	1
7				1440 / 714			ENTREE CART	8	20	1
9	20	1	REFRIGERATOR		1440 / 2520		CONVECTION OVEN	10	20	1
11	20	1	REFRIGERATOR			1440 / 1440	EXHAUST HOOD LIGHTING	12	20	1
13	20	1	BUFFET-5 OUTLETS	1200 / 1000			FUTURE EQUIP. OUTLET	14	20	1
15	20	1	ELEC. DRINKING FOUNTAIN		1200 / 1200		FUTURE EQUIP. OUTLET	16	20	1
17	20	1	FUTURE EQUIP. OUTLET			1200 / 1200	KITCHEN - 4 OUTLETS	18	20	1
19	20	1	FUTURE EQUIP. OUTLET	200 / 200			1 AUDITORIUM φ	20	20	1
21	20	1	KITCHEN PORCH OUTLETS		200 / 400		1 AUDITORIUM φ	22	20	1
23	20	1	TELEPHONE CABINET φ			1200 / 1200	SPARE	24	20	1
25	20	1	FIRE ALARM CONTROL PANEL	1200			SPACE	26	—	—
27	20	1	SPARE		1200		SPACE	28	—	—
29	—	—	SPACE				SPACE	30	—	—
31								32		
33								34		
35								36		
37								38		
39								40		
41								42		
			TOTAL	10/1440	8/1080	7/1087				

TOTAL CONNECTED LOAD 26.496 KVA
ESTIMATED DEMAND LOAD 24.096 KVA
DEMAND LINE AMPS 66.88

225A MLO; 120/208/3φ/4W FLUSH MOUNTED; GROUND BAR & LUG; 36 POLE

PANEL 'E' VOLTS 120/208 MAIN AMPS 150 PHASE 3 WIRES 4 MCB (MLO) 225 A.

CIRC NO.	FUSE SIZE	SWITCH SIZE	LOAD SERVED	PHASE LOAD V A A	B	C	LOAD SERVED	CIRC NO.	FUSE SIZE	SWITCH SIZE
1				719 / 1320				2		
3	20	3	AIR HANDLING UNIT #1 *		719 / 1320		AIR HANDLING UNIT #2 *	4	150	3
5						719 / 1320		6		
7				719 / 2123				8		
9	40	3	AIR HANDLING UNIT #3 *		719 / 2123		AIR HANDLING UNIT #4 *	10	150	3
11						719 / 2123		12		
13				1320 / 719				14		
15	150	3	1½ HP EXHAUST FAN		1320 / 719		3 HP CHILLED WATER PUMP	16	20	3
17						1320 / 719		18		
19				719 / 719				20		
21	20	3	EXIST. KITCHEN EXHAUST FAN		719 / 719		EXIST. KITCHEN EXHAUST FAN	22	20	3
23						719 / 719		24		
25	20	1	AHU #5, #6, & #7	1200 / 1200			SPARE	26	20	1
27	20	1	CHILLER CONTROLS		1200 / 1200		SPARE	28	20	1
29	20	1	HEAT TAPE			1200 / 1200	SPARE	30	20	1
31	—	—	SPACE	1200			SPARE	32	20	1
33	—	—	SPACE		1200		SPARE	34	20	1
35	—	—	SPACE			1200	SPARE	36	20	1
37								38		
39								40		
41								42		
			TOTAL	11/1749	11/1754	11/1754				

TOTAL CONNECTED LOAD 35.657 KVA
ESTIMATED DEMAND LOAD 28.457 KVA
DEMAND LINE AMPS 79.06

* LOW VOLTAGE TRANSFORMERS INCLUDED W/UNIT.

LIGHTING FIXTURE SCHEDULE

FIXT. SYMB. DESIG	TYPE	FINISH	LENS	LAMPS	FIXTURE VOLTAGE	MANUFACTURER & CATALOG NO.	REMARKS
A	FLUORESCENT TROFFER	WHITE	ACRYLIC PRISMATIC	2-F40 RS	120	LITHONIA 2GL 240 A12 120 (OR EQUAL)	
B	RECESSED INCANDESCENT	BLACK BAFFLE	—	1-175 W	120	HALO M1182 (OR EQUAL)	
C	RECESSED INCANDESCENT	BLACK BAFFLE	—	1-300 W M	120	LIGHTOLIER 7225-AD (OR EQUAL)	
D	RECESSED ADJUSTABLE INCAN.	BLACK BAFFLE		1-150 PAR	120	LIGHTOLIER 7249 (OR EQUAL)	W/STEP BAFFLE #5249-31
E	SURF.-MOUNTED FLUORESCENT	WHITE	ACRYLIC PRISMATIC	4-F40 RS	120	LITHONIA 2M 440 M A12 120	
F	SURF.-MOUNTED FLUORESCENT	WHITE	ACRYLIC PRISMATIC	2-F40 RS	120	LITHONIA 2GM 2 120 (OR EQUAL)	
G	SURF.-MOUNTED INCANDESCENT	ALUM	OPAL GLASS	2-100 W	120	HALO 2710 (OR EQUAL)	
H	RECESSED PROJECTOR LIGHT	WHITE	—	1-100 W T-4	120	LIGHTOLIER 7052 (OR EQUAL)	NOW T-4 CLEAR SYLVANIA TUNGSTEN-HALOGEN BULB
K	SURF. MOUNTED FLUORESCENT	WHITE	ACRYLIC PRISMATIC	2-40 W/T12 RS	120	DAYNE B244 (OR EQUAL)	
M	CEILING MTD. INCAND.	ALUM.	OPAL GLASS	1-100 W	120	HALO H2411 (OR EQUAL)	
N	WALL MTD. INCAND.	ALUM.	OPAL GLASS	1-100 W	120	HALO H2411 W/GI (OR EQUAL)	W/WIREGUARD
R	MAKE-UP LIGHT	WHITE	—	1-60 W G40	120	WIRELAND R240 (OR EQUAL)	PROVIDE ADAPTERS TO FIT G40 LAMP IF NECESSARY
X	EXIT	ALUM	PLASTIC	2-T20	120	LITHONIA EDGELITE EL (OR EQUAL)	W/BATTERY PACK
W	EMER. EQUIPMENT	WHITE	SHATTERPROOF ACRYLIC-PRISM.	1-25W QUARTZ	120	CHLORIDE 6PU-10 (OR EQUAL)	
Y	SURF.-MOUNTED FLUORESCENT	WHITE	ACRYLIC PRISMATIC	2-F40 RS	120	LITHONIA LB 240 A/120 (OR EQUAL)	SUITABLE FOR INSTALLATION ON COMBUSTIBLES CEILINGS
Z	INCANDESCENT STEPLIGHT	ALUM	TEMP. GLASS	1-25 W	120	SCHWARTZ 1-GRGW-1 (OR EQUAL)	MOUNT 24" AFF *

* COORDINATE INSTALLATION W/ CONTRACTING OFFICER SO THAT INTEGRITY OF FIRE WALL WILL BE MAINTAINED.

NOTE: IN Rooms CONTROLLED BY NIGHT SETBACK OF AHUs, FLUORESCENT FIXTURES SCHEDULED TO HAVE 60W LAMPS SHALL HAVE 35W LAMPS. COORDINATE W/MECH. CONTRACTOR. SEE SHEET E-3.

225A MLO; 120/208/3φ/4W FLUSH MOUNTED; GROUND BAR & LUG; 36 POLE

PANEL 'C' VOLTS 120/208 MAIN AMPS 150 PHASE 3 WIRES 4 MCB (MLO) 225 A.

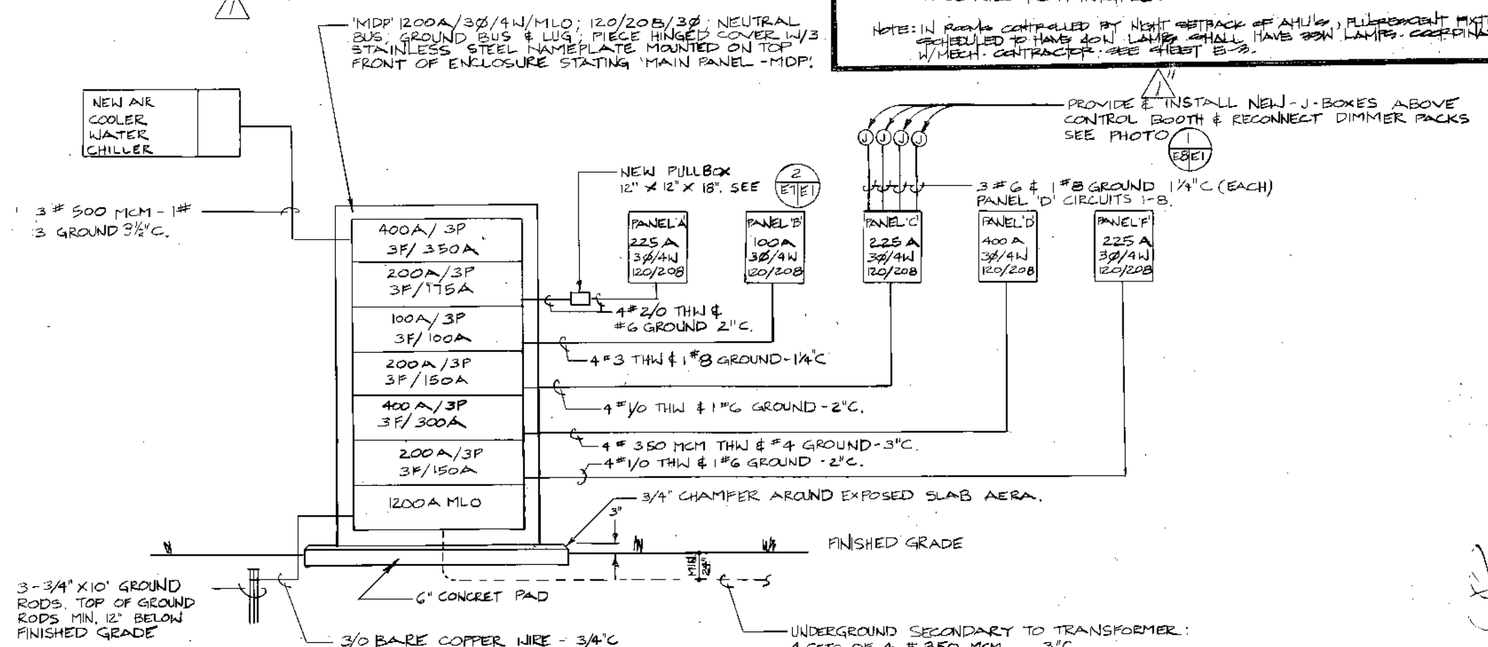
CIRC NO.	FUSE SIZE	SWITCH SIZE	LOAD SERVED	PHASE LOAD V A A	B	C	LOAD SERVED	CIRC NO.	FUSE SIZE	SWITCH SIZE
1	20	1	* LIGHTING: G.P., 4'S, 2'S, 2'S FIXTURES	1200 / 800			4 φ	2	20	1
3	20	1	2 STAGE φ		1200 / 600		SOUND MODULE	4	20	1
5	20	1	1 STAGE φ			1200 / 1200	DRESSING TABLE OUTLETS	6	20	1
7	20	1	2 STAGE φ	300 / 600			4 φ EXHAUST FAN φ **	8	20	1
9	20	1	2 STAGE φ		200 / 200		1 STAGE φ **	10	20	1
11	20	1	3 AUDITORIUM φ			200 / 200	1 STAGE φ	12	20	1
13	20	1	4 AUDITORIUM φ	1200 / 800			SOUND MODULE	14	20	1
15	40	1	DIMMER MODULE 3A		1200 / 2400		SOUND MODULE	16	20	1
17	40	1	DIMMER MODULE B			1200 / 2400	SEE NOTE BELOW	18	20	1
19	40	1	DIMMER MODULE 1A	1200 / 2400				20	20	1
21	40	1	DIMMER MODULE 2A		960 / 2600		MAKE-UP LIGHTS	22	20	1
23	20	1	GALAXY BAR LIGHTS			1200 / 1200	MAKE-UP LIGHTS	24	20	1
25	20	1	NEW EXTERIOR LIGHTS	1200 / 400			SPARE	26	20	1
27	15	1	NEW TIMECLOCK		1200 / 120		SPARE	28	20	1
29	20	1	SPARE			1200 / 1200	SPARE	30	20	1
31	—	—	SPACE				SPACE	32	—	—
33	—	—	SPACE				SPACE	34	—	—
35	—	—	SPACE				SPACE	36	—	—
37								38		
39								40		
41								42		
			TOTAL	13/1000	12/960	10/860				

TOTAL CONNECTED LOAD 21.920 KVA
ESTIMATED DEMAND LOAD 21.920 KVA
DEMAND LINE AMPS 90.89

* FOR CORRIDOR 110, MEN 111, WOMEN 112, ENTRY 113, DRESSING 114, CORRIDOR 115, & STAGE WING 116.

** FOR CORRIDOR 110, MEN 111, WOMEN 112, & DRESSING 114.

NOTE: THERE ARE 4 EXISTING REMOTE CIRCUITS THAT ARE TO BE TRACED OUT & REPLACE W/WIRE & SEND IT & CONNECTED TO NEW 20A/1P BREAKERS IN NEW PANEL 'C'.



NEW ELECTRICAL RISER DIAGRAM

NO SCALE

NOTES
1. SEE RENOVATED FLOOR PLAN & SWITCHBOARD LOCATIONS. 1/E2E4 & 2/E2E4 FOR NEW PANELBOARD E5/E4
2. FOR CONDUIT & WIRE SIZES OF NEW UNDERGROUND SECONDARY TO NEW MDP, SEE G/E2E4
3. FOR DETAIL OF TRANSFORMER SEE DETAIL G/E2E4

SYM	DO. NO.	ACTION	DATE	DESCRIPTION OF REVISION
			22MAR82	"DRAWING OF WORK AS BUILT."
	F4935	CO#00001	20NOV81	GENERAL REVISIONS
		AM#0001	13MAR81	ADDED "OR EQUAL" TO MANUF. FOR '2' TYPE FIXTURE
				REMOVED "THW" FROM FEEDER SIZE.

KILLEBREW/RUCKER/ASSOC., INC.
600 PETROLEUM BLDG.
WICHITA FALLS, TEXAS

U.S. ARMY ENGINEER DISTRICT, FORT WORTH
CORPS OF ENGINEERS
FORT WORTH, TEXAS

DESIGNED BY: GB
DRAWN BY: GB
CHECKED BY: GRP

FORT SAM HOUSTON
SAN ANTONIO, TEXAS

UPGRADE BUILDING 2652
REPAIR BUILDING 2652

ELECTRICAL PANEL SCHEDULE

SUBMITTED BY:
ENGINEER:

INV. NO. DACA63-81-B-0071 DATED: 2/13/81
CONTR. NO. DACA63-81-C-0080
DRAWING NUMBER
SHEET NO. ES.1 OF 8
SEQUENCE NO. 24

SUPERSEDES SHT ES OF 8

CONTR. NO. DACA63-81-C-0080

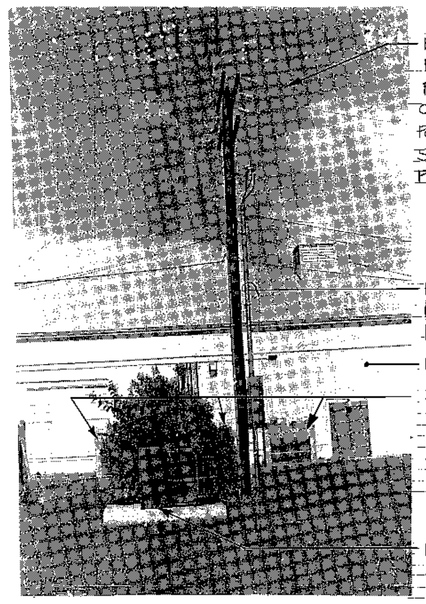


PHOTO 1 EG|E1
PAD MOUNTED TRANSFORMER

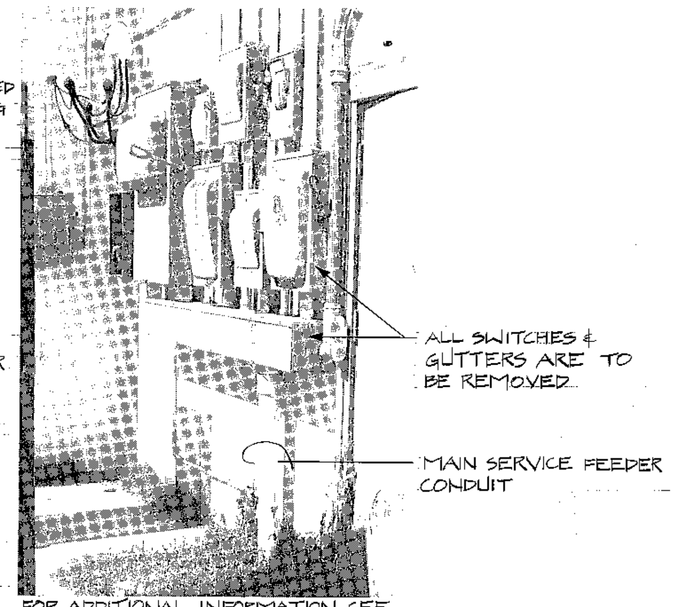


PHOTO 2 EG|E1
MAIN ELECTRICAL SERVICE ENTRANCE

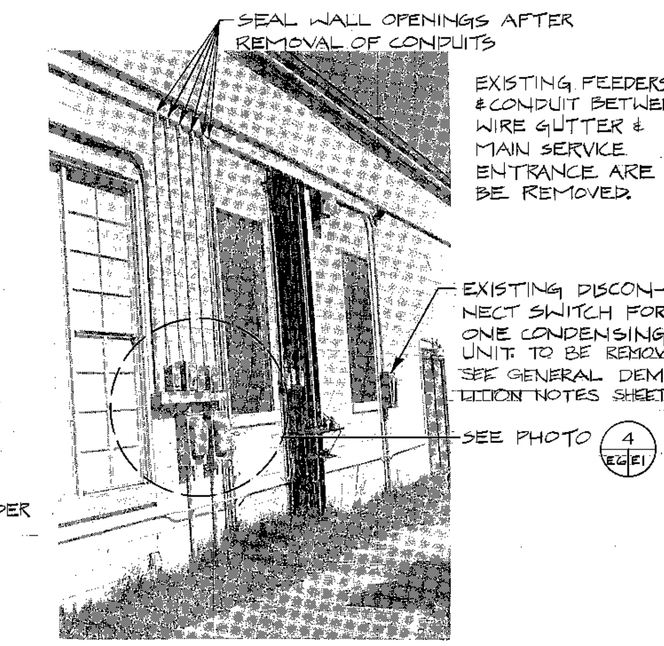


PHOTO 3 EG|E1
ELECTRICAL RISER FOR CONDENSING UNITS & ATTIC MOUNTED AIR HANDLING UNITS ETC.

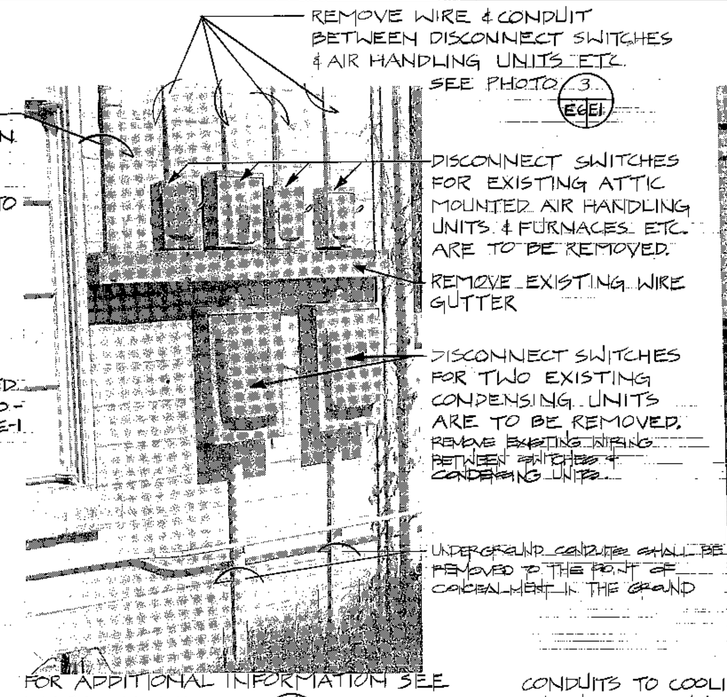


PHOTO 4 EG|E1
ELECTRICAL RISER FOR TWO EXISTING CONDENSING UNITS AND ATTIC MOUNTED AIR HANDLING UNITS ETC.

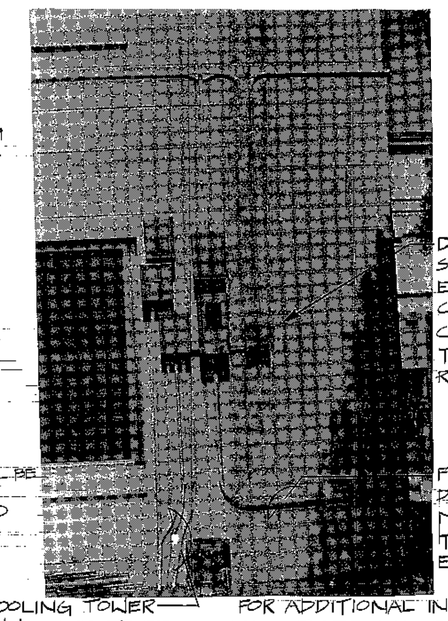


PHOTO 5 EG|E1
ELECTRICAL RISER FOR EXISTING COOLING TOWER & WINDOW AIR CONDITIONING UNITS.

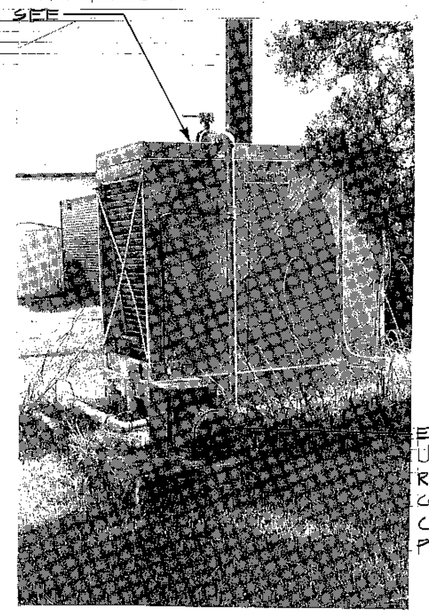


PHOTO 6 EG|E1
EXISTING WATER TOWER

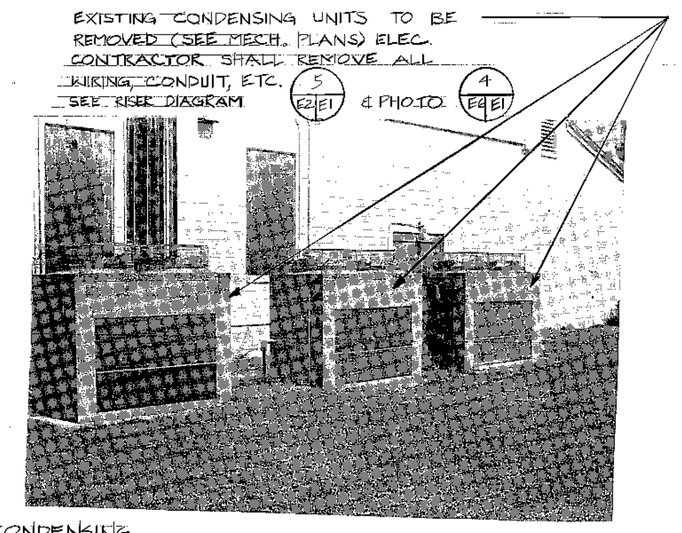


PHOTO 7 EG|E1
EXISTING AIR COOLED CONDENSING UNITS.

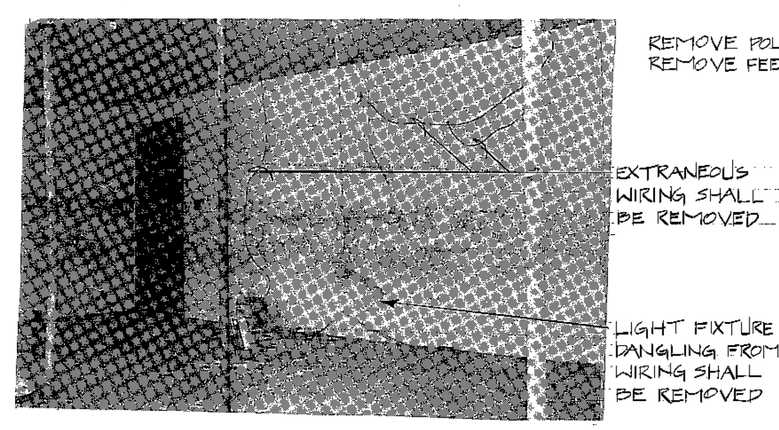


PHOTO 8 EG|E1
COVERED AREA AT KITCHEN ENTRANCE.

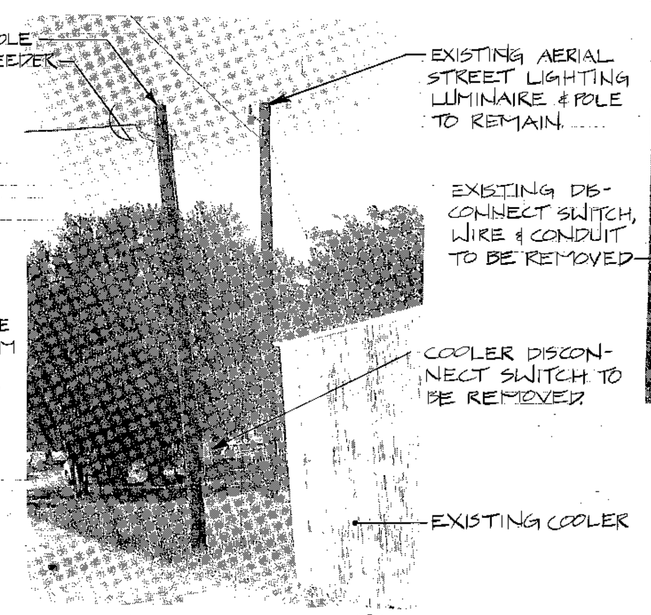


PHOTO 9 EG|E1
ELECTRIC SERVICE TO OUTDOOR COOLER
SEE PHOTO 10 EG|E1

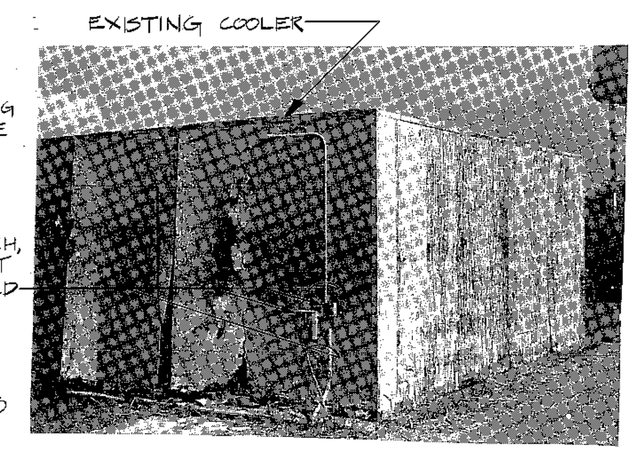


PHOTO 10 EG|E1
OUTDOOR COOLER
SEE PHOTO 9 EG|E1

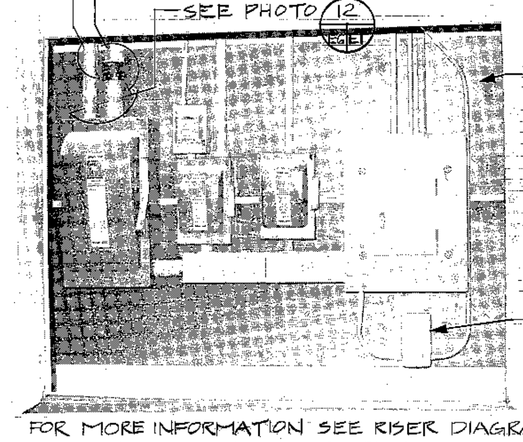


PHOTO 11 EG|E1
ELECTRICAL RISER IN ROOM 106

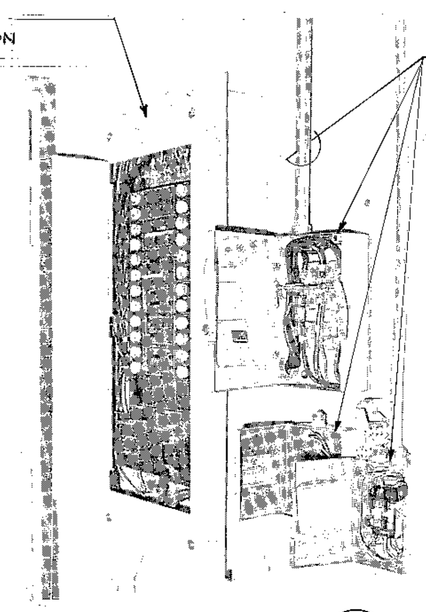


PHOTO 12 EG|E1

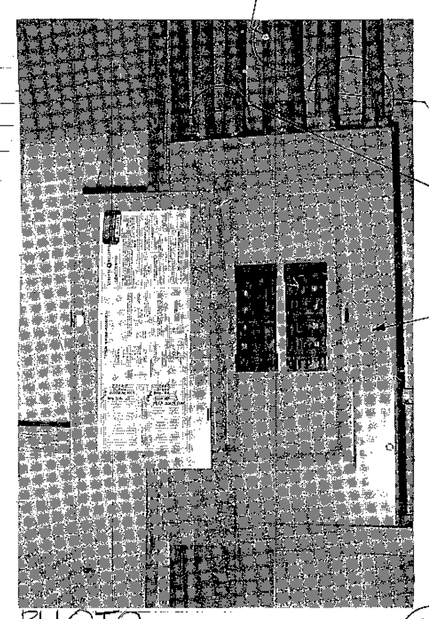


PHOTO 13 EG|E1
CONTROL BOOTH POWER PANEL

VOIDED

SYM.	DO. NO.	ACTION	DATE	DESCRIPTION OF REVISION
			22 MAR 82	DRAWING OF WORK

KILLEBREW/RUCKER/ASSOC., INC.
600 PETROLEUM BLDG.
WICHITA FALLS, TEXAS

U.S. ARMY ENGINEER DISTRICT, FORT WORTH
CORPS OF ENGINEERS
FORT WORTH, TEXAS

DESIGNED BY: CRP
DRAWN BY: DMD
CHECKED BY: RRD

FORT SAM HOUSTON
SAN ANTONIO, TEXAS

UPGRADE BUILDING 2652
REPAIR BUILDING 2652

ELECTRICAL ~ PHOTOS

SUBMITTED BY:
ENGINEER:

INV. NO. DACA63-81-B-0071 DATED: 2/13/82
CONTR. NO. DACA63-81-C-0080
DRAWING NUMBER
SHEET NO. EG OF 8
SEQUENCE NO. 251
CONTR. NO. DACA63-81-C-0080

PANEL 'A' VOLTS 120/240 MAIN AMPS — PHASE 1 WIRES 3 MLO 225 A.

CIRC NO.	TRIP AMPS	NO. POLES	LOAD SERVED	PHASE LOAD V.A A	PHASE LOAD V.A B	LOAD SERVED	CIRC NO.	TRIP AMPS	NO. POLES
1	20	1	* LIGHTING	1200 / 1200		EXISTING LOUNGE LIGHTING	2	20	1
3	20	1	6 OUTLETS		1200 / 1200	EXISTING LOUNGE LIGHTING	4	20	1
5	20	1	7 OUTLETS	1200 / 1200		EXISTING LOUNGE LIGHTING	6	20	1
7	20	1	EXHAUST FAN #2		660		8	20	1
9	20	1	EXHAUST FAN #3	660			10	20	1
11	20	1					12	20	1
13	20	1					14	20	1
15	20	1					16	20	1
17	20	1					18	20	1
19	20	1					20	20	1
21	20	1					22	20	1
23	20	1					24	20	1
25	20	1					26	20	1
27	20	1					28	20	1
29	20	1					30	20	1
31	20	1	EXISTING TIMECLOCK		1200		32	20	1
33	20	1	EXT. LIGHTING FIXTURES	1200			34	20	1
35	20	1	SPARE		1200 / 1200	SPARE	36	20	1
37	20	1	SPARE	1200		SPACE	38	—	—
39	20	1	SPARE		1200	SPACE	40	—	—
41	20	1	SPARE	1200		SPACE	42	—	—
			TOTAL	24,000	24,000				

TOTAL CONNECTED LOAD 48.2 KVA
ESTIMATED DEMAND LOAD 43.2 KVA
DEMAND LINE AMPS 200

SWF FORM 630-J REV 13 APR 72 PAGE 1

* REF. CORRIDOR 120, LAUNDRY 121, TOILET 123, STORAGE 124, DISHWASHING 125, FOOD THRU 126, BAR 127, & CORRIDOR 128.

NOTE: THERE ARE 24 EXISTING CIRCUITS THAT ARE TO BE RECONNECTED TO NEW PANEL 'A'. TRACE OUT EACH OF THESE CIRCUITS, RECONNECT TO NEW 20A/1P CIRCUIT BREAKERS & PROPERLY FILL IN PANELBOARD INDEX.

PANEL 'B' VOLTS 120/240 MAIN AMPS — PHASE 1 WIRES 3 MLO 225 A.

CIRC NO.	TRIP AMPS	NO. POLES	LOAD SERVED	PHASE LOAD V.A A	PHASE LOAD V.A B	LOAD SERVED	CIRC NO.	TRIP AMPS	NO. POLES
1	20	1	KITCHEN LIGHTING	1200 / 1200		4 WELL HOT FOOD	2	20	2
3	20	1	BUFFET LIGHTING		1200 / 1200	STATION	4		
5	20	2	FREEZER	1400 / 1414		COFFEE MAKER	6	20	1
7					1414 / 1414	ENTRE CART	8	20	1
9	20	1	REFRIGERATOR	1127 / 1125		CONVECTION OVEN	10	20	1
11	20	1	REFRIGERATOR		1200 / 1125	EXHAUST HOOD LIGHTING	12	20	1
13	20	1	BUFFET - 5 OUTLETS	1200 / 1200		FUTURE EQUIP. OUTLET	14	20	1
15	20	1	ELEC. DRINKING FOUNTAIN		1200 / 1200	FUTURE EQUIP. OUTLET	16	20	1
17	20	1	FUTURE EQUIP. OUTLET	1200 / 1200		KITCH - 4 OUTLETS	18	20	1
19	20	1	FUTURE EQUIP. OUTLET		200 / 1200	AUDITORIUM ϕ	20	20	1
21	20	1	KITCH. PERCH LIGHTING	200 / 200		AUDITORIUM ϕ	22	20	1
23	20	1	TELEPHONE CABINET ϕ		1200 / 1200	SPARE	24	20	1
25	20	1	FIRE ALARM CONTROL PANEL	1200		SPACE	26	—	—
27	—	—	SPARE		1200	SPACE	28	—	—
29	—	—	SPACE			SPACE	30	—	—
31							32		
33							34		
35							36		
37							38		
39							40		
41							42		
			TOTAL	12044	12682				

TOTAL CONNECTED LOAD 24.72 KVA
ESTIMATED DEMAND LOAD 24.72 KVA
DEMAND LINE AMPS 102.84

SWF FORM 630-J REV 13 APR 72 PAGE 1

PANEL 'C' VOLTS 120/240 MAIN AMPS — PHASE 1 WIRES 3 MLO 225 A.

CIRC NO.	TRIP AMPS	NO. POLES	LOAD SERVED	PHASE LOAD V.A A	PHASE LOAD V.A B	LOAD SERVED	CIRC NO.	TRIP AMPS	NO. POLES
1	20	1	LIGHTING: 5A, 5P, 4G, 2R, 2R	1000 / 1200		4 ϕ	2	20	1
3	20	1	2 STAGE ϕ		1200 / 1200	SOUND MODULE	4	20	1
5	20	1	STAGE ϕ	1200 / 1200		DRESSING TABLE WAREUP	6	20	1
7	20	1	2 STAGE ϕ		1200 / 1200	4 ϕ & EXHAUST PANEL	8	20	1
9	20	1	2 STAGE ϕ	1200		STAGE ϕ	10	20	1
11	20	1	3 AUDITORIUM ϕ		1200 / 1200	STAGE ϕ	12	20	1
13	20	1	4 AUDITORIUM ϕ	1200 / 1200		SOUND MODULE	14	20	1
15	40	1	DIMMER MODULE 1A		1200 / 3600	SOUND MODULE	16	20	1
17	40	1	DIMMER MODULE 2A	1200 / 3600		* SEE NOTE BELOW →	18	20	1
19	40	1	DIMMER MODULE 3A		1200 / 3600		20	20	1
21	40	1	DIMMER MODULE B	900 / 3600		MAKE-UP LIGHTS	22	20	1
23	20	1	SALAD BAR LIGHTS		900 / 1200	MAKE-UP LIGHTS	24	20	1
25	20	1	NEW EXTERIOR LIGHTS	1200 / 1200		SPARE	26	20	1
27	15	1	NEW TIMECLOCK		120 / 200	SPARE	28	20	1
29	20	1	SPARE	1200 / 1200		SPARE	30	20	1
31	—	—	SPACE			SPACE	32	—	—
33	—	—	SPACE			SPACE	34	—	—
35	—	—	SPACE			SPACE	36	—	—
37							38		
39							40		
41							42		
			TOTAL	21000	17900				

TOTAL CONNECTED LOAD 39.02 KVA
ESTIMATED DEMAND LOAD 34.22 KVA
DEMAND LINE AMPS 142.58

SWF FORM 630-J REV 13 APR 72 PAGE 1

* REF. CORRIDOR 110, MEN 111, WOMEN 112, ENTRY 113, DRESSING 114, CORRIDOR 115, & STAGE AREA 116.

** REF. CORRIDOR 110, MEN 111, WOMEN 112, & DRESSING 114.

NOTE: THERE ARE 8 EXISTING REMOTE CIRCUITS THAT ARE TO BE TRACED OUT & REPLACED W/WIRE & CIRCUIT & CONNECTED TO NEW 20A/1P BREAKERS IN NEW PANEL 'C'.

PANEL 'D' VOLTS 120/240 MAIN AMPS — PHASE 1 WIRES 3 MLO 400 A.

CIRC NO.	TRIP AMPS	NO. POLES	LOAD SERVED	PHASE LOAD V.A A	PHASE LOAD V.A B	LOAD SERVED	CIRC NO.	TRIP AMPS	NO. POLES
1	60	2	EXISTING STAGE DIMMER	5820 / 5820		EXISTING STAGE DIMMER	2	60	2
3					5820 / 5820		4		
5	60	2	EXISTING STAGE DIMMER	5820 / 5820		EXISTING STAGE DIMMER	6	60	2
7					5820 / 5820		8		
9	60	2	220V ϕ	5820 / 5820		220 V ϕ	10	60	2
11					5820 / 5820		12		
13	20	1	3 OUTLETS	600 / 600		3 HI FIXTURES	14	20	1
15	20	1	3 OUTLETS		1200 / 600		16	20	1
17	20	1	SPARE	1200 / 1200		SPARE	18	20	1
19	20	1	SPARE		1200 / 1200	SPACE	20	—	—
21	20	1	SPARE	1200 / 1200		SPACE	22	—	—
23	20	1	SPARE		1200 / 1200	SPACE	24	—	—
25							26		
27							28		
29							30		
31							32		
33							34		
35							36		
37							38		
39							40		
41							42		
			TOTAL	37620	37320				

TOTAL CONNECTED LOAD 74.94 KVA
ESTIMATED DEMAND LOAD 67.74 KVA
DEMAND LINE AMPS 282.25

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PANEL 'F' VOLTS 120/240 MAIN AMPS — PHASE 3 WIRES 4 MLO 100 A.

CIRC NO.	TRIP AMPS	NO. POLES	LOAD SERVED	PHASE LOAD VA A	PHASE LOAD VA B	PHASE LOAD VA C	LOAD SERVED	CIRC NO.	TRIP AMPS	NO. POLES
1				719 / 1200				2		
3	20	3	AIR HANDLING UNIT #1 *		719 / 1200			4	15	3
5						719 / 1200	AIR HANDLING UNIT #2 *	6		
7				719 / 1200				8		
9	40	3	AIR HANDLING UNIT #3 *		719 / 1200			10	15	3
11						719 / 1200	AIR HANDLING UNIT #4 *	12		
13				1200 / 719				14		
15	15	3	1½ HP EXHAUST FAN		1200 / 719			16	20	3
17						1200 / 719	3 HP CHILLED WATER PUMP	18		
19				719 / 719				20		
21	20	3	EXISTING KITCHEN EXHAUST FAN		719 / 719			22	20	3
23						719 / 719	EXISTING KITCHEN EXHAUST FAN	24		
25	20	1	AHU #3, #4, #5	1200 / 1200				26	20	1
27	20	1	CHILLER CONTROLS		1200 / 1200		SPARE	28	20	1
29	20	1	HEAT TAPE			1200 / 1200	SPARE	30	20	1
31	—	—	SPACE	1200			SPARE	32	20	1
33	—	—	SPACE		1200		SPARE	34	20	1
35	—	—	SPACE			1200	SPARE	36	20	1
37								38		
39								40		
41								42		
			TOTAL	11749	11954	11954				

TOTAL CONNECTED LOAD 35.657 KVA
ESTIMATED DEMAND LOAD 35.657 KVA
DEMAND LINE AMPS 99.24

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* LOW VOLTAGE TRANSFORMERS INCLUDED W/UNIT.

NOTE: METAL CONDUIT MAY SERVE AS EQUIP. GROUNDING CONDUCTOR

PANEL MDP VOLTS 240 MAIN AMPS — PHASE 3 WIRES 4 MCB/MLO 1200 A.

CIRC NO.	FUSE SIZE	SWITCH SIZE + NO. POLES	LOAD SERVED	PHASE LOAD V A A	PHASE LOAD V A B	PHASE LOAD V A C	LOAD SERVED	CIRC NO.	FUSE SIZE	SWITCH SIZE
1				17942				2		
3	35A	60A/3P	AIR COOLED WATER CHILLER		17942			4		
5						17942		6		
7	2P 200A	200A 2P	PANEL 'A'	14200				8		
9					11000			10		
11	—	—	BLANK					12		
13	2P 100A	100A 2P	PANEL 'B'	7662				14		
15					6662			16		
17	—	—	BLANK					18		
19	2P 100A	100A 2P	PANEL 'C'	6662				20		
21					11662			22		
23	—	—	BLANK					24		
25	2P 400A	400A 3P	PANEL 'D'	37620				26		
27	350	2P			37320			28		
29	—	—	BLANK					30		
31				11954				32		
33	3P 100A	100A 3P	PANEL 'F'		11954			34		
35						11954		36		
37								38		
39								40		
41								42		
			TOTAL	93218	92600	87489				

TOTAL CONNECTED LOAD 273.012 KVA
ESTIMATED DEMAND LOAD 252.718 KVA
DEMAND AT 240V: A - 972.3, B - 974.3, C - 729.1

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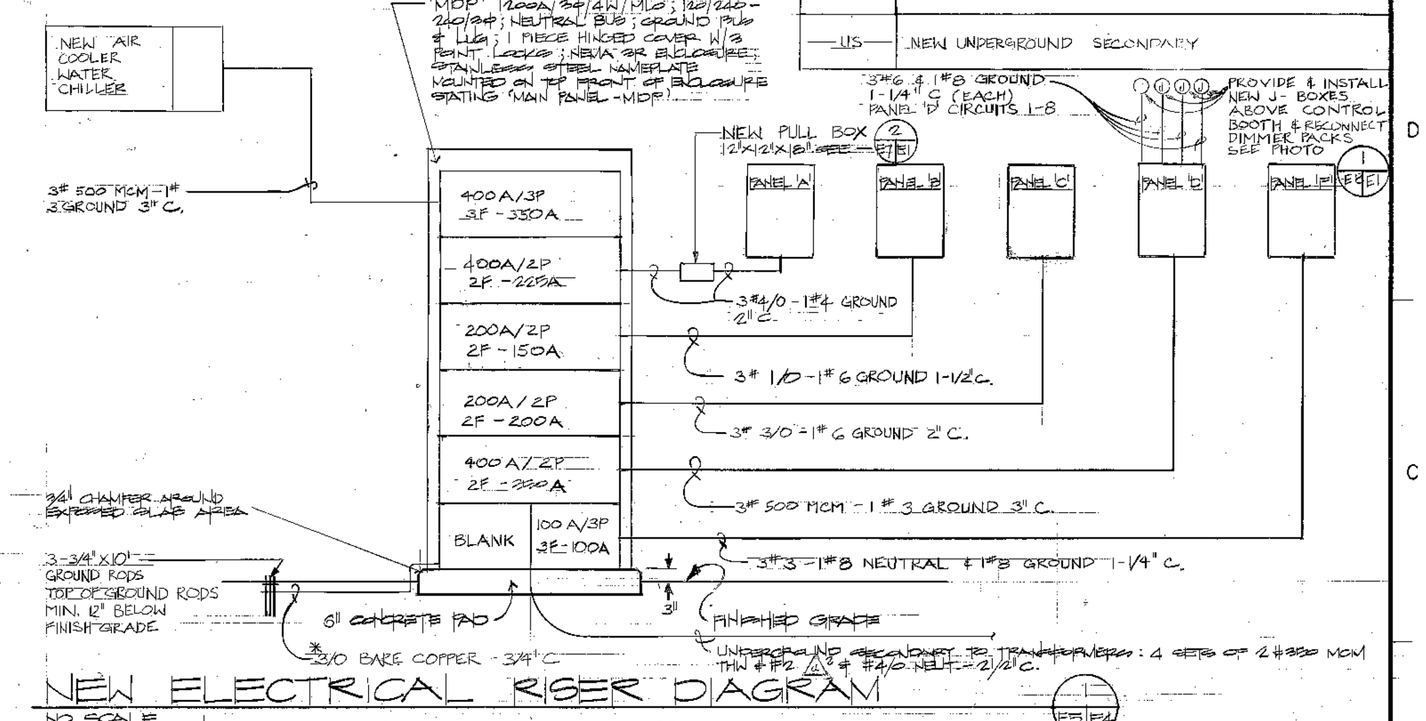
LIGHTING FIXTURE SCHEDULE

FIXT. SYMB. DESIG.	TYPE	FINISH	LENS	LAMPS	FIXTURE VOLTAGE	MANUFACTURER & CATALOG NO.	REMARKS
A	FLUORESCENT TROFFER	WHITE	ACRYLIC PRISMATIC	2-F40 T12	120	LITHONIA 2GL-240 A12 120 (OR EQUAL)	
B	RECESSED INCANDESCENT	BLACK BAFFLE	—	1-175W	120	HALO M1182 (OR EQUAL)	
C	RECESSED INCANDESCENT	BLACK BAFFLE	—	1-300W M	120	LIGHTOLIER 7259 (OR EQUAL)	
D	RECESSED ADJUSTABLE NAVY	BLACK BAFFLE	—	1-100W PAR	120	LIGHTOLIER 7249 (OR EQUAL)	WATER BAFFLE #56931
E	SURF. MOUNTED FLUORESCENT	WHITE	ACRYLIC PRISMATIC	4-F40 T12	120	LITHONIA 2GM 440 M A12 120 (OR EQUAL)	
F	SURF. MOUNTED FLUORESCENT	WHITE	ACRYLIC PRISMATIC	2-F40 T12	120	LITHONIA 2GM 240 M A12 120 (OR EQUAL)	
G	SURF. MOUNTED INCANDESCENT	ALUM.	GLASS	2-100W	120	HALO 2710 (OR EQUAL)	
H	RECESSED FACIECLIFT LIGHT	WHITE	—	1-F32W T-8	120	LIGHTOLIER 7259 (OR EQUAL)	NEW T-8 CLEAN ST. VANIA ENERGY SAVING BULB
K	SURF. MOUNTED FLUORESCENT	WHITE	ACRYLIC PRISMATIC	2-40W/T12	120	DAVIS 8340 (OR EQ.)	
M	CEILING MTD. INT. INCAND.	ALUM.	GLASS	1-100W	120	HALO 2410 (OR EQ.)	
N	WALL MTD. INT. INCAND.	ALUM.	GLASS	1-100W	120	HALO 2411 WS (OR EQ.)	W/WIREGUARD
R	WALLPAK LIGHT	WHITE	—	1-100W G40	120	W. SHARP 2120P	PROVIDE ADAPTERS TO FIT G40 LAMP IF NECESSARY
X	EXIT	ALUM.	PLASTIC	2-T20	120	LITHONIA LE SERIES EL (OR EQUAL)	W/BATTERY PACK
Z	SYMB. 2GRF8W	WHITE	EXT. TRAVERTINE MARBLE PATTERN	1-25W QUARTZ	120	CHLORIDE SPU 10 SR (OR EQUAL)	
Y	SURF. MOUNTED FLUORESCENT	WHITE	ACRYLIC PRISMATIC	2-F40 T12	120	LITHONIA LP 2GM A 120 SR (OR EQUAL)	SUITABLE FOR INSTALLATION ON COMBUSTIBLE CEILINGS
Z	INCANDESCENT STRIPLIGHT	ALUM.	TEMP. GLASS	1-25W	120	[illegible] (OR EQUAL)	MOUNT 2½' AFF ← COORDINATE INSTALLATION W/ CONTRACTOR AFTER AS THE INTEGRITY OF FIRE WALL WILL BE MAINTAINED

NOTES: 1. ROOMS CONTROLLED BY VERT. SETBACK OF AHU'S, FLUORESCENT FIXTURES SCHEDULED TO HAVE NON-LAMPS SHALL HAVE NON-LAMPS. COORDINATE W/MECH. CONTRACTOR. SEE SHEET E-2.

ELECTRICAL SYMBOLS LEGEND

A a	FLUORESCENT LIGHT FIXTURE. CAPITAL LETTER DENOTES TYPE IN FIXTURE SCHEDULE. SMALL LETTERS DENOTES SWITCH WITH SAME LETTER OPERATES THIS FIXTURE. CIRCLE IS OUTLET BOX.
C c	INCANDESCENT OR MERCURY VAPOR LIGHT FIXTURE. CAPITAL LETTER DENOTES TYPE IN FIXTURE SCHEDULE. SMALL LETTER DENOTES SWITCH WITH SAME LETTER OPERATES THIS FIXTURE.
WP	DUPLEX RECEPTACLE. 15A., 125V., GROUNDED TYPE MOUNTED 12 INCHES ABOVE FLOOR UNLESS OTHERWISE NOTED. WP INDICATES WEATHERPROOF.
▲	TELEPHONE JACK — UT — UNDERGROUND TELEPHONE CONDUIT
$a	SINGLE POLE WALL SWITCH MOUNT 4'-0" ABOVE FLOOR. SMALL LETTER DENOTES SWITCH OPERATES FIXTURE HAVING SAME LETTER.
$2 $3 $4 $K $P	SAME AS SWITCH ABOVE WITH THE FOLLOWING EXCEPTIONS: 2 DENOTES 2 POLE, 3 DENOTES 3 WAY, 4 DENOTES 4 WAY, K DENOTES KEY OPERATED, P DENOTES PILOT LIGHT.
M 5	MOTOR. NUMERAL INDICATES HORSEPOWER.
⊠	MOTOR CONTROLLER.
R	SAFETY SWITCH. UNFUSED UNLESS OTHERWISE NOTED. R INDICATES RAINTIGHT.
	PANELBOARD, LIGHTING
C	CONDUIT & CONDUCTORS CONCEALED IN WALL, CEILING OR FLOOR WITH NEUTRAL, HOT, SWITCH AND GROUND CONDUCTORS INDICATED RESPECTIVELY. NO CROSS LINES INDICATES 2 CONDUCTORS. T INDICATES TRAVELER.
C	CONDUIT & CONDUCTORS RUN EXPOSED, WITH NEUTRAL, HOT, SWITCH AND GROUND CONDUCTORS INDICATED RESPECTIVELY. NO CROSS LINES INDICATES 2 CONDUCTORS. T INDICATES TRAVELER.
	BRANCH CIRCUIT HOME RUN TO PANELBOARD WITH CIRCUIT NUMBERS SHOWN. NUMBERS OF ARROWS INDICATES NUMBER OF CIRCUITS.
J	JUNCTION BOX
GFI	GROUND FAULT INTERRUPTER TYPE OUTLET
MDP	MAIN DISTRIBUTION PANELBOARD
GFGI	GOVERNMENT FURNISHED, GOVERNMENT INSTALLED
—US—	EXISTING UNDERGROUND SECONDARY
	PICTURE DESIGNATION - ARROW DESIGNATES DIRECTION PICTURE WAS TAKEN
	REF. NOTE DESIGNATION
—US—	NEW UNDERGROUND SECONDARY



NEW ELECTRICAL RISER DIAGRAM

NO SCALE

NOTES: SEE RENOVATED FLOOR PLAN & SWITCHBOARD LOCATIONS. (1 E4/E4) & (2 E4/E4) FOR NEW PANELBOARD

2. REF. CONDUIT & WIRE SIZES OF NEW UNDERGROUND SECONDARY TO NEW MDP SEE (1 E2/E4)

SYM.	NO.	ACTION	DATE	DESCRIPTION OF REVISION
			22 MAR 82	"DRAWING OF WORK AS BUILT"
2		AM#2001	19 MAR. 81	ADDED REQUIRED TO MANUF. REF. 'Z' TYPE FIXTURE
1				REVISED 'THW' FROM FREEZE SIZE

VOIDED

KILLEBREW/RUCKER/ASSOC., INC.
600 PETROLEUM BLDG.
WICHITA FALLS, TEXAS

U.S. ARMY ENGINEER DISTRICT, FORT WORTH
CORPS OF ENGINEERS
FORT WORTH, TEXAS

DESIGNED BY: RMR
DRAWN BY: DVD
CHECKED BY: CRF

FORT SAM HOUSTON
SAN ANTONIO, TEXAS

UPGRADE BUILDING 2652
REPAIR BUILDING 2652

ELECTRICAL PANEL SCHEDULE

SUBMITTED BY:
ENGINEER:

INV. NO. DACA63-81-B-0071 DATED: 2/3/81
CONTR. NO. DACA63-81-C-0080
DRAWING NUMBER SHEET NO. E5 OF 8
SEQUENCE NO. 24A

CONTR. NO. DACA63-81-C-0080

WATER PUMP SCHEDULE

UNIT DESIGNATION		PUMP NO. 1
QUANTITY REQUIRED		ONE
LOCATION		CONCRETE PAD OUTSIDE
OPERATING WEIGHT LBS. (APPROX.)		220
PUMP SERVICE		CHILLED WATER
PUMP TYPE		END SUCTION
CAPACITY	WATER FLOW GPM	76
	TOTAL DYNAMIC HEAD, FT. OF H_2O	75
	MAX. PUMP SPEED, RPM	1750
	DESIGN EFFICIENCY	53%
MOTOR TYPE		OPEN DRIP PROOF
MOTOR STARTER	MOTOR SIZE, HP (APPROX.)	3
	ELECTRICAL SERVICE AVAILABLE	230-3-60
	MAGNETIC STARTER TYPE	SEE NOTE 1
	STARTER FURNISHED BY	MECHANICAL CONTRACTOR
ACCESSORIES	BASE	STEEL
	COUPLING GUARD	YES
	SHAFT SEAL	YES
	SUCTION AND DISCHARGE GAGE TAPPING	YES
	FITTINGS	BRONZE
EQUAL TO MANUFACTURED BY		THRUSH
SIMILAR TO MODEL NO.		PF 1½ x 1½ x 9

NOTES:

1. MAGNETIC ACROSS THE LINE WITH START/STOP PUSH BUTTON AND A RED RUNNING LIGHT.

OMIT

AIR COOLED CHILLER SCHEDULE

UNIT DESIGNATION		CHILLER NO. 1
QUANTITY REQUIRED		ONE
LOCATION		CONC. PAD OUTSIDE
APPROX. OPERATING WEIGHT		3500 LBS.
CHILLER	G.P.M.	76
	MAX. PRESS. DROP	12
	ENT. WATER TEMP. °F	57
	LEAVE WATER TEMP. °F	45
	SCALE FACTOR	0.0005
	NOMINAL TONS. MIN.	37.9
	AIR ENT. TO CONDENSER	~~105°F~~, 99°F
	BTU/HR.	454,800
COMPRESSOR	ELECTRICAL SERVICE AVAILABLE	230-3-60
	FLA/LRA (MAX.)	180/900
	No. OF COMPRESSORS (MIN.)	TWO
	COMPRESSOR TYPE	RECIPROCATING ACCESSIBLE HERMETRIC
	STARTER TYPE *	MAGNETIC ACROSS THE LINE
CONDENSER	CONDENSER FAN TYPE	~~PROPELLER~~
	QUANTITY (MAX.)	~~4 FAN & MOTOR~~
	COND. FAN HP (APPROX.)	1 H.P. (EACH)
	ELEC. SERVICE AVAILABLE	230-3-60
	ENTERING AIR TEMP	99°F
EQUAL TO MANUFACTURED BY		TECHNICAL SYSTEM INC.
SIMILAR TO MODEL NO.		CD-98

* FURNISH ~~BY MECHANICAL CONTRACTOR~~ WITH START/STOP PUSH BUTTON.

NOTE: CHILLER & PUMP SHALL BE INTERLOCKED.

NOTE:
PROVIDE MAINTAINED CONTACTS DIVICES ON CHILLER, PUMP NO. 1 AND AIR HANDLING UNITS 1, 2, 3 & 4.
SEE SPECIFICATIONS FOR STARTERS NEMA CLASSIFICATIONS.

AIR HANDLING UNIT SCHEDULE

UNIT DESIGNATION		AHU-1	AHU-2	AHU-3	AHU-4	FCU-5	FCU-6	FCU-7
EQUAL TO MANUFACTURED BY		THERMAL				CARRIER		
SIMILAR TO MODEL NO.		61	41	81	41	42CF-5	42CG-7	42CG-3
PREHEATING COIL		N/A	N/A	N/A	N/A	N/A	N/A	N/A
AREA SERVED		BUFFET	STAGE	AUDITORIUM	LOUNGE	1ST. FLR.	2ND. FLR.	
LOCATION		ATTIC SPACE					BAR	WOMEN LOUNGE
CONDENSATE DRAIN SIZE		1"	1"	1¼"	1¼"	¾"	1"	¾"
OPERATING WEIGHT (APPROX. LBS.)		750	500	1200	500	110	180	80
FAN	TOTAL AIR, CFM	2425	1630	4000	2100	400	900	210
	OUTSIDE VENT AIR, CFM (MIN.)	500	300	1200	800	NOT APPLICABLE		
	EXTERNAL STATIC PRESSURE, IN OF H_2O (APPROX.)	1.20	1.20	1.20	1.20	.35	.2	.2
	FAN MOTOR HP. (MAX.)	3 H.P.	1½	5	1½	*225 WTS	*500 WTS	*170 WTS
	ELECTRICAL SERVICE AVAILABLE	230-3-60				120-1-60		
COOLING COIL	GRAND TOTAL CAPACITY, BTU/HR. (MIN.)	97,120	52,000	183,000	118,680	12,000	29,350	5310
	TOTAL SENSIBLE CAPACITY, BTU/HR. (MIN.)	92,285	39,230	109,400	69,400	8300	15,000	3710
	ENTERING AIR TEMP. °F, DB/WB	80.3/67.3	80.2/67.0	81.0/68.2	85.2/69.5	78/63	78/65	78/65
	LEAVING AIR TEMP. °F, DB/WB	~~58.5/56.0~~	~~58.2/57.1~~	~~60.2/55.5~~	~~56.4/54.2~~			
	TOTAL GPM	16.2	8.7	30.5	19.2	2.0	7.5	0.9
	WATER ENT./LEAVE TEMP. °F	46/58	46/58	46/58	46/58	46/58	46/56	46/58
	FACE VELOCITY, FPM/DESIGN (MAX.) ***	430/550	470/550	530/550	510/600	400/180	400/285	350/200
	CONTROL VALVE/MAX. P.D.	200% OF COOLING COIL PRESSURE DROP						
	MAX. PRESS. DROP	2.5 FT. PER ROW OF COIL				5	4	5
FILTER	FACE VELOCITY, FPM MAX.	400	400	400	480	180	260	150
	QUANTITY AND SIZE (APPROX.)	(2) 20x25	(2) 16x20	(4) 16x20	(2) 16x20	10x37	10x30	16x28
	TYPE AND THICKNESS	THROW AWAY				ONE INCH THROW AWAY		

STARTER SHALL BE MAGNETIC ACROSS THE LINE WITH H.O.A. SWITCH.
* PROVIDED WITH PERMANENT SPLIT CAPACITOR MOTOR.
** PROVIDED WITH ELIMINATORS.
*** MAX. 11 FIN PER INCHES.

OMIT

HEATING & AIR CONDITIONING DESIGN CONDITIONS

LOCATION	SUMMER	WINTER
OUTSIDE	97 °FDB/ 76°FWB	30 °F DB
INSIDE	78 °FDB/ %RH	68 °F DB
ELEVATION	760 FT. ABOVE SEA LEVEL	DAILY RANGE 19 °F
DEGREE DAYS	N/A	1570

VENTILATION FAN SCHEDULE

FAN Nº	1	2 & 3	4
FAN TYPE	PRV	IN-LINE	IN-LINE
LOCATION	ROOF	ATTIC	ATTIC
MIN. OPENING REQUIRED	10x10	N/A	N/A
TOTAL AIR CFM	300	1000	7000
TOTAL S.P. (IN. OF WATER)	.25	.25	.25
DESIGN RPM	1585	1360	331
DESIGN H.P.	1/22	1/6	1½
TYPE OF DRIVE	V-BELT	V-BELT	V-BELT
EQUAL TO MANUFACTURED BY	ACME		
SIMILAR TO MODEL NO.	PRL75	PXB-12D	PXB-30H
ELECTRICAL SERVICE	SEE ELECTRICAL PLANS		
ACCESSORIES: FIRE STAT	YES	YES	YES
DISCONNECT SWITCH	YES	YES	YES
BACK DRAFT DAMPER	YES	N/A	N/A
BIRD SCREEN (1/4")	YES	N/A	N/A
PREFABRICATED CURB	YES	N/A	N/A
12 MIN. TIMER SWITCH	N/A	YES	N/A

OMIT

AIR DISTRIBUTION SCHEDULE

	DESIGNATION	DESCRIPTION	ACCESSORIES	REMARKS
A	SIDEWALL REG. (S.W.R.)	DIFFUSER SIDE WALL REGISTER (SUPPLY AIR)	OPPOSED BLADE DAMPER	ALUMINUM AIRFOIL LOUVER, STEEL FRAME WITH TWO SETS OF LOUVERS PARALLEL TO SHORT DIMENSION (SEE PLANS FOR AIR PATTERN).
B	C.D.	CEILING DIFFUSER (SUPPLY AIR)	OPPOSED BLADE DAMPER	EXTRUDED ALUMINUM LOUVERS, MANUALLY ADJUSTABLE, O.B.D. ADJUSTABLE FROM FACE OF DAMPER (SEE PLANS FOR AIR PATTERN).
C	CEILING EXH. REG.	TOILET EXHAUST	OPPOSED BLADE DAMPER	1-1/4 INCH BORDER, ALUMINUM WITH 45° LOUVERS AT 1/2 INCH SPACE WITH ONE SET OF FIXED LOUVERS PARALLEL TO LONG DIMENSION.
D	CEILING GRILLE (C.G.)	CEILING RETURN AIR GRILLE (R.A.G.)	N/A	
E	WALL RETURN GRILLE	RETURN AIR	N/A	1-1/4 INCH BORDER, ALUMINUM, WITH 45° LOUVERS AT 1/2 INCH SPACE WITH ONE SET OF FIXED LOUVERS PARALLEL TO SHORT DIMENSION.
F	DOOR GRILLE	RETURN AIR	N/A	EXTRUDED FRAME, ALUMINUM INVERTED "V" TYPE, LOUVER DEPTH 7/8" SPACING 1/2" PROVIDED WITH AUXILIARY FRAME.
G	WALL LOUVER (W.L.)	AIR INTAKE OR AIR RELIEF	1/4 WIRE MESH	EXTRUDED ALUMINUM 6063 T-5 ALLOY 0.085 THICK 4 INCH LOUVERS WITH TWO WEATHER STOPS AND FLANGE FRAME.

EXAMPLE
12"x 18" CD-B 4W
840 CFM
SIZE = 12" x 18"
TYPE = C.D.
DESIG. = B
THROW PATTERN = 4W

DUCT FURNACE NOTES:

1. DESIGNED PILOT WITH ELECTRONIC FLAME SUPERVISION WITH 100% LOCKOUT ON FLAME FAILURE.
2. ALL FURNACES TO HAVE #409 (AISI) STAINLESS STEEL HEAT EXCHANGERS.

DUCT FURNACE SCHEDULE

UNIT DESIGNATION		1	2	3	4
QUANTITY REQUIRED		ONE	ONE	ONE	ONE
LOCATION		ATTIC SPACE			
OPERATING WEIGHT (MAX.)		160	160	160	160
TOTAL STATIC PRESS. (IN. OF WG)		0.05	0.15	0.15	0.15
TOTAL AIR CFM		500	1350	13	800
FLUE SIZE INCHES ø		5	5	6	6
VENTER		N/A	N/A	N/A	N/A
ELEC. SERVICE		CONTROL 24 VOLTS (120-1ø0)			
GAS	BTU/HR (MIN. OUTPUT) *	41,000	26,000	78,800	70,000
	CU. FT. INPUT	34	34	103	91
	PILOT SYSTEM	SEE NOTE 1			
	PIPE SIZE	1/2	1/2	1/2	1/2
EQUAL TO MANUFACTURED BY		REZNOR			
SIMILAR TO MODEL NO.		HX-75	HX-75	HX-100	HX-100

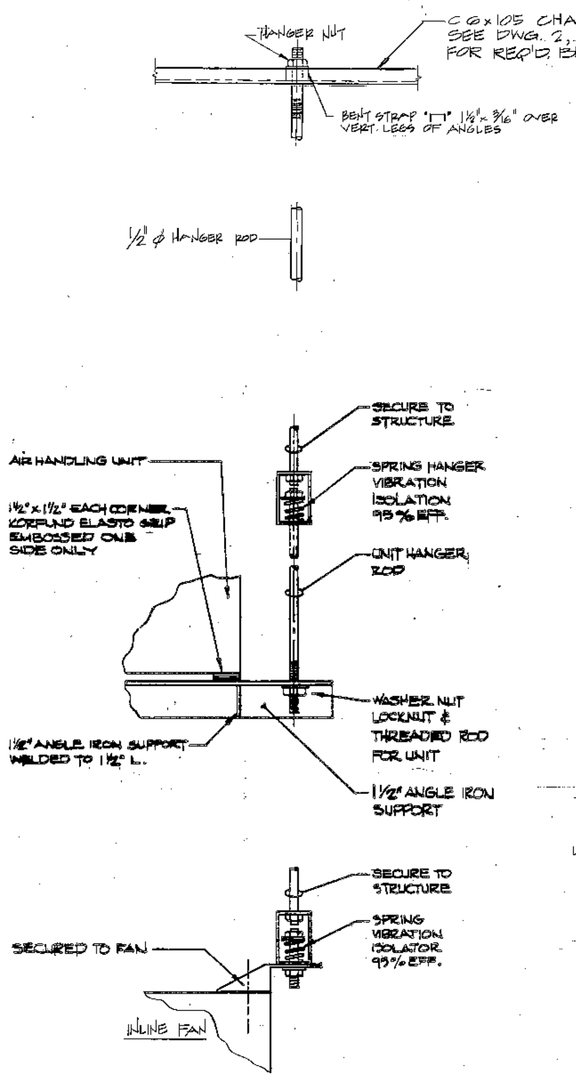
* MINIMUM REQUIRED
NOTE: FURNACE SHALL BE DESIGNED FOR 100% O.A.

OMIT

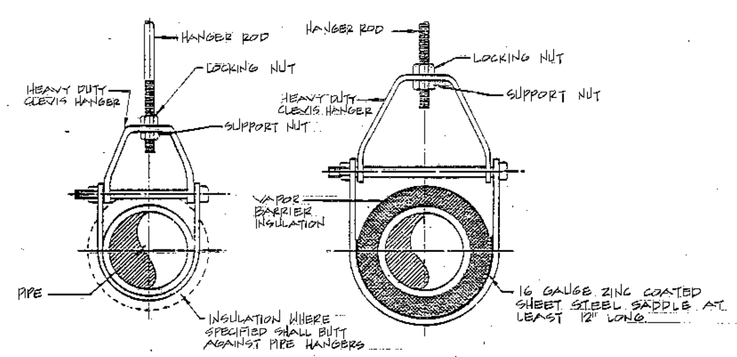
3/6 (*) STATIC PRESSURE CONSISTS OF DUCTWORK, DIFFUSERS, BUT STATIC PRESSURE DOES NOT INCLUDE APPARATUS LOSSES (COILS, FILTER CASINGS) IN A.H.U.

GENERAL NOTES:

1. ALL DUCT WORK SIZES SHOWN ARE INSIDE DIMENSIONS.
2. DIRECTIONAL ARROWS INDICATE AIR FLOW.
3. FIRE DAMPERS AS ~~APPROVED BY LOCAL GOVERNING AUTHORITIES~~ SPECIFIED SHALL BE INSTALLED IN ALL DUCTWORK PASSING THROUGH A FLOOR, ROOF, FIRE PARTITION OR FIRE WALL (PROVIDE ACCESS) AS SHOWN ON THE DRAWINGS.
4. CONTRACTOR SHALL PROVIDE POSITIVE MEANS FOR BALANCING EACH INDIVIDUAL AIR CONDITIONING SUPPLY AIR OUTLET, AS PER SCHEDULES AND DRAWINGS.
5. EQUIPMENT QUANTITIES GIVEN ON SCHEDULES ARE MINIMUM REQUIRED CONTRACTOR IS RESPONSIBLE FOR PROVIDING EQUIPMENT THAT MEETS THESE MINIMUM QUANTITIES.
6. ALL A/C AND VENTILATION SYSTEMS SHALL BE BALANCED FOR AIR QUANTITIES SHOWN AGAINST THE ACTUAL INSTALLED STATIC PRESSURE OF EACH SYSTEM.
7. CONTRACTOR SHALL COORDINATE WITH BUILDING STRUCTURE FOR ALL ROOF AND FLOOR OPENING.
8. CONDENSATE PIPING SHALL BE PITCHED 1/8" PER FEET MIN., WHERE HORIZONTAL RUNS ARE EMPLOYED WITH A PITCH OF LESS THAN 1" PER 10 FEET INCREASE TO ONE PIPE SIZE LARGER.
9. ALL HORSEPOWER RATINGS ARE APPROXIMATE.
10. ALL 5 HORSEPOWER AND LARGER MOTORS WILL HAVE A .95 OR GREATER POWER FACTOR.



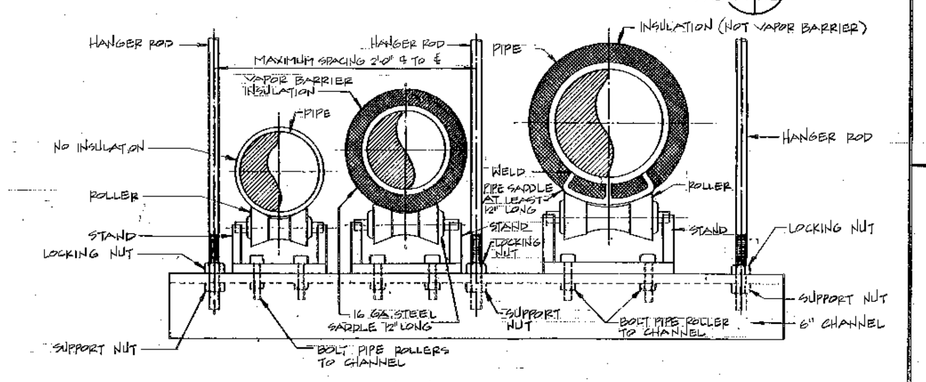
HANGING DETAIL 1
NO SCALE M-3 & 4 | M-2



CLEVIS HANGER
NO VAPOR BARRIER INSULATION

CLEVIS HANGER
NO VAPOR BARRIER INSULATION

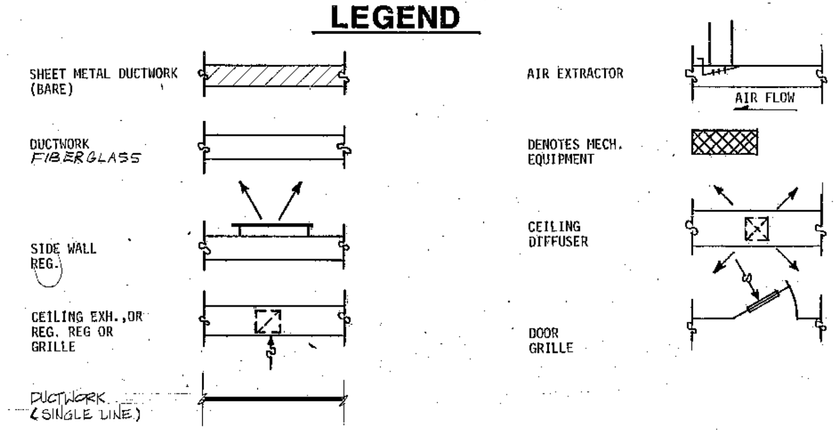
PIPE HANGING DETAIL 2
NO SCALE M-3 & 4 | M-2



TRAPEZE HANGER
ALL MULTIPLE HORIZONTAL RUNS WITH OR WITHOUT VAPOR BARRIERS INSULATION

PIPE SUPPORT DETAIL 3
NO SCALE M-3 & 4 | M-2

LEGEND



VOIDED

SYM.	DO. NO.	ACTION.	DATE	DESCRIPTION OF REVISION
			22MAR82	"REVISED AS BUILT"
6	AM#0001		15MAR81	SCHEDULE REVISIONS

KILLEBREW/RUCKER/ASSOC., INC.
600 PETROLEUM BLDG.
WICHITA FALLS, TEXAS

U.S. ARMY ENGINEER DISTRICT, FORT WORTH
CORPS OF ENGINEERS
FORT WORTH, TEXAS

DESIGNED BY: J.A.G.
DRAWN BY: K.G.
CHECKED BY: J.A.O.
SUBMITTED BY:
ENGINEER:

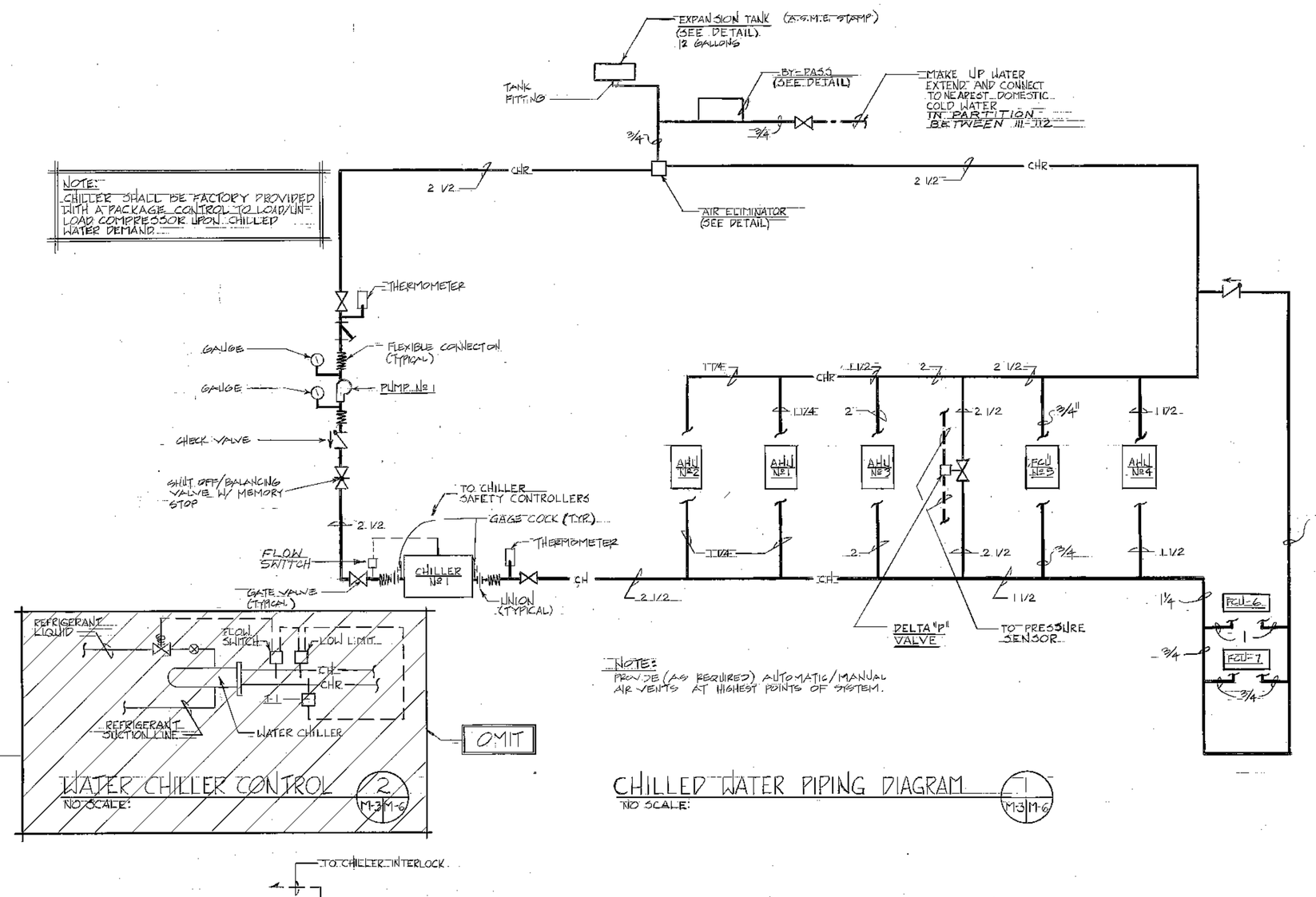
FORT SAM HOUSTON
SAN ANTONIO, TEXAS

UPGRADE BUILDING 2652
REPAIR BUILDING 2652

MECH. SCHEDULES & DETAILS

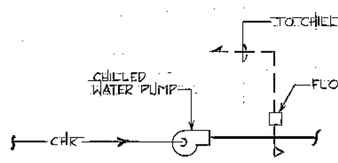
INV. NO. DACA63-81-B-0071 | DATED: 2/13/81
CONTR. NO. DACA63-81-C-0080
DRAWING NUMBER | SHEET NO. M-2 OF 8 | SEQUENCE NO. 13A

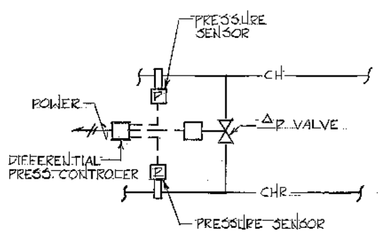
CONTR. NO. DACA63-81-C-0080



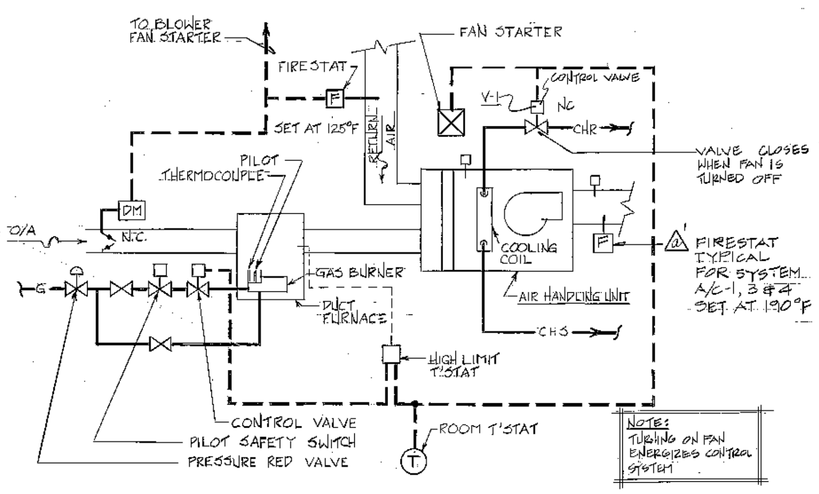
CHILLED WATER PIPING DIAGRAM (1) M-3 | M-6
NO SCALE:

WATER CHILLER CONTROL (2) M-3 | M-6
NO SCALE:





CHILLED WATER PUMP AND BY-PASS (3) M-3 | M-6
NO SCALE:



TYPICAL FOR A/C SYSTEMS 1 THRU 4 (4) M-3 | M-6
NO SCALE:

I. CONTROL SYSTEM DESCRIPTION:

SYSTEM WILL PROVIDE COMPLETE CONTROL WITH A SINGLE UNIT PER ZONE AS SHOWN ON PLANS. (HEATING OR COOLING MODE,) MANUALLY SELECTED.

II. SEQUENCES:

A. CONTROL PANEL

HEATING MODE
ZONE THERMOSTAT SHALL BE MANUALLY SET ON HEATING MODE.

THE TIME CLOCK WILL SET THE SYSTEM ON/OFF AS PROGRAMED.

AN OUTSIDE SENSOR SHALL OVERRIDE THE ON TIME SCHEDULE ON DROP OF TEMPERATURE TO PREVENT FREEZING IN THE BUILDING, ALLOWING THE FOUR AIR HANDLING UNITS AND PUMP No 1 TO START. CHILLER SWITCH SHALL BE SET IN THE OFF POSITION TO PREVENT THE CHILLER FROM STARTING. ON A CALL FOR HEATING, SPACE THERMOSTAT SHALL START AIR HANDLING BLOWER FAN AND OPERATE THE GAS FIRED DUCT FURNACE TO MAINTAIN SPACE TEMPERATURE.

COOLING MODE
ZONE THERMOSTAT SHALL BE MANUALLY SET ON COOLING MODE.

CHILLER
ON A CALL FOR COOLING (CHILLED WATER), THE REFRIGERANT SHALL BE START/STOP. BY A SOLENOID VALVE IN THE REFRIGERANT LIQUID LINE, CONTROLED BY A CHILLED WATER THERMOSTAT. INTERLOCKS TO IMPROVE CHILLED WATER FLOW AND PREVENT FREEZE UP ARE REQUIRED. THE OPERATING THERMOSTAT T1 IN THE CHILLED WATER RETURN WILL OPEN THE SOLENOID VALVE IN THE REFRIGERANT LIQUID LINE PROVIDING THAT THE WATER IS FLOWING AT NOT NEAR FREEZING. A FLOW SWITCH IN THE CHILLED WATER SUPPLY LINE SHALL BE INTERLOCKED TO PREVENT CHILLER FROM OPERATING UNTIL FLOW IN THE SYSTEM HAS BEEN PROVEN.

B. SYSTEM AHU-1 THRU 4
ON A CALL FOR COOLING THE FLOW OF CHILLED WATER THROUGH THE COOLING COIL SHALL BE CONTROLLED BY A VALVE V1 WHICH SHALL BE MODULATED BY SPACE THERMOSTAT.

C. SYSTEM FCU-5 THRU 7
ON A CALL FOR COOLING THE FLOW OF CHILLED WATER THRU THE COOLING COIL SHALL BE CONTROLLED BY A VALVE WHICH SHALL BE OPERATED IN TWO POSITION BY SPACE THERMOSTAT.

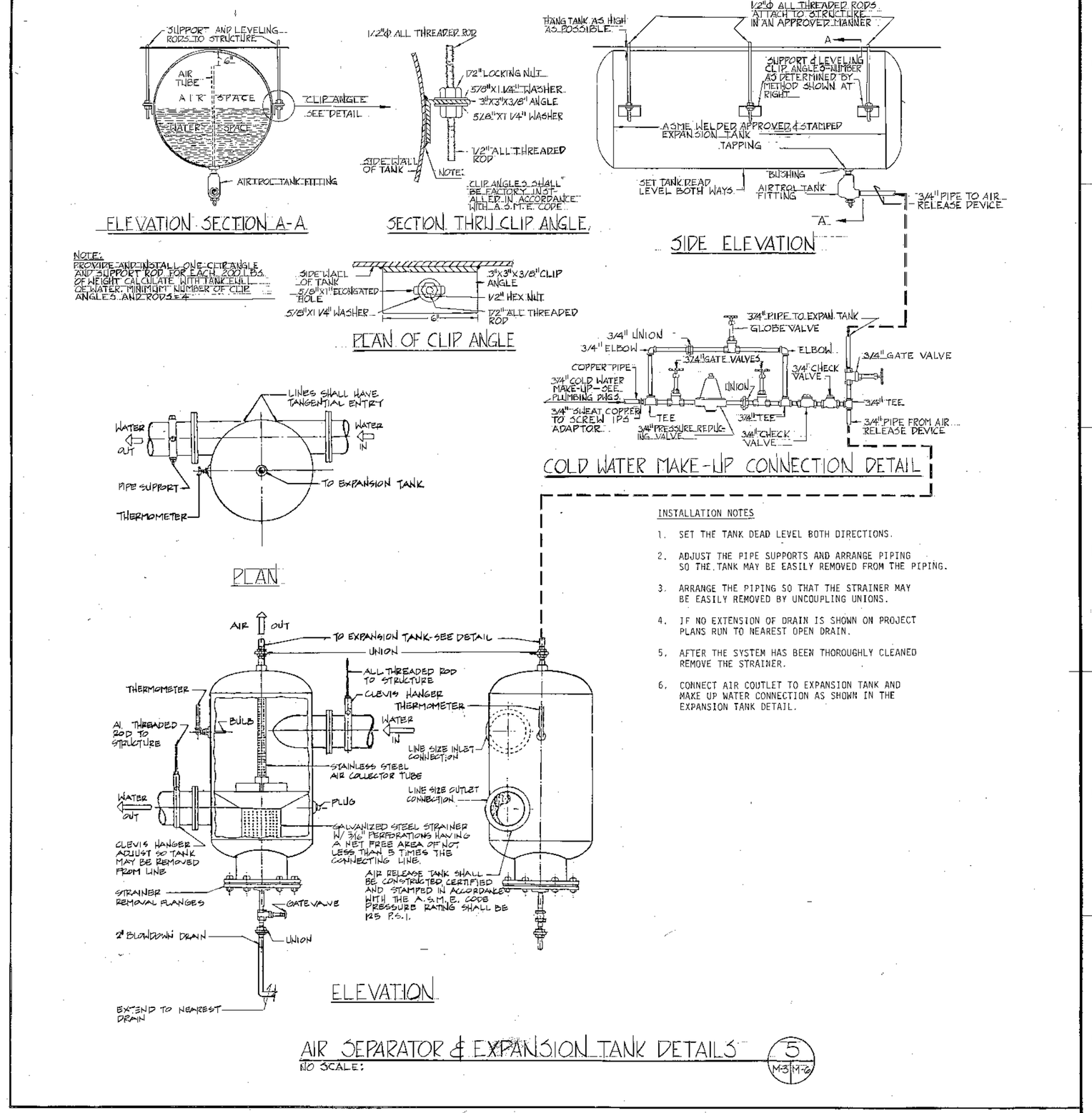
D. PRESSURE DIFFERENTIAL VALVE
A DUAL IN-PUT DIFFERENTIAL PRESSURE CONTROLLER WILL MONITOR THE DIFFERENCE IN PRESSURE BETWEEN THE SUPPLY AND RETURN LINES. THIS DIFFERENTIAL CONTROLLER SHALL MODULATE A MOTORIZED TWO-WAY VALVE (ΔP VALVE) TO MAINTAIN A CONSTANT SYSTEM DIFFERENTIAL, PROVIDING ADEQUATE BY-PASS TO MAINTAIN PUMP INSTALLED TOTAL DYNAMIC HEAD.

E. FIRESTAT
ON A RISE OF TEMPERATURE FIRESTAT SHALL DE-ENERGIZE THE FAN (AS INDICATED). SET FIRESTAT AT
F. 125° F OTHERWISE NOTED. PROVIDE A SET OF AUXILIARY CONTACTS FOR FUTURE FIRE ALARM SYSTEM. CONTROLS REQUIRE A MANUAL RESET TO RESTART FAN.

G. FAN No 1
SHALL OPERATE ALL THE TIME WHEN ENERGIZED.

H. FAN No 2 & 3
SPACE EXHAUST SHALL OPERATE ONLY WHEN ENERGIZED THRU A 15 MINUTE TIMER SWITCH.

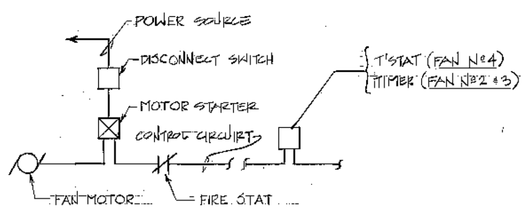
I. FAN No 4
ATTIC VENTILATION SHALL OPERATE ONLY WHEN ENERGIZED THROUGH A THERMOSTAT. PROVIDE A WALL SWITCH AS SHOWN ON DRAWINGS (WINTER OFF/SUMMER ON.)



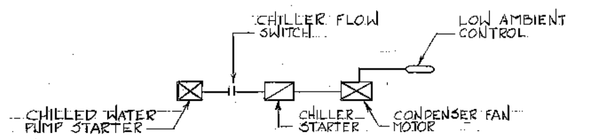
INSTALLATION NOTES

1. SET THE TANK DEAD LEVEL BOTH DIRECTIONS.
2. ADJUST THE PIPE SUPPORTS AND ARRANGE PIPING SO THE TANK MAY BE EASILY REMOVED FROM THE PIPING.
3. ARRANGE THE PIPING SO THAT THE STRAINER MAY BE EASILY REMOVED BY UNCOUPLING UNIONS.
4. IF NO EXTENSION OF DRAIN IS SHOWN ON PROJECT PLANS RUN TO NEAREST OPEN DRAIN.
5. AFTER THE SYSTEM HAS BEEN THOROUGHLY CLEANED REMOVE THE STRAINER.
6. CONNECT AIR OUTLET TO EXPANSION TANK AND MAKE UP WATER CONNECTION AS SHOWN IN THE EXPANSION TANK DETAIL.

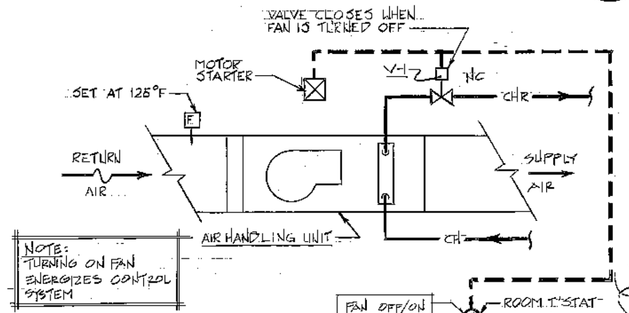
AIR SEPARATOR & EXPANSION TANK DETAILS (5) M-3 | M-6
NO SCALE:



FAN CONTROL DIAGRAM (6) M-3 | M-6



TEMPERATURE CONTROL DIAGRAM (7) M-3 | M-6
NO SCALE:



TYP. FOR A/C SYSTEM 5,6,7 (8) M-3 | M-6
NO SCALE:

SYM	DR. NO.	ACTION	DATE	DESCRIPTION OF REVISION
			22MAR82	"REVISED AS BUILT"
4	AM#00001		13 MAR.81	ADDED FIRESTAT TO DWG. NO 4

KILLEBREW/RUCKER/ASSOC., INC.
600 PETROLEUM BLDG.
WICHITA FALLS, TEXAS

U.S. ARMY ENGINEER DISTRICT, FORT WORTH
CORPS OF ENGINEERS
FORT WORTH, TEXAS

DESIGNED BY: J.A.C.
DRAWN BY: K.G.
CHECKED BY: J.A.C.
SUBMITTED BY:
ENGINEER:

FORT SAM HOUSTON
SAN ANTONIO, TEXAS

UPGRADE BUILDING 2652
REPAIR BUILDING 2652
CHILLED WATER PIPING DIAGRAMS & DETAILS

INV. NO. DACA63-81-B-0071 DATED: 2/3/81
CONTR. NO DACA63-81-C-0080
DRAWING NUMBER: M-5 OF 8
SEQUENCE NO. 17

CONTR. NO. DACA63-81-C-0080